3-1-1b

1. If $y=\sqrt{3x^2+1}$, then $\frac{dy}{dx}=$

(A) $\frac{1}{2\sqrt{6x}}$

(B) $\frac{1}{2\sqrt{3x^2+1}}$

(C) $\frac{3x}{\sqrt{3x^2+1}}$ ✓

(D) $3x\sqrt{3x^2+1}$

Answer C

Correct. A combination of the power rule and the chain rule is used to compute the derivative.
$\frac{dy}{dx}=\frac{d}{dx}\left(3x^2+1\right)^{\frac{1}{2}}=\frac{1}{2}\left(3x^2+1\right)^{-\frac{1}{2}}\cdot\frac{d}{dx}\left(3x^2+1\right)=\frac{1}{2}\left(3x^2+1\right)^{-\frac{1}{2}}\cdot(6x)=\frac{3x}{\sqrt{3x^2+1}}$

2. If $f(x)=\sqrt{x^4+8}$, then $f'(-1)=$

(A) $-\frac{4}{3}$

(B) $-\frac{2}{3}$ ✓

(C) $-\frac{1}{6}$

(D) $\frac{1}{6}$

Answer B

Correct. A combination of the power rule and the chain rule is used to compute the derivative.
$f'(x)=\frac{d}{dx}\left(x^4+8\right)^{\frac{1}{2}}=\frac{1}{2}\left(x^4+8\right)^{-\frac{1}{2}}\cdot\frac{d}{dx}\left(x^4+8\right)=\frac{1}{2}\left(x^4+8\right)^{-\frac{1}{2}}\cdot\left(4x^3\right)=\frac{2x^3}{\sqrt{x^4+8}}$
$f'(-1)=\frac{2(-1)}{\sqrt{9}}=-\frac{2}{3}$

3. If $f(x)=\sin\left(e^{3x}\right)$, then $f'(x)=$

(A) $\cos\left(e^{3x}\right)$

(B) $\cos\left(3e^{3x}\right)$

(C) $e^{3x}\cos\left(e^{3x}\right)$

(D) $3e^{3x}\cos\left(e^{3x}\right)$ ✓

Answer D

Correct. The chain rule must be used twice in doing the differentiation, as follows.
$f'(x)=\cos\left(e^{3x}\right)\cdot\frac{d}{dx}\left(e^{3x}\right)=\cos\left(e^{3x}\right)\cdot e^{3x}\cdot\frac{d}{dx}(3x)=\cos\left(e^{3x}\right)\cdot e^{3x}\cdot 3$

3-1-1b

4. An object moves along the x-axis with initial position $x(0) = 2$. The velocity of the object at time $t \geq 0$ is given by $v(t) = \sin\left(\frac{\pi}{3}t\right)$.

What is the acceleration of the object time $t = 4$?

Part A

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

0	1

The student response earns all of the following points:

1 point is earned for the answer

$$a(4) = v'(4) = \frac{\pi}{3}\cos\left(\frac{4\pi}{3}\right)$$
$$= -\frac{\pi}{6} \text{ or } -0.523 \text{ or } -0.524$$

3-1-1b

5. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of mosquitoes in a field after a major rainfall is modeled by the function M defined by $M(t) = -t^3 + 12t^2 + 144t$, where t is the number of days after the rainfall ended and $0 \le t \le 18$.

(a) Using correct units, interpret the meaning of $M'(15) = -171$ in the context of the problem.

(b) Based on the model, what is the absolute maximum number of mosquitoes in the field over the time interval $0 \le t \le 18$? Justify your answer.

(c) For what values of t is the rate of change of the number of mosquitoes in the field increasing?

(d) For $0 \le t \le 18$, the number of bats in the field is modeled by the differentiable function B, where B is a function of the number of mosquitoes in the field. Based on the models, write an expression for the rate of change of the number of bats in the field at time $t = a$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1

The student response accurately includes a correct interpretation of $M\prime(15) = -171$

Solution:

At time $t = 15$ days, the number of mosquitoes in the field is decreasing at a rate of 171 mosquitoes per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $M'(t)$ with a maximum of one computational error.

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4

3-1-1b

The student response accurately includes all four of the criteria below.

- ☐ $M'(t)$
- ☐ sets $M'(t) = 0$
- ☐ identifies $t = 12$ as a candidate
- ☐ answer with justification

Solution:

$M'(t) = -3t^2 + 24t + 144 = -3\left(t^2 - 8t - 48\right) = -3(t-12)(t+4)$

$M'(t) = 0 \ \Rightarrow \ t = 12$

Because $M'(t)$ changes from positive to negative at $t = 12$, M has a relative maximum at $t = 12$.

Because $t = 12$ is the only critical point of M on the interval $0 \leq t \leq 18$, that is the location of a relative maximum. It is also the location of the absolute maximum of M on the interval $0 \leq t \leq 18$.

$M(12) = -12^3 + 12\left(12^2\right) + 144(12) = 144(12) = 1728$

The absolute maximum number of mosquitoes in the field over the time interval $0 \leq t \leq 18$ is $M(12) = 1728$.

Part C

The third point may be earned based on a consistent answer using an incorrect $M''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ considers $M''(t)$
- ☐ expression for $M''(t)$
- ☐ answer

Solution:

$M''(t) = -6t + 24 = -6(t-4)$

$M''(t) = -6(t-4) > 0 \ \Rightarrow \ t < 4$

The rate of change of the number of mosquitoes in the field is increasing for $0 \leq t \leq 4$.

Part D

The point is earned with $B'(M(a)) \cdot M'(a)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1b

0	1

The student response accurately includes a correct expression.

Solution:

The number of bats in the field at time $t = a$ is given by $B\left(M\left(a\right)\right)$.

Therefore, the rate of change of the number of bats in the field at time $t = a$ is given by $B'\left(M\left(a\right)\right) \cdot M'\left(a\right) = B'\left(-a^3 + 12a^2 + 144a\right) \cdot \left(-3a^2 + 24a + 144\right)$.

3-1-1b

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The number of fish in a small bay is modeled by the function F defined by $F(t) = 10\left(t^3 - 12t^2 + 45t + 100\right)$, where t is measured in days and $0 \le t \le 8$.

(a) Using correct units, interpret the meaning of $F'(4) = -30$ in the context of the problem.

(b) Based on the model, what is the absolute minimum number of fish in the bay over the time interval $0 \le t \le 8$? Justify your answer.

(c) For what values of t, $0 \le t \le 8$, is the rate of change of the number of fish in the bay decreasing?

(d) For $0 \le t \le 8$, the number of pelicans flying near the bay is modeled by the differentiable function P, where P is a function of the number of fish in the bay. Based on the models, write an expression for the rate of change of the number of pelicans flying near the bay at time $t = c$.

Part A

The point requires a time reference, rate, and units.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1 ✓

The student response accurately includes a correct interpretation of $F\prime(4) = -30$.

Solution:

At time $t = 4$ days, the number of fish in the bay is decreasing at a rate of 30 fish per day.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $F'(t)$ with a maximum of one computational error.

The fourth point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3	4 ✓

3-1-1b

The student response accurately includes all four of the criteria below.

- ☐ $F'(t)$
- ☐ sets $F'(t) = 0$
- ☐ identifies $t = 5$ as a candidate
- ☐ answer with justification

Solution:

$F'(t) = 10\left(3t^2 - 24t + 45\right) = 10 \cdot 3\left(t^2 - 8t + 15\right) = 30(t-3)(t-5)$

$F'(t) = 0 \ \Rightarrow \ t = 3 \text{ and } t = 5$

Because $F'(t)$ changes from positive to negative at $t = 3$ and from negative to positive at $t = 5$, consider $t = 0$ and $t = 5$ for the location of the absolute minimum. F has a relative minimum at $t = 5$.

$F(0) = 1000$

$F(5) = 10\left(5^3 - 12(5)^2 + 45(5) + 100\right) = 1500$

The absolute minimum number of fish in the bay over the time interval

$0 \le t \le 8$ is $F(0) = 1000$.

Part C

The third point may be earned based on a consistent answer using an incorrect $F''(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ considers $F''(t)$
- ☐ expression for $F''(t)$
- ☐ answer

Solution:

$F''(t) = 10(6t - 24) = 60(t-4)$

$F''(t) = 60(t-4) < 0 \ \Rightarrow \ t < 4$

The rate of change of the number of fish in the bay is decreasing for $0 \le t \le 4$.

Part D

The point is earned with $P'(F(c)) \cdot F'(c)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1b

0	1

The student response accurately includes a correct expression.

Solution:

The number of pelicans flying near the bay at time $t = c$ is given by $P\left(F\left(c\right)\right)$.

Therefore, the rate of change of the number of pelicans flying near the bay at time $t = c$ is given by $P'\left(F\left(c\right)\right) \cdot F'\left(c\right) = P'\left(10\left(c^3 - 12c^2 + 45c + 100\right)\right) \cdot \left(10\left(3c^2 - 24c + 45\right)\right)$.

3-1-1b

7. Let *f* be the function given by $f(x) = \frac{x}{\sqrt{x^2-4}}$.

(a) Find the domain of *f*.

(b) Write an equation for each vertical asymptote to the graph of *f*.

(c) Write an equation for each horizontal asymptote to the graph of *f*.

(d) Find *f′*(*x*).

Part A

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$\boxed{x < -2 \text{ or } x > 2}$
OR $|x| > 2$

0	1	2

The student response earns all of the following points:

1 point is earned for $x^2 - 4 > 0$

1 point is earned for correct solution

$\boxed{x < -2 \text{ or } x > 2}$
OR $|x| > 2$

Part B

1 point is earned for finding value of *x*

$\boxed{x = 2, x = -2}$

0	1

The student response earns all of the following points:

1 point is earned for finding value of *x*

$\boxed{x = 2, x = -2}$

3-1-1b

Part C

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \to +\infty} \frac{x}{\sqrt{x^2-4}} = 1$$
$$\lim_{x \to -\infty} \frac{1}{\sqrt{x^2-4}} = -1$$
$$\boxed{y = 1, y = -1}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for analysis in both direction

1 point is earned for 1st consistent solution

1 point is earned for 2nd consistent solution

$$\lim_{x \to +\infty} \frac{x}{\sqrt{x^2-4}} = 1$$
$$\lim_{x \to -\infty} \frac{1}{\sqrt{x^2-4}} = -1$$
$$\boxed{y = 1, y = -1}$$

Part D

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$f'(x) = \frac{\sqrt{x^2-4} - x\left[\frac{1}{2}(x^2-4)^{-1/2} 2x\right]}{x^2-4}$$
$$= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4}$$
$$= \frac{-4}{(x^2-4)^{3/2}}$$

0	1	2	3

3-1-1b

The student response earns all of the following points:

3 points are earned for finding $f'(x)$

2 points are **deducted** for Chain rule error, Quotient (Product) rule error; and calculus error in exponent

1 point is **deducted** for all other errors

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4}-x\left[\frac{1}{2}\left(x^2-4\right)^{-1/2}2x\right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4}-\frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{\left(x^2-4\right)^{3/2}} \end{aligned}$$

3-1-1b

8. Let f be the function given by $f(x)$=ln(x / x−1)

Find the value of the derivative of f at x=−1 .

Part B

2 point is earned for finding the derivative

2 point is deducted for error in chain,product, quotient or derivative of ln.

1 point is deducted for all other error

1 point is earned for evaluating his/her f'(x) at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$
$$= \frac{-1}{x(x-1)}$$
or
$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$
$$f'(-1) = -\frac{1}{2}$$

0	1	2	3

The student response earns three of the following points:

2 point is earned for finding the derivative

2 point is deducted for error in chain,product, quotient or derivative of ln.

1 point is deducted for all other error

1 point is earned for evaluating his/her f'(x) at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$
$$= \frac{-1}{x(x-1)}$$
or
$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$
$$f'(-1) = -\frac{1}{2}$$

3-1-1b

9. Let f be the function given by $f(x) = \ln(x / x - 1)$.

(a) *What is the domain of f?*

(b) Find the value of the derivative of f at $x = -1$.

(c) Write an expression for $f^{-1}(x)$, where f^{-1} denotes the inverse function of f.

Part A

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$\frac{x}{x-1} > 0$

$x > 0$ and $x - 1 > 0 \Rightarrow x > 1$

$x < 0$ and $x - 1 < 0 \Rightarrow x < 0$

$\boxed{x < 0 \text{ or } x > 1}$

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $\frac{x}{x-1} > 0$

1 point is earned for first interior point

1 point is earned for second interior point

1 point is **deducted** for each incorrect point

$\frac{x}{x-1} > 0$

$x > 0$ and $x - 1 > 0 \Rightarrow x > 1$

$x < 0$ and $x - 1 < 0 \Rightarrow x < 0$

$\boxed{x < 0 \text{ or } x > 1}$

Part B

2 point is earned for finding the derivative

2 point is **deducted** for error in chain,product, quotient or derivative of ln.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f'(x)$ at $x = -1$

3-1-1b

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$
$$= \frac{-1}{x(x-1)}$$
or
$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$
$$\boxed{f'(-1) = -\frac{1}{2}}$$

0	1	2	3 ✓

The student response earns all of the following points:

2 point is earned for finding the derivative

2 point is **deducted** for error in chain,product, quotient or derivative of ln.

1 point is **deducted** for all other error

1 point is earned for evaluating his/her $f(x)$ at $x = -1$

$$f'(x) = \frac{x-1}{x} \cdot \frac{(x-1)-x}{(x-1)^2}$$
$$= \frac{-1}{x(x-1)}$$
or
$$\ln|x| - \ln|x-1| \Rightarrow f'(x) = \frac{1}{x} - \frac{1}{x-1}$$
$$\boxed{f'(-1) = -\frac{1}{2}}$$

Part C

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$$y = \ln\left(\frac{x}{x-1}\right)$$
$$e^y = \frac{x}{x-1}$$
$$x(e^y - 1) = e^y$$
$$x = \frac{e^y}{e^y-1}$$
$$\boxed{f^{-1}(x) = \frac{e^x}{e^x-1}}$$

3-1-1b

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for correctly introducing exponential

1 point is earned for solving for x in terms of e^y

1 point is earned for expressing solution in terms of x

$y = \ln\left(\frac{x}{x-1}\right)$
$e^y = \frac{x}{x-1}$
$x(e^y - 1) = e^y$
$x = \frac{e^y}{e^y-1}$
$\boxed{f^{-1}(x) = \frac{e^x}{e^x-1}}$

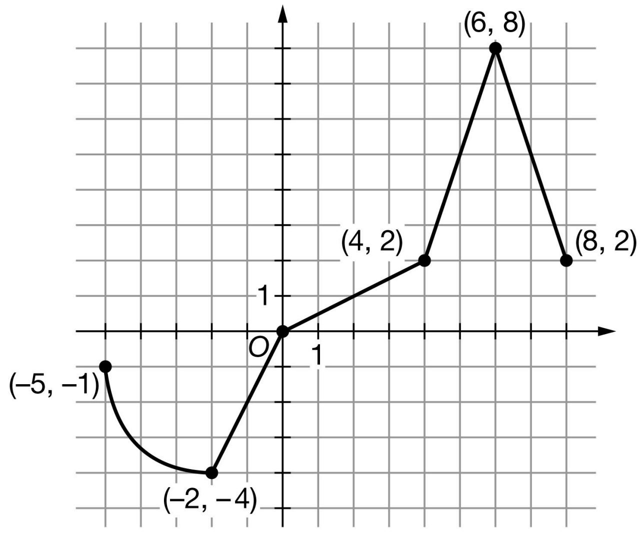
3-1-1b

10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of a quarter circle and four line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^{x} g(t)\, dt$ for $-5 \leq x \leq 8$.

(a) Find the value of $h'(4)$ or explain why it does not exist.

(b) At what value of x does h attain its minimum value on the interval $-5 \leq x \leq 8$? Justify your answer.

(c) Find the x-coordinate of all points of inflection of the graph of h for $-5 < x < 8$. Justify your answer.

(d) Let k be the function defined by $k(x) = (g(x))^3$. Find the value of $k'(7)$. Show the computations that lead to your answer.

(e) Find $\lim_{x \to -2} \frac{h(x)}{3x+6}$. Give a reason for your answer.

(f) Find the value of $h(6)$.

(g) For what value of r does the series $30r + 30r^2 + \cdots + 30r^n + \cdots$ equal the value of $h(6)$ found in part (f)? Show the computations that lead to your answer.

(h) Does the Mean Value Theorem guarantee a value of c, for $-5 < c < 8$, such that $h''(c)$ is equal to the average rate of change of h' over the interval $-5 \leq x \leq 8$? Why or why not?

(i) What is the value of $\int_{-5}^{-2} \sqrt{1 + (g'(x))^2}\, dx$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer of 2.

The response can earn the point without showing any supporting work: the correct answer is enough.

3-1-1b

Even with a correct answer, a response that includes false claims ("says too much") will not earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x)$

$h'(4) = g(4) = 2$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by stating that $x = 0$ is a critical point.

Second point:

Eligibility: The response must earn the first point to be eligible for the second point.

The second point is earned, for example, by a statement like "g changes from negative to positive at $x = 0$."

This verification requires the explicit connection of $h' = g$. The point cannot be earned for " h' changes from negative to positive at $x = 0$."

A response of " $h' = g$, and h' changes from negative to positive at $x = 0$" would earn the second point. Note: The connection "$h' = g$" might have been made in part (a). Whenever the connection is made, it carries through the rest of this problem.

Language like "g changes at $x = 0$" is ambiguous and will not earn the point.

Words such as "the graph" are interpreted to be referring to g, the graph given in the prompt.

The terms "the function" or "the derivative" are ambiguous and cannot earn this point.

Third point:

This requires justifying $x = 0$ as the location of an absolute minimum.

Eligibility: The response must earn the second point to be eligible for the third point.

The response may earn the third point by appealing to the uniqueness of the critical point at $x = 0$. For example: " $x = 0$ is the only critical point of h in the interval $[-5,\ 8]$, so the minimum there has to be an absolute minimum."

The point is not earned for, "$h' = g$ is negative to the left of $x = 0$ and positive to the right of $x = 0$." It's not clear whether this refers to all points to the left and right of $x = 0$, or only to points near $x = 0$.

Alternate solution: Candidates Test

The absolute minimum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \;\Rightarrow\; x = 0$

3-1-1b

$h(-5) = 3 + \frac{9\pi}{4}$, $h(0) = -4$, and $h(8) = 20$, so the absolute minimum of h is at $x = 0$.

The first point is earned by considering $x = 0$.

The second point is earned by stating that the absolute minimum occurs at $x = 0$.

The third point is earned by presenting the correct values or signs of $h(-5)$, $h(0)$, and $h(8)$.

General comments:

Intervals and inequalities may be stated with or without endpoints included, e.g., $[-5,\ 8]$ or $(-5,\ 8)$.

“Typographical” errors such as “$h'(x) = g(t)$” will not be penalized.

A response that contains false statements (“says too much”), no matter how many, will fail to earn only one of the points that would otherwise be earned in this part.

Look out for ‘wrong derivative’ errors! For example: A response stating that “g changes from decreasing to increasing at $x = 0$” is not a correct justification of a minimum.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Identification of $x = 0$ as critical point
- ☐ Verification of relative minimum at $x = 0$
- ☐ Justification of absolute minimum at $x = 0$

Solution:

The absolute minimum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \ \Rightarrow \ x = 0$

Because $h'(x) = g(x)$ changes from negative to positive at $x = 0$, $x = 0$ is the location of a relative minimum of h.

Because $x = 0$ is the only critical point of h on the interval $-5 \le x \le 8$, $x = 0$ is also the location of the absolute minimum of h on the interval $-5 \le x \le 8$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for listing the x-values -2 and 6.

All of the justification is wrapped up in the second point.

If other candidates are mentioned, they must be ruled out by valid reasoning.

The points of inflection for the graph of h are not $(-2,\ -4)$ and $(6,\ 8)$ (these are points on the graph of g). Responses that make this claim will not earn the first point, but might still earn the second point.

Second point:

3-1-1b

This point requires the connection $h\prime = g$. Remember that the connection may have been made in a previous part.

Eligibility: The response must identify the x-coordinate of at least one of the correct points of inflection and no incorrect x-values to be eligible for the second point.

The justification must be in terms of g. Statements about h are formulas, not applied to our problem.

The point is earned for the following responses:

· g changes from increasing to decreasing or vice versa at these points.

· g' changes from positive to negative or vice versa at these points.

· g has a relative maximum or minimum at these points.

The point is not earned for the following responses:

· h changes concavity at $x = -2$ and $x = 6$.

· h'' changes sign at $x = -2$ and $x = 6$.

· Claims that $g'(-2) = 0$ or $g'(6) = 0$. These values do not exist.

· A response that claims g' changes from increasing to decreasing.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ x-values
- ☐ Justification

Solution:

$h'(x) = g(x)$

$h' = g$ changes from decreasing to increasing at $x = -2$.

$h' = g$ changes from increasing to decreasing at $x = 6$.

Points of inflection for the graph of h occur at $x = -2$ and $x = 6$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for differentiating k correctly, whether evaluated at $x = 7$ or not.

This "chain rule" point can also be earned by using the product rule (correctly, twice). This can occur if the student rewrites $k(x) = g(x) \cdot g(x) \cdot g(x)$, or $k(x) = (g(x))^2 \cdot g(x)$.

Second point:

The correct answer is -225. This answer does not need to be simplified but if it is, the simplification must be correct.

3-1-1b

The response "$3\left(5^2\right)(-3)$" without other supporting work earns the answer point, but does not show enough computation to earn the chain rule point. In all other cases, the response must earn the first point to be eligible for the second point.

It is not necessary to simplify numeric answers.

The response $3 \cdot (g(7))^2 \cdot g'(7) = 3 \cdot 5^2 \cdot (-3)$ earns both points.

The response $3 \cdot (g(7))^2 \cdot g'(7)$ earns the first point, but not the second. The student must evaluate g and $g\prime$.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Chain rule
- [ ] Answer

Solution:

$$k'(x) = 3 \cdot (g(x))^2 \cdot g'(x)$$

$$k'(7) = 3 \cdot (g(7))^2 \cdot g'(7) = 3 \cdot 5^2 \cdot (-3) = -225$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by conveying that both the numerator and denominator each approach 0.

The response does not need to reference continuity or differentiability.

Responses including "$\text{limit} = \frac{0}{0}$" will not earn the first point.

Note: $\frac{0}{0}$ may appear in disguise, as in "$\text{limit} = \frac{h(-2)}{3(-2)+6}$." Responses that include this will not earn the first point.

Other presentations of the limits, such as "$\to \frac{0}{0}$" or "$=$ indeterminate form $\frac{0}{0}$" will earn the first point.

Simply evaluating the numerator and denominator separately will earn this point since both are continuous functions.

Second point:

The point is earned by presenting a limit of a fraction whose numerator and denominator are clearly attempts at derivatives of $h(x)$ and $3x + 6$ respectively.

The point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Third point:

The point is earned for the correct answer, which equals $\frac{-4}{3}$. The answer does not need to be simplified.

3-1-1b

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Limits of numerator and denominator equal 0
- [ ] Application of L'Hospital's Rule
- [ ] Answer

Solution:

$h' = g$ is continuous. $\Rightarrow$ h is continuous.
$\Rightarrow \lim\limits_{x \to -2} h(x) = h(-2)$ and $\lim\limits_{x \to -2} h'(x) = h'(-2)$.

$$\lim_{x \to -2} h(x) = h(-2) = \int_{-2}^{-2} g(t)\, dt = 0$$

$$\lim_{x \to -2} (3x + 6) = 3(-2) + 6 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to -2} \frac{h(x)}{3x+6} = \lim_{x \to -2} \frac{h'(x)}{3}$$

$$= \frac{h'(-2)}{3} = \frac{g(-2)}{3} = \frac{-4}{3}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by presenting numeric expressions identifiable as attempts to compute area.

Second point:

The point is earned for the correct answer, which equals 10, supported by appropriate work. The answer does not need to be simplified.

The answer "$-4 + 4 + 10$," by itself, earns both points.

The unsupported answer "10" does not earn either point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Areas
- [ ] Answer

Solution:

3-1-1b

$h(6) = \int_{-2}^{6} g(t)\ dt = -4 + 4 + 10 = 10$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for the sum of the series: $\frac{30r}{1-r}$.

The sum may appear in other forms, for example, $\frac{30}{1-r} - 30$. (The constant may be moved to the other side of the equation before the sum appears in closed form.)

Second point:

This point is earned for finding the value of r that is consistent with the answer for $h(6)$ found in part (f).

Eligibility: The response must earn the first point to be eligible for the second point.

The correct answer $r = \frac{1}{4}$ earns the point if the correct answer 10 is given in part (f).

An answer $a > -15$ in part (f) is consistent with $r = \frac{a}{30+a}$.

An answer $a \le -15$ in part (f) is consistent with no solution (since $|r| \ge 1$).

If no answer appears in part (f), the point cannot be earned unless $h(6)$ is computed in part (g).

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Sum of geometric series
- [ ] Answer

Solution:

$$30r + 30r^2 + \cdots + 30r^n + \cdots = \frac{30r}{1-r} \text{ for } |r| < 1$$

$$10 = \frac{30r}{1-r} \ \Rightarrow\ 10 - 10r = 30r \ \Rightarrow\ 10 = 40r \ \Rightarrow\ r = \frac{10}{40} = \frac{1}{4}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the answer "no" with the reason that g is not differentiable.

The response doesn't need to name the interval. If an interval is presented, it can be open or closed, but cannot contradict the graph of g.

The response must name g, or "the graph," as the function that is non-differentiable. The name h' is sufficient if the connection $h' = g$ has been made earlier.

3-1-1b

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

No, the Mean Value Theorem does *not* guarantee the existence of a value of c with the stated properties because $h' = g$ is not differentiable for at least one point in $-5 < x < 8$.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response earns the point by referring to arc length, or by working with the arc length of some portion of a circle.

Use your judgment to determine whether or not the response conveys an attempt to compute the length of the curve.

Second point:

The point is earned for the correct answer $\frac{3\pi}{2}$, supported by appropriate reasoning.

The unsupported answer $\frac{3\pi}{2}$ does not earn the point.

Alternate solution:

On the interval $[-5, \ -2]$, we have:

$$g\left(x\right) = -1 - \sqrt{9 - \left(x+2\right)^2}$$

$$g'\left(x\right) = \frac{x+2}{\sqrt{9-(x+2)^2}}$$

$$1 + \left(g'\left(x\right)\right)^2 = \frac{9}{9-(x+2)^2} = \frac{1}{1-\left(\frac{x+2}{3}\right)^2}$$

$$\int_{-5}^{-2} \sqrt{1 + \left(g'\left(x\right)\right)^2}\, dx = 3\sin^{-1}\left(\frac{x+2}{3}\right)\Big|_{-5}^{-2} = 3\sin^{-1}\left(0\right) - 3\sin^{-1}\left(-1\right) = \frac{3\pi}{2}$$

This alternate solution earns the first point for an antiderivative of the form $A\sin^{-1}\left(u\right) + C$, where

· A is a nonzero real number,

· C may be an arbitrary constant, or any real number, or may be omitted, and

· u may be indeterminate or a declared function of x.

The second point is earned for the correct answer $\frac{3\pi}{2}$, supported by appropriate reasoning.

3-1-1b

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Recognizes arc length
- [ ] Answer

Solution:

$\int_{-5}^{-2} \sqrt{1 + (g'(x))^2}\, dx$ is the length of the graph of g from $x = -5$ to $x = -2$, which is the circumference of a quarter circle with radius 3.

$\frac{1}{4}(2\pi \cdot 3) = \frac{3\pi}{2}$

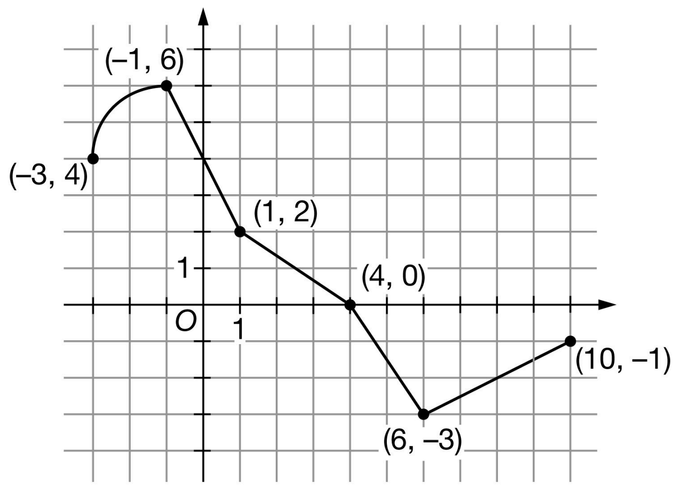
3-1-1b

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-3 \leq x \leq 10$ consists of a quarter circle and four line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_1^x g(t)\,dt$ for $-3 \leq x \leq 10$.

(a) Find the value of $h'(-1)$ or explain why it does not exist.

(b) At what value of x does h attain its maximum value on the interval $-3 \leq x \leq 10$? Justify your answer.

(c) Find the x-coordinate of all points of inflection of the graph of h for $-3 < x < 10$. Justify your answer.

(d) Let k be the function defined by $k(x) = 3(g(x))^2 + 7$. Find the value of $k'(0)$. Show the computations that lead to your answer.

(e) Find $\lim_{x \to 1} \frac{h(x)}{9x-9}$. Give a reason for your answer.

(f) Find the value of $h(8)$.

(g) For what value of r does the series $25r + 25r^2 + \cdots + 25r^n + \cdots$ equal the value of $h(8)$ found in part (f)? Show the computations that lead to your answer.

(h) Does the Mean Value Theorem guarantee a value of c, for $-3 < c < 10$, such that $h''(c)$ is equal to the average rate of change of h' over the interval $-3 \leq x \leq 10$? Why or why not?

(i) What is the value of $\int_{-3}^{-1} \sqrt{1 + (g'(x))^2}\,dx$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer of 6.

The response can earn the point without showing any supporting work: the correct answer is enough.

3-1-1b

Even with a correct answer, a response that includes false claims (“says too much”) will not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x)$

$h'(-1) = g(-1) = 6$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by stating that $x = 4$ is a critical point.

Second point:

Eligibility: The response must earn the first point to be eligible for the second point.

The second point is earned, for example, by a statement like “g changes from positive to negative at $x = 4$.”

This verification requires the explicit connection of $h\prime= g$. The point cannot be earned for “h' changes from positive to negative at $x = 4$.”

A response of “$h' = g$, and h' changes from positive to negative at $x = 4$” would earn the second point. Note: The connection “$h' = g$” might have been made in part (a). Whenever the connection is made, it carries through the rest of this problem.

Language like “g changes at $x = 4$” is ambiguous and will not earn the point.

Words such as “the graph” are interpreted to be referring to g, the graph given in the prompt.

The terms “the function” or “the derivative” are ambiguous and cannot earn this point.

Third point:

This requires justifying $x = 4$ as the location of an absolute maximum.

Eligibility: The response must earn the second point to be eligible for the third point.

The response may earn the third point by appealing to the uniqueness of the critical point at $x = 4$. For example: “$x = 4$ is the only critical point of h in the interval $[-3,\ 10]$, so the maximum there has to be an absolute maximum.”

The point is not earned for, “$h' = g$ is positive to the left of $x = 4$ and negative to the right of $x = 4$.” It’s not clear whether this refers to all points to the left and right of $x = 4$, or only to points near $x = 4$.

Alternate solution: Candidates Test

The absolute maximum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \ \Rightarrow\ x = 4$

3-1-1b

$h(-3) = -16 - \pi$, $h(4) = 3$, and $h(10) = -8$, so the absolute maximum of h is at $x = 4$.

The first point is earned by considering $x = 4$.

The second point is earned by stating that the absolute maximum occurs at $x = 4$.

The third point is earned by presenting the correct values or signs of $h(-3)$, $h(4)$, and $h(10)$.

General comments:

Intervals and inequalities may be stated with or without endpoints included, e.g., $[-3,\ 4]$ or $(-3,\ 4)$.

"Typographical" errors such as "$h'(x) = g(t)$" will not be penalized.

A response that contains false statements ("says too much"), no matter how many, will fail to earn only one of the points that would otherwise be earned in this part.

Look out for 'wrong derivative' errors! For example: A response stating that "g changes from increasing to decreasing at $x = 4$" is not a correct justification of a maximum.

✓

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Identification of $x = 4$ as critical point
- ☐ Verification of relative maximum at $x = 4$
- ☐ Justification of absolute maximum at $x = 4$

Solution:

The absolute maximum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \ \Rightarrow \ x = 0$

Because $h'(x) = g(x)$ changes from positive to negative at $x = 4$, $x = 4$ is the location of a relative maximum of h.

Because $x = 4$ is the only critical point of h on the interval $-3 \leq x \leq 10$, $x = 4$ is also the location of the absolute maximum of h on the interval $-3 \leq x \leq 10$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for listing the x-values -1 and 6.

All of the justification is wrapped up in the second point.

If other candidates are mentioned, they must be ruled out by valid reasoning.

The points of inflection for the graph of h are not $(-1,\ 6)$ and $(6,\ -3)$ (these are points on the graph of g). Responses that make this claim will not earn the first point, but might still earn the second point.

Second point:

3-1-1b

This point requires the connection $h\prime = g$. Remember that the connection may have been made in a previous part.

Eligibility: The response must identify the x-coordinate of at least one of the correct points of inflection and no incorrect x-values to be eligible for the second point.

The justification must be in terms of g. Statements about h are formulas, not applied to our problem.

The point is earned for the following responses:

· g changes from increasing to decreasing or vice versa at these points.

· g' changes from positive to negative or vice versa at these points.

· g has a relative maximum or minimum at these points.

The point is not earned for the following responses:

· h changes concavity at $x = -1$ and $x = 6$.

· h'' changes sign at $x = -1$ and $x = 6$.

· Claims that $g'(-1) = 0$ or $g'(6) = 0$. These values do not exist.

· A response that claims g' changes from increasing to decreasing.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] x-values
- [ ] Justification

Solution:

$h'(x) = g(x)$

$h' = g$ changes from increasing to decreasing at $x = -1$.

$h' = g$ changes from decreasing to increasing at $x = 6$.

Points of inflection for the graph of h occur at $x = -1$ and $x = 6$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for differentiating k correctly, whether evaluated at $x = 0$ or not.

This “chain rule” point can also be earned by using the product rule.

Second point:

The correct answer is -48. This answer does not need to be simplified but if it is, the simplification must be correct.

3-1-1b

The response "$6\,(4)\,(-2)$ " without other supporting work earns the answer point, but does not show enough computation to earn the chain rule point. In all other cases, the response must earn the first point to be eligible for the second point.

It is not necessary to simplify numeric answers.

The response $6 \cdot g\,(0) \cdot g'\,(0) = 6 \cdot 4 \cdot (-2)$ earns both points.

The response $6 \cdot g\,(0) \cdot g'\,(0)$ earns the first point, but not the second. The student must evaluate g and g'.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$k'\,(x) = 6 \cdot g\,(x) \cdot g'\,(x)$

$k'\,(0) = 6 \cdot g\,(0) \cdot g'\,(0) = 6 \cdot 4 \cdot (-2) = -48$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by conveying that both the numerator and denominator each approach 0.

The response does not need to reference continuity or differentiability.

Responses including "$\text{limit} = \frac{0}{0}$ " will not earn the first point.

Note: $\frac{0}{0}$ may appear in disguise, as in "$\text{limit} = \frac{h(1)}{9(1)-9}$. " Responses that include this will not earn the first point.

Other presentations of the limits, such as "$\to \frac{0}{0}$ " or "$= \text{ indeterminate form } \frac{0}{0}$ " will earn the first point.

Simply evaluating the numerator and denominator separately will earn this point since both are continuous functions.

Second point:

The point is earned by presenting a limit of a fraction whose numerator and denominator are clearly attempts at derivatives of $h\,(x)$ and $9x - 9$ respectively.

The point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Third point:

The point is earned for the correct answer, which equals $\frac{2}{9}$. The answer does not need to be simplified.

3-1-1b

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Limits of numerator and denominator equal 0
- [ ] Application of L'Hospital's Rule
- [ ] Answer

Solution:

$h' = g$ is continuous. $\Rightarrow$ h is continuous.
$\Rightarrow \lim\limits_{x\to1} h(x) = h(1)$ and $\lim\limits_{x\to1} h'(x) = h'(1)$.

$$\lim_{x\to1} h(x) = h(1) = \int_1^1 g(t)dt = 0$$

$$\lim_{x\to1} (9x-9) = 9 \cdot 1 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to1} \frac{h(x)}{9x-9} = \lim_{x\to1} \frac{h'(x)}{9}$$

$$= \frac{h'(1)}{9} = \frac{g(1)}{9} = \frac{2}{9}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by presenting numeric expressions identifiable as attempts to compute area.

Second point:

The point is earned for the correct answer, which equals -5, supported by appropriate work. The answer does not need to be simplified.

The answer "$3-3-5$, " by itself, earns both points.

The unsupported answer "-5 " does not earn either point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Areas
- [ ] Answer

Solution:

3-1-1b

$h(8) = \int_1^8 g(t)\, dt = 3 - 3 - 5 = -5$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for the sum of the series: $\frac{25r}{1-r}$.

The sum may appear in other forms, for example, $\frac{25}{1-r} - 25$. (The constant may be moved to the other side of the equation before the sum appears in closed form.)

Second point:

This point is earned for finding the value of r that is consistent with the answer for $h(8)$ found in part (f).

Eligibility: The response must earn the first point to be eligible for the second point.

The correct answer $r = -\frac{1}{4}$ earns the point if the correct answer -5 is given in part (f).

An answer $a > -12.5$ in part (f) is consistent with $r = \frac{a}{25+a}$.

An answer $a \leq -12.5$ in part (f) is consistent with no solution (since $|r| \geq 1$).

If no answer appears in part (f), the point cannot be earned unless $h(8)$ is computed in part (g).

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Sum of geometric series
- [ ] Answer

Solution:

$25r + 25r^2 + \cdots + 25r^n + \cdots = \frac{25r}{1-r}$ for $|r| < 1$

$-5 = \frac{25r}{1-r} \Rightarrow -5 + 5r = 25r \Rightarrow -5 = 20r \Rightarrow r = -\frac{5}{20} = -\frac{1}{4}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the answer "no" with the reason that g is not differentiable.

The response doesn't need to name the interval. If an interval is presented, it can be open or closed, but cannot contradict the graph of g.

The response must name g, or "the graph," as the function that is non-differentiable. The name h' is sufficient if the connection $h' = g$ has been made earlier.

3-1-1b

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

No, the Mean Value Theorem does *not* guarantee the existence of a value of c with the stated properties because $h' = g$ is not differentiable for at least one point in $-3 < x < 10$.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response earns the point by referring to arc length, or by working with the arc length of some portion of a circle.

Use your judgment to determine whether or not the response conveys an attempt to compute the length of the curve.

Second point:

The point is earned for the correct answer π, supported by appropriate reasoning.

The unsupported answer π does not earn the point.

Alternate solution:

On the interval $[-3, \ -1]$, we have:

$$g\left(x\right) = 4 + \sqrt{4 - \left(x+1\right)^2}$$

$$g'\left(x\right) = -\frac{x+1}{\sqrt{4-\left(x+1\right)^2}}$$

$$1 + \left(g'\left(x\right)\right)^2 = \frac{4}{4-\left(x+1\right)^2} = \frac{1}{1-\left(\frac{x+1}{2}\right)^2}$$

$$\int_{-3}^{-1} \sqrt{1 + \left(g'\left(x\right)\right)^2}\, dx = 2\sin^{-1}\left(\frac{x+1}{2}\right)\Big|_{-3}^{-1} = 2\sin^{-1}\left(0\right) - 2\sin^{-1}\left(-1\right) = \pi$$

This alternate solution earns the first point for an antiderivative of the form $A\sin^{-1}\left(u\right) + C$, where

· A is a nonzero real number,

· C may be an arbitrary constant, or any real number, or may be omitted, and

· u may be indeterminate or a declared function of x.

The second point is earned for the correct answer π, supported by appropriate reasoning.

0	1	2

3-1-1b

The student response accurately includes **both** of the criteria below.

- ☐ Recognizes arc length
- ☐ Answer

Solution:

$\int_{-3}^{-1} \sqrt{1+(g'(x))^2}\, dx$ is the length of the graph of g from $x=-3$ to $x=-1$, which is the circumference of a quarter circle with radius 2.

$\frac{1}{4}(2\pi \cdot 2) = \pi$

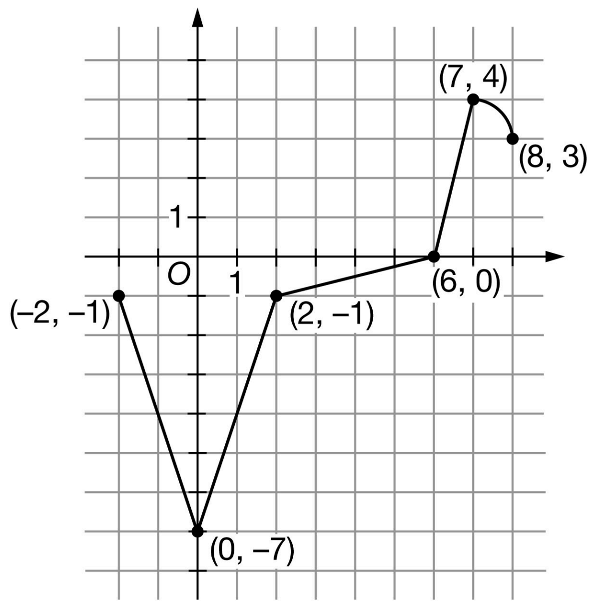
3-1-1b

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-2 \leq x \leq 8$ consists of a quarter circle and four line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_0^x g(t)\,dt$ for $-2 \leq x \leq 8$.

(a) Find the value of $h'(2)$ or explain why it does not exist.

(b) At what value of x does h attain its minimum value on the interval $-2 \leq x \leq 8$? Justify your answer.

(c) Find the x-coordinate of all points of inflection of the graph of h for $-2 < x < 8$. Justify your answer.

(d) Let k be the function defined by $k(x) = 6(g(x))^2 + 2$. Find the value of $k'(1)$. Show the computations that lead to your answer.

(e) Find $\lim_{x \to 0} \frac{x+h(x)}{\sin x}$. Give a reason for your answer.

(f) Find the value of $h(6)$.

(g) For what value of r does the series $32r + 32r^2 + \cdots + 32r^n + \cdots$ equal the value of $h(6)$ found in part (f)? Show the computations that lead to your answer.

(h) Does the Mean Value Theorem guarantee a value of c, for $-2 < c < 8$, such that $h''(c)$ is equal to the average rate of change of h' over the interval $-2 \leq x \leq 8$? Why or why not?

(i) What is the value of $\int_7^8 \sqrt{1 + (g'(x))^2}\,dx$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1b

The point is earned for the correct answer of -1.

The response can earn the point without showing any supporting work: the correct answer is enough.

Even with a correct answer, a response that includes false claims (“says too much”) will not earn the point.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = g(x)$

$h'(2) = g(2) = -1$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by stating that $x = 6$ is a critical point.

Second point:

Eligibility: The response must earn the first point to be eligible for the second point.

The second point is earned, for example, by a statement like “g changes from negative to positive at $x = 6$.”

This verification requires the explicit connection of $h' = g$. The point cannot be earned for “h' changes from negative to positive at $x = 6$.”

A response of “$h' = g$, and h' changes from negative to positive at $x = 6$” would earn the second point. Note: The connection “$h' = g$” might have been made in part (a). Whenever the connection is made, it carries through the rest of this problem.

Language like “g changes at $x = 6$” is ambiguous and will not earn the point.

Words such as “the graph” are interpreted to be referring to g, the graph given in the prompt.

The terms “the function” or “the derivative” are ambiguous and cannot earn this point.

Third point:

This requires justifying $x = 6$ as the location of an absolute minimum.

Eligibility: The response must earn the second point to be eligible for the third point.

The response may earn the third point by appealing to the uniqueness of the critical point at $x = 6$. For example: “$x = 6$ is the only critical point of h in the interval $[-2, \ 8]$, so the minimum there has to be an absolute minimum.”

The point is not earned for, “$h' = g$ is negative to the left of $x = 6$ and positive to the right of $x = 6$.” It’s not clear whether this refers to all points to the left and right of $x = 6$, or only to points near $x = 6$.

Alternate solution: Candidates Test

3-1-1b

The absolute minimum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \;\Rightarrow\; x = 6$

$h(-2) = 8$, $h(6) = -10$, and $h(8) = \frac{\pi}{4} - 5$, so the absolute minimum of h is at $x = 6$.

The first point is earned by considering $x = 6$.

The second point is earned by stating that the absolute minimum occurs at $x = 6$.

The third point is earned by presenting the correct values or signs of $h(-2)$, $h(6)$, and $h(8)$.

General comments:

Intervals and inequalities may be stated with or without endpoints included, e.g., $[-2,\ 6]$ or $(-2,\ 6)$.

"Typographical" errors such as "$h'(x) = g(t)$" will not be penalized.

A response that contains false statements ("says too much"), no matter how many, will fail to earn only one of the points that would otherwise be earned in this part.

Look out for 'wrong derivative' errors! For example: A response stating that "g changes from decreasing to increasing at $x = 6$" is not a correct justification of a minimum.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Identification of $x = 6$ as critical point
- ☐ Verification of relative minimum at $x = 6$
- ☐ Justification of absolute minimum at $x = 6$

Solution:

The absolute minimum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \;\Rightarrow\; x = 6$

Because $h'(x) = g(x)$ changes from negative to positive at $x = 6$, $x = 6$ is the location of a relative minimum of h.

Because $x = 6$ is the only critical point of h on the interval $-2 \le x \le 8$, $x = 6$ is also the location of the absolute minimum of h on the interval $-2 \le x \le 8$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for listing the x-values 0 and 7.

All of the justification is wrapped up in the second point.

If other candidates are mentioned, they must be ruled out by valid reasoning.

3-1-1b

The points of inflection for the graph of h are not $(0, \ -7)$ and $(7, \ 4)$ (these are points on the graph of g). Responses that make this claim will not earn the first point, but might still earn the second point.

Second point:

This point requires the connection $h\prime = g$. Remember that the connection may have been made in a previous part.

Eligibility: The response must identify the x-coordinate of at least one of the correct points of inflection and no incorrect x-values to be eligible for the second point.

The justification must be in terms of g. Statements about h are formulas, not applied to our problem.

The point is earned for the following responses:

· g changes from increasing to decreasing or vice versa at these points.

· g' changes from positive to negative or vice versa at these points.

· g has a relative maximum or minimum at these points.

The point is not earned for the following responses:

· h changes concavity at $x = 0$ and $x = 7$.

· h'' changes sign at $x = 0$ and $x = 7$.

· Claims that $g'\ (0) = 0$ or $g'\ (7) = 0$. These values do not exist.

· A response that claims g' changes from increasing to decreasing.

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ x-values
- ☐ Justification

Solution:

$h'\ (x) = g\ (x)$

$h' = g$ changes from decreasing to increasing at $x = 0$.

$h' = g$ changes from increasing to decreasing at $x = 7$.

Points of inflection for the graph of h occur at $x = 0$ and $x = 7$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for differentiating k correctly, whether evaluated at $x = 1$ or not.

This “chain rule” point can also be earned by using the product rule.

3-1-1b

Second point:

The correct answer is -144. This answer does not need to be simplified but if it is, the simplification must be correct.

The response "$12\,(-4)\,(3)$" without other supporting work earns the answer point, but does not show enough computation to earn the chain rule point. In all other cases, the response must earn the first point to be eligible for the second point.

It is not necessary to simplify numeric answers.

The response $12 \cdot g\,(1) \cdot g'\,(1) = 12 \cdot \;(-4) \cdot 3$ earns both points.

The response $12 \cdot g\,(1) \cdot g'\,(1)$ earns the first point, but not the second. The student must evaluate g and g'.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$k'\,(x) = 12 \cdot g\,(x) \cdot g'\,(x)$

$k'\,(1) = 12 \cdot g\,(1) \cdot g'\,(1) = 12 \cdot \;(-4) \cdot 3 = -144$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by conveying that both the numerator and denominator each approach 0.

The response does not need to reference continuity or differentiability.

Responses including "$\text{limit} = \frac{0}{0}$" will not earn the first point.

Note: $\frac{0}{0}$ may appear in disguise, as in "$\text{limit} = \frac{h\,(0)}{\sin\,0}$." Responses that include this will not earn the first point.

Other presentations of the limits, such as "$\to \frac{0}{0}$" or "$=$ indeterminate form $\frac{0}{0}$" will earn the first point.

Simply evaluating the numerator and denominator separately will earn this point since both are continuous functions.

Second point:

The point is earned by presenting a limit of a fraction whose numerator and denominator are clearly attempts at derivatives of $x + h\,(x)$ and $\sin\,x$ respectively.

The point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Third point:

The point is earned for the correct answer, which equals -6. The answer does not need to be simplified.

3-1-1b

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is continuous. $\Rightarrow$ h is continuous.
$\Rightarrow \lim\limits_{x \to 0} h(x) = h(0)$ and $\lim\limits_{x \to 0} h'(x) = h'(0)$.

$$\lim_{x \to 0} (x + h(x)) = 0 + h(0) = 0 + \int_0^0 g(t)dt = 0$$

$$\lim_{x \to 0} \sin x = \sin 0 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 0} \frac{x+h(x)}{\sin x} = \lim_{x \to 0} \frac{1+h'(x)}{\cos x}$$

$$= \frac{1+h'(0)}{\cos 0} = \frac{1+g(0)}{\cos 0} = \frac{1+(-7)}{1} = -6$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by presenting numeric expressions identifiable as attempts to compute area.

Note that supporting work may appear in part (b); we look there if we see an unsupported answer, correct or incorrect, in part (f). An incorrect answer from part (b), with supporting work, may be imported into part (f) and earns both points.

In case of conflict, work presented in part (f) supersedes work from part (b).

Second point:

The point is earned for the correct answer, which equals -10, supported by appropriate work. The answer does not need to be simplified.

The answer "$-8 - 2$, " by itself, earns both points.

The unsupported answer "-10 " does not earn either point.

0	1	2

The student response accurately includes **both** of the criteria below.

Areas

3-1-1b

- [ ]
- [ ] Answer

Solution:

$$h(6) = \int_0^6 g(t)\,dt = -8 - 2 = -10$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for the sum of the series: $\frac{32r}{1-r}$.

The sum may appear in other forms, for example, $\frac{32}{1-r} - 32$. (The constant may be moved to the other side of the equation before the sum appears in closed form.)

Second point:

This point is earned for finding the value of r that is consistent with the answer for $h(6)$ found in part (f).

Eligibility: The response must earn the first point to be eligible for the second point.

The correct answer $r = -\frac{5}{11}$ earns the point if the correct answer -10 is given in part (f) or part (b).

An answer $a > -16$ in part (f) is consistent with $r = \frac{a}{32+a}$.

An answer $a \leq -16$ in part (f) is consistent with no solution (since $|r| \geq 1$).

If no answer appears in part (f) or part (b), the point cannot be earned unless $h(6)$ is computed in part (g).

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Sum of geometric series
- [ ] Answer

Solution:

$$32r + 32r^2 + \cdots + 32r^n + \cdots = \frac{32r}{1-r} \text{ for } |r| < 1$$

$$-10 = \frac{32r}{1-r} \Rightarrow -10 + 10r = 32r \Rightarrow -10 = 22r \Rightarrow r = -\frac{10}{22} = -\frac{5}{11}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the answer “no” with the reason that g is not differentiable.

3-1-1b

The response doesn't need to name the interval. If an interval is presented, it can be open or closed, but cannot contradict the graph of g.

The response must name g, or "the graph," as the function that is non-differentiable. The name h' is sufficient if the connection $h' = g$ has been made earlier.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

No, the Mean Value Theorem does *not* guarantee the existence of a value of c with the stated properties because $h' = g$ is not differentiable for at least one point in $-2 < x < 8$.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response earns the point by referring to arc length, or by working with the arc length of some portion of a circle.

Use your judgment to determine whether or not the response conveys an attempt to compute the length of the curve.

Second point:

The point is earned for the correct answer $\frac{\pi}{2}$, supported by appropriate reasoning.

The unsupported answer $\frac{\pi}{2}$ does not earn the point.

Alternate solution:

On the interval $[7,\ 8]$, we have:

$$g\left(x\right) = 3 + \sqrt{1 - \left(x - 7\right)^2}$$

$$g'\left(x\right) = -\frac{x-7}{\sqrt{1-\left(x-7\right)^2}}$$

$$1 + \left(g'\left(x\right)\right)^2 = \frac{1}{1-\left(x-7\right)^2}$$

$$\int_7^8 \sqrt{1 + \left(g'\left(x\right)\right)^2}\, dx = \sin^{-1}\left(x - 7\right)\Big|_7^8 = \sin^{-1}\left(1\right) - \sin^{-1}\left(0\right) = \frac{\pi}{2}$$

This alternate solution earns the first point for an antiderivative of the form $A\sin^{-1}\left(u\right) + C$, where

· A is a nonzero real number,

· C may be an arbitrary constant, or any real number, or may be omitted, and

· u may be indeterminate or a declared function of x.

The second point is earned for the correct answer $\frac{\pi}{2}$, supported by appropriate reasoning.

3-1-1b

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Recognizes arc length
- [ ] Answer

Solution:

$\int_7^8 \sqrt{1+(g'(x))^2}\,dx$ is the length of the graph of g from $x=7$ to $x=8$, which is the circumference of a quarter circle with radius 1.

$\frac{1}{4}(2\pi \cdot 1) = \frac{\pi}{2}$

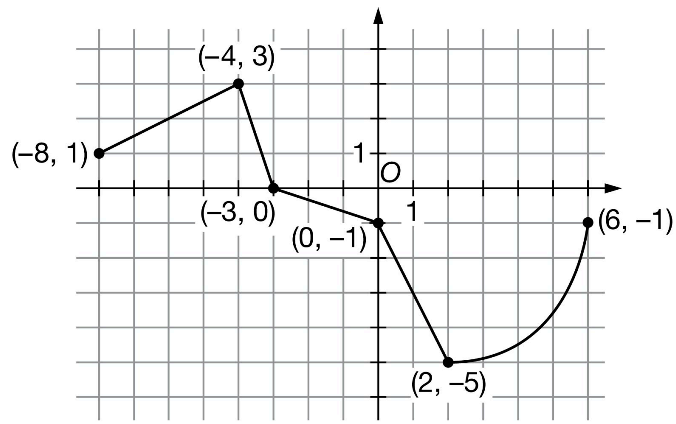
3-1-1b

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-8 \leq x \leq 6$ consists of a quarter circle and four line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-4}^{x} g(t)\,dt$ for $-8 \leq x \leq 6$.

(a) Find the value of $h'(2)$ or explain why it does not exist.

(b) At what value of x does h attain its maximum value on the interval $-8 \leq x \leq 6$? Justify your answer.

(c) Find the x-coordinate of all points of inflection of the graph of h for $-8 < x < 6$. Justify your answer.

(d) Let k be the function defined by $k(x) = 5(g(x))^2 + 7$. Find the value of $k'(1)$. Show the computations that lead to your answer.

(e) Find $\lim\limits_{x \to -4} \frac{h(x)}{2x+8}$. Give a reason for your answer.

(f) Find the value of $h(2)$.

(g) For what value of r does the series $36r + 36r^2 + \cdots + 36r^n + \cdots$ equal the value of $h(2)$ found in part (f)? Show the computations that lead to your answer.

(h) Does the Mean Value Theorem guarantee a value of c, for $-8 < c < 6$, such that $h''(c)$ is equal to the average rate of change of h' over the interval $-8 \leq x \leq 6$? Why or why not?

(i) What is the value of $\int_{2}^{6} \sqrt{1 + (g'(x))^2}\,dx$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the correct answer of -5.

The response can earn the point without showing any supporting work: the correct answer is enough.

Even with a correct answer, a response that includes false claims ("says too much") will not earn the point.

3-1-1b

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x)$

$h'(2) = g(2) = -5$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by stating that $x = -3$ is a critical point.

Second point:

Eligibility: The response must earn the first point to be eligible for the second point.

The second point is earned, for example, by a statement like "g changes from positive to negative at $x = -3.$ "

This verification requires the explicit connection of $h' = g$. The point cannot be earned for "h' changes from positive to negative at $x = -3.$ "

A response of "$h' = g$, and h' changes from positive to negative at $x = -3$ " would earn the second point. Note: The connection "$h' = g$ " might have been made in part (a). Whenever the connection is made, it carries through the rest of this problem.

Language like "g changes at $x = -3$ " is ambiguous and will not earn the point.

Words such as "the graph" are interpreted to be referring to g, the graph given in the prompt.

The terms "the function" or "the derivative" are ambiguous and cannot earn this point.

Third point:

This requires justifying $x = -3$ as the location of an absolute maximum.

Eligibility: The response must earn the second point to be eligible for the third point.

The response may earn the third point by appealing to the uniqueness of the critical point at $x = -3$. For example: "$x = -3$ is the only critical point of h in the interval $[-8, \ 6]$, so the maximum there has to be an absolute maximum."

The point is not earned for, "$h' = g$ is positive to the left of $x = -3$ and negative to the right of $x = -3.$ " It's not clear whether this refers to all points to the left and right of $x = -3$, or only to points near $x = -3.$

Alternate solution: Candidates Test

The absolute maximum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \ \Rightarrow \ x = -3$

3-1-1b

$h(-8) = -8$, $h(-3) = \frac{3}{2}$, and $h(6) = -10 - 4\pi$, so the absolute maximum of h is at $x = -3$.

The first point is earned by considering $x = -3$.

The second point is earned by stating that the absolute maximum occurs at $x = -3$.

The third point is earned by presenting the correct values or signs of $h(-8)$, $h(-3)$, and $h(6)$.

General comments:

Intervals and inequalities may be stated with or without endpoints included, e.g., $[-8,\ 3]$ or $(-8,\ 3)$.

"Typographical" errors such as "$h'(x) = g(t)$ " will not be penalized.

A response that contains false statements ("says too much"), no matter how many, will fail to earn only one of the points that would otherwise be earned in this part.

Look out for 'wrong derivative' errors! For example: A response stating that "g changes from increasing to decreasing at $x = -3$ " is not a correct justification of a maximum.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Identification of $x = -3$ as critical point
- ☐ Verification of relative maximum at $x = -3$
- ☐ Justification of absolute maximum at $x = -3$

Solution:

The absolute maximum will occur at a critical point where $h'(x) = 0$ or at an endpoint.

$h'(x) = g(x) = 0 \;\Rightarrow\; x = -3$

Because $h'(x) = g(x)$ changes from positive to negative at $x = -3$, $x = -3$ is the location of a relative maximum of h.

Because $x = -3$ is the only critical point of h on the interval $-8 \le x \le 6$, $x = -3$ is also the location of the absolute maximum of h on the interval $-8 \le x \le 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for listing the x-values -4 and 2.

All of the justification is wrapped up in the second point.

If other candidates are mentioned, they must be ruled out by valid reasoning.

The points of inflection for the graph of h are not $(-4,\ 3)$ and $(2,\ -5)$ (these are points on the graph of g). Responses that make this claim will not earn the first point, but might still earn the second point.

Second point:

3-1-1b

This point requires the connection $h\prime = g$. Remember that the connection may have been made in a previous part.

Eligibility: The response must identify the x-coordinate of at least one of the correct points of inflection and no incorrect x-values to be eligible for the second point.

The justification must be in terms of g. Statements about h are formulas, not applied to our problem.

The point is earned for the following responses:

· g changes from increasing to decreasing or vice versa at these points.

· g' changes from positive to negative or vice versa at these points.

· g has a relative maximum or minimum at these points.

The point is not earned for the following responses:

· h changes concavity at $x = -4$ and $x = 2$.

· h'' changes sign at $x = -4$ and $x = 2$.

· Claims that $g'(-4) = 0$ or $g'(2) = 0$. These values do not exist.

· A response that claims g' changes from increasing to decreasing.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] x-values
- [ ] Justification

Solution:

$h'(x) = g(x)$

$h' = g$ changes from increasing to decreasing at $x = -4$.

$h' = g$ changes from decreasing to increasing at $x = 2$.

Points of inflection for the graph of h occur at $x = -4$ and $x = 2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for differentiating k correctly, whether evaluated at $x = 1$ or not.

This "chain rule" point can also be earned by using the product rule.

Second point:

The correct answer is 60.This answer does not need to be simplified but if it is, the simplification must be correct.

3-1-1b

The response "$10\,(-3)\,(-2)$" without other supporting work earns the answer point, but does not show enough computation to earn the chain rule point. In all other cases, the response must earn the first point to be eligible for the second point.

It is not necessary to simplify numeric answers.

The response $10 \cdot g\,(1) \cdot g'\,(1) = 10\,(-3)\,(-2)$ earns both points.

The response $10 \cdot g\,(1) \cdot g'\,(1)$ earns the first point, but not the second. The student must evaluate g and g'.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Chain rule
- ☐ Answer

Solution:

$k'\,(x) = 10 \cdot g\,(x) \cdot g'\,(x)$

$k'\,(1) = 10 \cdot g\,(1) \cdot g'\,(1) = 10 \cdot \ (-3) \cdot \ (-2) = 60$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by conveying that both the numerator and denominator each approach 0.

The response does not need to reference continuity or differentiability.

Responses including "$\text{limit} = \frac{0}{0}$" will not earn the first point.

Note: $\frac{0}{0}$ may appear in disguise, as in "$\text{limit} = \frac{h(-4)}{2(-4)+8}$."

Responses that include this will not earn the first point.

Other presentations of the limits, such as "$\rightarrow \frac{0}{0}$" or "$=$ indeterminate form $\frac{0}{0}$" will earn the first point.

Simply evaluating the numerator and denominator separately will earn this point since both are continuous functions.

Second point:

The point is earned by presenting a limit of a fraction whose numerator and denominator are clearly attempts at derivatives of $h\,(x)$ and $2x + 8$ respectively.

The point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Third point:

The point is earned for the correct answer, which equals $\frac{3}{2}$. The answer does not need to be simplified.

3-1-1b

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

$h' = g$ is continuous. $\Rightarrow$ h is continuous.
$\Rightarrow \lim\limits_{x \to -4} h(x) = h(-4)$ and $\lim\limits_{x \to -4} h'(x) = h'(-4)$.

$$\lim_{x \to -4} h(x) = h(-4) = \int_{-4}^{-4} g(t)dt = 0$$

$$\lim_{x \to -4} (2x+8) = 2 \cdot (-4) + 8 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to -4} \frac{h(x)}{2x+8} = \lim_{x \to -4} \frac{h'(x)}{2}$$

$$= \frac{h'(-4)}{2} = \frac{g(-4)}{2} = \frac{3}{2}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned by presenting numeric expressions identifiable as attempts to compute area.

Second point:

The point is earned for the correct answer, which equals -6, supported by appropriate work. The answer does not need to be simplified.

The answer "$\frac{3}{2} - \frac{3}{2} - 6$, " by itself, earns both points.

The unsupported answer "-6 " does not earn either point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Areas
- ☐ Answer

Solution:

3-1-1b

$h(2) = \int_{-4}^{2} g(t)\, dt = \frac{3}{2} - \frac{3}{2} - 6 = -6$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The point is earned for the sum of the series: $\frac{36r}{1-r}$.

The sum may appear in other forms, for example, $\frac{36}{1-r} - 36$. (The constant may be moved to the other side of the equation before the sum appears in closed form.)

Second point:

This point is earned for finding the value of r that is consistent with the answer for $h(2)$ found in part (f).

Eligibility: The response must earn the first point to be eligible for the second point.

The correct answer $r = -\frac{1}{5}$ earns the point if the correct answer -6 is given in part (f).

An answer $a > -18$ in part (f) is consistent with $r = \frac{a}{36+a}$.

An answer $a \leq -18$ in part (f) is consistent with no solution (since $|r| \geq 1$).

If no answer appears in part (f), the point cannot be earned unless $h(2)$ is computed in part (g).

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Sum of geometric series
- ☐ Answer

Solution:

$$36r + 36r^2 + \cdots + 36r^n + \cdots = \frac{36r}{1-r} \text{ for } |r| < 1$$

$$-6 = \frac{36r}{1-r} \Rightarrow -6 + 6r = 36r \Rightarrow -6 = 30r \Rightarrow r = -\frac{6}{30} = -\frac{1}{5}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The point is earned for the answer "no" with the reason that g is not differentiable.

The response doesn't need to name the interval. If an interval is presented, it can be open or closed, but cannot contradict the graph of g.

The response must name g, or "the graph," as the function that is non-differentiable. The name h' is sufficient if the connection $h' = g$ has been made earlier.

3-1-1b

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

No, the Mean Value Theorem does *not* guarantee the existence of a value of c with the stated properties because $h' = g$ is not differentiable for at least one point in $-8 < x < 6$.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response earns the point by referring to arc length, or by working with the arc length of some portion of a circle.

Use your judgment to determine whether or not the response conveys an attempt to compute the length of the curve.

Second point:

The point is earned for the correct answer 2π, supported by appropriate reasoning.

The unsupported answer 2π does not earn the point.

Alternate solution:

On the interval $[2,\ 6]$, we have:

$$g\left(x\right) = -1 - \sqrt{16 - \left(x-2\right)^2}$$

$$g'\left(x\right) = \frac{x-2}{\sqrt{16-\left(x-2\right)^2}}$$

$$1 + \left(g'\left(x\right)\right)^2 = \frac{16}{16-\left(x-2\right)^2} = \frac{1}{1-\left(\frac{x-2}{4}\right)^2}$$

$$\int_2^6 \sqrt{1 + \left(g'\left(x\right)\right)^2}\, dx = 4\sin^{-1}\left(\frac{x-2}{4}\right)\Big|_2^6 = 4\sin^{-1}\left(1\right) - 4\sin^{-1}\left(0\right) = 2\pi$$

This alternate solution earns the first point for an antiderivative of the form $A\sin^{-1}\left(u\right) + C$, where

· A is a nonzero real number,

· C may be an arbitrary constant, or any real number, or may be omitted, and

· u may be indeterminate or a declared function of x.

The second point is earned for the correct answer 2π, supported by appropriate reasoning.

0	1	2

3-1-1b

The student response accurately includes **both** of the criteria below.

- ☐ Recognizes arc length
- ☐ Answer

Solution:

$\int_2^6 \sqrt{1+(g'(x))^2}\,dx$ is the length of the graph of g from $x=2$ to $x=6$, which is the circumference of a quarter circle with radius 4.

$\frac{1}{4}(2\pi \cdot 4) = 2\pi$

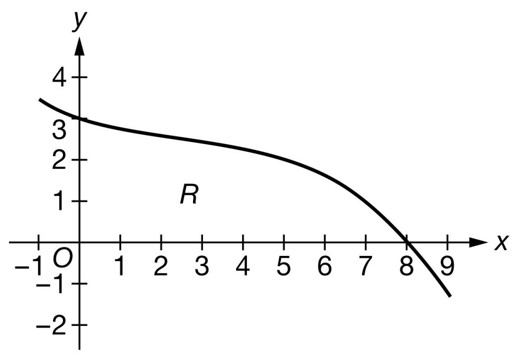
3-1-1b

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

The graph of g', the derivative of the twice-differentiable function g, is shown above for $-1 < x < 9$. Let R be the region in the first quadrant bounded by the graph of g' and the x-axis from $x = 0$ to $x = 8$. It is known that $g(0) = -7$, $g(8) = 9$, and $\int_0^8 g(x)\,dx = 20.8$.

(a) Find all values of x in the interval $-1 < x < 9$, if any, at which g has a critical point. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find the area of region R. Show the computations that lead to your answer.

(c) Must there exist a value of c, for $0 < c < 8$, such that $g''(c)$ is equal to the average rate of change of g' over the interval $0 \le x \le 8$? Justify your answer.

(d) Let h be the function defined by $h(x) = \int_0^{x^3} g(t)\,dt$. Find $h'(2)$. Show the computations that lead to your answer.

(e) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with height x and base in the region R. The volume of the solid is given by $\int_0^8 A(x)\,dx$. Write an expression for $A(x)$.

(f) What is the volume of the solid described in part (e)? Show the computations that lead to your answer.

(g) Evaluate $\lim_{x \to 8} \frac{g(x)-x-1}{\sin x - \sin 8}$. Give a reason for your answer.

(h) Find the value of $\int_0^8 \frac{g''(x)}{g'(x)}\,dx$ or show that it does not exist.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The response needs to identify $x = 8$ (and no other x-values). This can be done by presenting an ordered pair but in this case, the y-coordinate must be correct.

The following responses would earn the first point:

3-1-1b

· $x = 8$

· $(8,\ 9)$

The following responses would not earn the first point:

· $(8,\ 0)$

· $x = -1, x = 8, x = 9$

Second point:

The justification must discuss g' changing from positive to negative at $x = 8$.

The point cannot be earned for just stating that g' changes sign or changes at $x = 8$.

Note: Only referencing "the slope" or "the derivative" is too ambiguous to earn the point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Critical point at $x = 8$
- [ ] Relative maximum with justification

Solution:

g has a critical point where $g'(x) = 0$.

$g'(x) = 0 \ \Rightarrow\ x = 8$

$g'(x)$ changes from positive to negative at $x = 8$.

Therefore, g has a relative maximum at $x = 8$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response needs to apply the Fundamental Theorem of Calculus.

Here are some examples that earn the first point:

· $\int_0^8 g'(x)\ dx = g(x)\big|_0^8$

· $g(x)\big|_0^8$

· $g(8) - g(0)$

The first point would not be earned for $\int_0^8 g(x)\ dx = g(x)\big|_0^8 = g(8) - g(0)$.

Second point:

3-1-1b

The response must use the values of $g\,(8)$ and $g\,(0)$ to earn this point.

This response earns both points: $g\,(8) - g\,(0) = 9 - (-7)$.

A response of $9 + 7$ earns the second point, but not the first.

A response of 16 with no supporting work earns no points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$$\text{Area} = \int_0^8 g'\,(x)\;dx = g\,(8) - g\,(0)$$

$$= 9 - (-7) = 16$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must convey that g' is continuous as a consequence of g' being differentiable.

· “Since g' is differentiable, g' is continuous” or “ g' is differentiable and therefore continuous” would earn the first point.

If there is no implication involving g' being differentiable, then the first point is not earned.

· For example: “ g' is differentiable and continuous” does not earn this point.

Second point:

The response needs to be Yes. Further, the response must reference the MVT either directly or by stating the conclusion of the theorem in words. Since g is twice differentiable, the response does not need to say that g' is differentiable.

However, the response must reference the continuity of g' in order to earn this second point. The response must reference the difference quotient (or average rate of change) from the MVT either in words or with an equation.

Here are some examples that earn the second point, but not the first:

· Since g' is differentiable and continuous, then by the MVT there exists a value c such that $g''(c)$ is the average rate of change of g' on $[a,\ b]$.

· Since g' is continuous, then there must exist a value c such that $g''(c)$ is the average rate of change of g' on $[0,\ 8]$.

· Since g' is continuous, then there must exist a value c such that $g''(c) = \frac{g'(8) - g'(0)}{8-0}$.

These responses do not earn any points in part (c):

· Since g' is continuous then yes.

3-1-1b

Yes, by MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

g is twice differentiable. $\Rightarrow$ g' is differentiable on $0 \le x \le 8$.
$\Rightarrow$ g' is continuous on $0 \le x \le 8$.

Therefore, the Mean Value Theorem can be applied to g' on the interval $0 \le x \le 8$ to guarantee that there exists a value of c, for $0 < c < 8$, such that $g''(c)$ equals the average rate of change of g' over the interval $0 \le x \le 8$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First two points:

These two points may be awarded in either order, or simultaneously.

We must see $g\left(x^3\right)$ for the FTC point and $3x^2$ for the chain rule point. If there is a $g(0)$ present as part of the derivative of h or in the calculation of $h'(2)$, then the FTC point is not earned.

Examples:

· $h'(x) = g\left(x^3\right) \cdot 3x^2$ earns the FTC point and chain rule point.

· $g\left(x^3\right)$ earns the FTC point but not the chain rule point.

· $g\left(x^3\right) - g(0)$ does not earn either point.

· $g\left(x^3\right) \cdot 3x^2 - g(0)$ earns the chain rule point but not the FTC point.

Third point:

To be eligible for this point, the response must have already earned both the FTC point and the chain rule point. In addition, the response must use the value of $g(8)$. Answer must be our answer.

There is an alternate version of the FTC that is correct: $g\left(x^3\right) \cdot 3x^2 - g(0) \cdot 0$ would earn at least the first two points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Chain rule

3-1-1b

☐ Answer

Solution:

$h'(x) = g(x^3) \cdot 3x^2$

$h'(2) = g(2^3) \cdot 3 \cdot 2^2 = g(8) \cdot 3 \cdot 4 = 9 \cdot 12 = 108$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Both of these responses earn this point:

· $A(x) = x \cdot g\prime(x)$

· $x \cdot g\prime(x)$

Some responses give the formula for $A(x)$ indirectly. In this situation, the connection must be clear and correct.

This response earns the point: $\int_0^8 A(x)\, dx = \int_0^8 x \cdot g'(x)\, dx.$

These responses do not earn the point:

· $\int_0^8 x \cdot g'(x)\, dx$

· $\int_0^8 A(x)\, dx = A(x) = x \cdot g\prime(x)$

If the correct expression is used in part (f) but is never shown in part (e), the response does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$A(x) = x \cdot g'(x)$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The integration by parts point is earned for correctly performing integration by parts regardless of the limits of integration (those are part of the answer point). This is like an “indefinite integration by parts” point. Details of the $u \cdot v'$ decomposition are not required.

Here are some example of responses that earn the first point:

3-1-1b

- $\int_0^8 x\,g\prime\,(x)\;dx = x\,g\,(x)\,|_0^8 - \int_0^8 g\,(x)\;dx$
- $\int x\,g\prime\,(x)\;dx = x\,g\,(x) - \int g\,(x)\;dx$

Second point:

A numerical answer must be presented, although it does not need to be simplified.

$8\,\cdot\,g\,(8) - 20.8$ does not earn the answer point.

$8\,\cdot\,9 - 0 - 20.8$ is the minimal computation that must be shown in order to earn both points. If the 0 is missing from this expression, the first point for integration by parts is not earned.

If the response uses an expression from part (e) that was incorrect, then we can read for both points in part (f) only if the integration requires integration by parts.

To be eligible for the answer point the integration by parts point must be earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$u = x \qquad dv = g'\,(x)\;dx$
$du = dx \quad v = g\,(x)$

$$\int_0^8 x\,\cdot\,g'\,(x)\;dx = x\,\cdot\,g\,(x)\,|_0^8 - \int_0^8 g\,(x)\;dx$$

$$= 8\,\cdot\,g\,(8) - 0\,\cdot\,g\,(0) - \int_0^8 g\,(x)\;dx$$
$$= 8\,\cdot\,9 - 0\,\cdot\,(-7) - 20.8 = 51.2$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

To earn the first point, the response must convey that both the numerator and the denominator approach 0.

Any response that includes “ $\text{limit} = \frac{0}{0}$” does not earn the first point.

Presentations, such as “ $\to \frac{0}{0}$” or “ $\text{limit} = \text{indeterminate form}\ \frac{0}{0}$” can earn the first point as long as the response does not have “ $\text{limit} = \frac{0}{0}$”.

If the response writes the limit as a quotient and puts an arrow from the numerator to 0 and a separate arrow from the denominator to 0 then the point is earned.

3-1-1b

A response can earn the point by evaluating the numerator at $x = 8$, and the denominator at $x = 8$ without limit notation because both the numerator and denominator are continuous functions.

Second point:

The point is earned for attempting to apply L'Hospital's Rule by differentiating the numerator and the denominator.

The point is not earned for applying the quotient rule.

Errors in differentiating will not earn the answer point.

The second point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Each of these responses earns the second point:

- $\lim\limits_{x \to 8} \frac{g\prime(x)-1}{\cos x}$
- $\lim\limits_{x \to 8} \frac{g\prime(x)-1}{\cos x - \cos 8}$
- $\lim\limits_{x \to 8} \frac{g(x)-x-1}{\sin x - \sin 8} = \frac{g\prime(x)-1}{\cos x}$

Note: If the response shown in this last bullet is equated to our answer, the response earns only the second point, because of equating terms that are not equivalent. If the correct answer is given but not set equal to this expression, the response earns the second two points.

Third point:

The answer must be correct. The answer does not need to be simplified, but if it is, it must be correct. Either 6.872 or 6.873 are acceptable.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ g' is continuous. $\Rightarrow$ g is continuous. $\Rightarrow$ $\lim\limits_{x \to 8} g(x) = g(8)$ and $\lim\limits_{x \to 8} g'(x) = g'(8)$.

$$\lim_{x \to 8} (g(x) - x - 1) = g(8) - 8 - 1 = 9 - 8 - 1 = 0$$

$$\lim_{x \to 8} (\sin x - \sin 8) = \sin 8 - \sin 8 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 8} \frac{g(x)-x-1}{\sin x - \sin 8} = \lim_{x \to 8} \frac{g'(x)-1}{\cos x}$$

$$= \frac{g'(8)-1}{\cos 8} = \frac{0-1}{\cos 8} = -\frac{1}{\cos 8}$$

3-1-1b

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must include a correct one-sided limit that defines the improper integral in order to earn this point. Here is an example:

$$\cdot \lim_{b\to 8^-} \int_0^b \frac{g''(x)}{g\prime(x)}\, dx$$

Second point:

The point is earned for finding the antiderivative of $\dfrac{g''(x)}{g'(x)}$. No limit notation is required and limits of integration do not affect this point. This point can be earned if the first was not earned.

This response earns the second point: $\displaystyle\int_0^8 \frac{g''(x)}{g'(x)}\, dx = \ln|g'(x)|\,|_{x=0}^{x=8}$

Note that details of the substitution do not need to be provided.

The antiderivative must have absolute values.

Third point:

The point is earned for correctly evaluating the one sided limit of the anti-derivative.

This response earns the third point: $\displaystyle\lim_{b\to 8^-} \ln|g'(b)| - \ln 3$ does not exist.

A response of $-\infty$ earns the point.

Note: If there is no limit in part (h) then the response could only earn the antiderivative point if that is found correctly. It could not earn the first or third points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Improper integral
- ☐ Antiderivative
- ☐ Answer with reason

Solution:

$$\int_0^8 \frac{g''(x)}{g'(x)}\, dx = \lim_{b\to 8^-} \int_0^b \frac{g''(x)}{g'(x)}\, dx$$

$$= \lim_{b\to 8^-} [\ln|g'\ (x)|]_0^b = \lim_{b\to 8^-} [\ln\ g'\ (b) - \ln\ g'\ (0)]$$

$$= \lim_{b\to 8^-} [\ln\ g'\ (b) - \ln\ 3]$$

Because $\displaystyle\lim_{b\to 8^-} \ln\ g'\ (b)$ does not exist, the value of the integral does not exist.

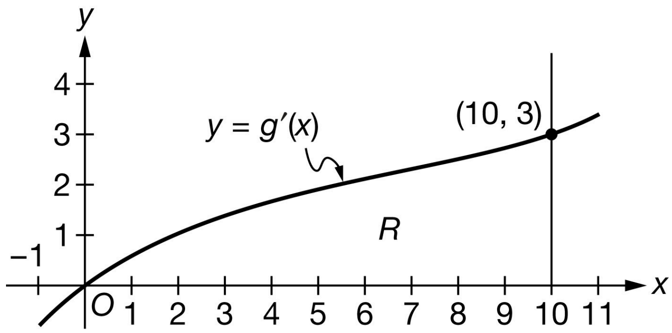
3-1-1b

15. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g', the derivative of the twice-differentiable function g, is shown above for $-1 < x < 11$. Let R be the region in the first quadrant bounded by the graph of g', the x-axis, and the vertical line $x = 10$. It is known that $g(0) = -2$, $g(10) = 16$, and $\int_0^{10} g(x)\,dx = 48$.

(a) Find all values of x in the interval $-1 < x < 11$, if any, at which g has a critical point. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find the area of region R. Show the computations that lead to your answer.

(c) Must there exist a value of c, for $0 < c < 10$, such that $g''(c)$ is equal to the average rate of change of g' over the interval $0 \le x \le 10$? Justify your answer.

(d) Let h be the function defined by $h(x) = \int_0^{x^2+1} g(t)\,dt$. Find $h'(3)$. Show the computations that lead to your answer.

(e) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with height x and base in the region R. The volume of the solid is given by $\int_0^{10} A(x)\,dx$. Write an expression for $A(x)$.

(f) What is the volume of the solid described in part (e)? Show the computations that lead to your answer.

(g) Evaluate $\lim_{x\to 0} \frac{g(x)+5x+2}{3\sin x}$. Give a reason for your answer.

(h) Find the value of $\int_0^{10} \frac{g''(x)}{g'(x)}\,dx$ or show that it does not exist.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The response needs to identify $x = 0$ (and no other x-values). This can be done by presenting an ordered pair but in this case, the y-coordinate must be correct.

The following responses would earn the first point:

· $x = 0$

· $(0,\ -2)$

3-1-1b

The following responses would not earn the first point:

· $(0,\ 0)$

· $x = -1, x = 0, x = 11$

Second point:

The justification must discuss g' changing from negative to positive at $x = 0$.

The point cannot be earned for just stating that g' changes sign or changes at $x = 0$.

Note: Only referencing “the slope” or “the derivative” is too ambiguous to earn the point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Critical point at $x = 0$
- [ ] Relative minimum with justification

Solution:

g has a critical point where $g'\ (x) = 0$.

$g'\ (x) = 0\ \Rightarrow\ x = 0$

$g'\ (x)$ changes from negative to positive at $x = 0$.

Therefore, g has a relative minimum at $x = 0$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response needs to apply the Fundamental Theorem of Calculus.

Here are some examples that earn the first point:

· $\int_0^{10} g'\ (x)\ dx = g\ (x)\big|_0^{10}$

· $g\ (x)\big|_0^{10}$

· $g\ (10) - g\ (0)$

The first point would not be earned for $\int_0^{10} g\ (x)\ dx = g\ (x)\big|_0^{10} = g\ (10) - g\ (0)$.

Second point:

The response must use the values of $g\ (10)$ and $g\ (0)$ to earn this point.

This response earns both points: $g\ (10) - g\ (0) = 16 - (-2)$.

3-1-1b

A response of $16 + 2$ earns the second point, but not the first.

A response of 18 with no supporting work earns no points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$$\text{Area} = \int_0^{10} g'(x)\,dx = g(10) - g(0)$$

$$= 16 - (-2) = 18$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must convey that g' is continuous as a consequence of g' being differentiable.

· "Since g' is differentiable, g' is continuous" or "g' is differentiable and therefore continuous" would earn the first point.

If there is no implication involving g' being differentiable, then the first point is not earned.

· For example: "g' is differentiable and continuous" does not earn this point.

Second point:

The response needs to be Yes. Further, the response must reference the MVT either directly or by stating the conclusion of the theorem in words. Since g is twice differentiable, the response does not need to say that g' is differentiable.

However, the response must reference the continuity of g' in order to earn this second point. The response must reference the difference quotient (or average rate of change) from the MVT either in words or with an equation.

Here are some examples that earn the second point, but not the first:

· Since g' is differentiable and continuous, then by the MVT there exists a value c such that $g''(c)$ is the average rate of change of g' on $[a,\ b]$.

· Since g' is continuous, then there must exist a value c such that $g''(c)$ is the average rate of change of g' on $[0,\ 10]$.

· Since g' is continuous, then there must exist a value c such that $g''(c) = \frac{g'(10)-g'(0)}{10-0}$.

These responses do not earn any points in part (c):

· Since g' is continuous then yes.

· Yes, by MVT.

3-1-1b

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

g is twice differentiable. $\Rightarrow$ g' is differentiable on $0 \leq x \leq 10$.

$\Rightarrow$ g' is continuous on $0 \leq x \leq 10$.

Therefore, the Mean Value Theorem can be applied to g' on the interval $0 \leq x \leq 10$ to guarantee that there exists a value of c, for $0 < c < 10$, such that $g''(c)$ equals the average rate of change of g' over the interval $0 \leq x \leq 10$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First two points:

These two points may be awarded in either order, or simultaneously.

We must see $g\left(x^2+1\right)$ for the FTC point and $2x$ for the chain rule point. If there is a $g(0)$ present as part of the derivative of h or in the calculation of $h'(3)$, then the FTC point is not earned.

Examples:

· $h'(x) = g\left(x^2+1\right) \cdot 2x$ earns the FTC point and chain rule point.

· $g\left(x^2+1\right)$ earns the FTC point but not the chain rule point.

· $g\left(x^2+1\right) - g(0)$ does not earn either point.

· $g\left(x^2+1\right) \cdot 2x - g(0)$ earns the chain rule point but not the FTC point.

Third point:

To be eligible for this point, the response must have already earned both the FTC point and the chain rule point. In addition, the response must use the value of $g(10)$. Answer must be our answer.

There is an alternate version of the FTC that is correct: $g\left(x^2+1\right) \cdot 2x - g(0) \cdot 0$ would earn at least the first two points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

3-1-1b

Solution:

$h'(x) = g(x^2+1) \cdot 2x$

$h'(3) = g(3^2+1) \cdot 2 \cdot 3 = g(10) \cdot 6 = 16 \cdot 6 = 96$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Both of these responses earn this point:

· $A(x) = x \cdot g'(x)$

· $x \cdot g'(x)$

Some responses give the formula for $A(x)$ indirectly. In this situation, the connection must be clear and correct.

This response earns the point: $\int_0^{10} A(x)\, dx = \int_0^{10} x \cdot g'(x)\, dx.$

These responses do not earn the point:

· $\int_0^{10} x \cdot g'(x)\, dx$

· $\int_0^{10} A(x)\, dx = A(x) = x \cdot g'(x)$

If the correct expression is used in part (f) but is never shown in part (e), the response does not earn this point.

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$A(x) = x \cdot g'(x)$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The integration by parts point is earned for correctly performing integration by parts regardless of the limits of integration (those are part of the answer point). This is like an "indefinite integration by parts" point. Details of the $u \cdot v'$ decomposition are not required.

Here are some example of responses that earn the first point:

· $\int_0^{10} x\, g'(x)\, dx = x\, g(x)|_0^{10} - \int_0^{10} g(x) dx$

3-1-1b

$$\cdot \int x\,g'(x)\ dx = x\,g(x) - \int g(x)dx$$

Second point:

A numerical answer must be presented, although it does not need to be simplified.

$10 \cdot g(10) - 48$ does not earn the answer point.

$10 \cdot 16 - 0 - 48$ is the minimal computation that must be shown in order to earn both points. If the 0 is missing from this expression, the first point for integration by parts is not earned.

If the response uses an expression from part (e) that was incorrect, then we can read for both points in part (f) only if the integration requires integration by parts.

To be eligible for the answer point the integration by parts point must be earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$\begin{array}{ll} u = x & dv = g'(x)\ dx \\ du = dx & v = g(x) \end{array}$$

$$\int_0^{10} x \cdot g'(x)\ dx = x \cdot g(x)\big|_0^{10} - \int_0^{10} g(x)\ dx$$

$$= 10 \cdot g(10) - 0 \cdot g(0) - \int_0^{10} g(x)\ dx$$

$$= 10 \cdot 16 - 0 \cdot\ (-2) - 48 = 112$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

To earn the first point, the response must convey that both the numerator and the denominator approach 0.

Any response that includes "$\text{limit} = \frac{0}{0}$" does not earn the first point.

Presentations, such as "$\rightarrow \frac{0}{0}$" or "$\text{limit} = \text{indeterminate form } \frac{0}{0}$" can earn the first point as long as the response does not have "$\text{limit} = \frac{0}{0}$".

If the response writes the limit as a quotient and puts an arrow from the numerator to 0 and a separate arrow from the denominator to 0 then the point is earned.

A response can earn the point by evaluating the numerator at $x = 0$, and the denominator at $x = 0$ without limit notation because both the numerator and denominator are continuous functions.

3-1-1b

Second point:

The point is earned for attempting to apply L'Hospital's Rule by differentiating the numerator and the denominator.

The point is not earned for applying the quotient rule.

Errors in differentiating will not earn the answer point.

The second point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Each of these responses earns the second point:

- $\lim\limits_{x \to 0} \frac{g\prime(x)+5}{3\cos\,x}$
- $\lim\limits_{x \to 0} \frac{g(x)+5x+2}{3\sin\,x} = \frac{g\prime(x)+5}{3\cos\,x}$

Note: If the response shown in this last bullet is equated to our answer, the response earns only the second point, because of equating terms that are not equivalent. If the correct answer is given but not set equal to this expression, the response earns the second two points.

Third point:

The answer must be correct. The answer does not need to be simplified, but if it is, it must be correct. Either 1.666 or 1.667 are acceptable.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ g' is continuous. $\Rightarrow$ g is continuous. $\Rightarrow$ $\lim\limits_{x \to 0} g\left(x\right) = g\left(0\right)$ and $\lim\limits_{x \to 0} g'\left(x\right) = g'\left(0\right)$.

$\lim\limits_{x \to 0} \left(g\left(x\right) + 5x + 2\right) = g\left(0\right) + 0 + 2 = -2 + 0 + 2 = 0$

$\lim\limits_{x \to 0} 3\sin\,x = 3\sin\,0 = 0$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 0} \frac{g(x)+5x+2}{3\sin\,x} = \lim_{x \to 0} \frac{g'(x)+5}{3\cos\,x}$$

$$= \frac{g'(0)+5}{3\cos\,0} = \frac{0+5}{3} = \frac{5}{3}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

3-1-1b

The response must include a correct one-sided limit that defines the improper integral in order to earn this point. Here is an example:

· $\lim_{b \to 0^+} \int_b^{10} \frac{g''(x)}{g'(x)} dx$

Second point:

The point is earned for finding the antiderivative of $\frac{g''(x)}{g'(x)}$. No limit notation is required and limits of integration do not affect this point. This point can be earned if the first was not earned.

This response earns the second point: $\int_0^{10} \frac{g''(x)}{g'(x)} dx = \ln|g'(x)|\,|_{x=0}^{x=10}$

Note that details of the substitution do not need to be provided.

The antiderivative must have absolute values.

Third point:

The point is earned for correctly evaluating the one sided limit of the anti-derivative.

This response earns the third point: $\ln 3 - \lim_{b \to 0^+} \ln|g'(b)|$ does not exist.

A response of ∞ earns the point.

Note: If there is no limit in part (h) then the response could only earn the antiderivative point if that is found correctly. It could not earn the first or third points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Improper integral
- ☐ Antiderivative
- ☐ Answer with reason

Solution:

$$\int_0^{10} \frac{g''(x)}{g'(x)} dx = \lim_{b \to 0^+} \int_b^{10} \frac{g''(x)}{g'(x)} dx$$

$$= \lim_{b \to 0^+} [\ln|g'(x)|]_b^{10} = \lim_{b \to 0^+} [\ln g'(10) - \ln g'(b)]$$

$$= \lim_{b \to 0^+} [\ln 3 - \ln g'(b)]$$

Because $\lim_{b \to 0^+} \ln g'(b)$ does not exist, the value of the integral does not exist.

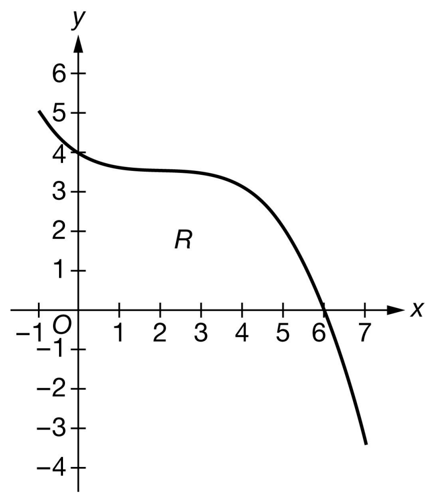
3-1-1b

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

The graph of g', the derivative of the twice-differentiable function g, is shown above for $-1 < x < 7$. Let R be the region in the first quadrant bounded by the graph of g' and the x-axis from $x = 0$ to $x = 6$. It is known that $g(0) = -7$, $g(6) = 11$, and $\int_0^6 g(x)\,dx = 20.4$.

(a) Find all values of x in the interval $-1 < x < 7$, if any, at which g has a critical point. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find the area of region R. Show the computations that lead to your answer.

(c) Must there exist a value of c, for $0 < c < 6$, such that $g''(c)$ is equal to the average rate of change of g' over the interval $0 \le x \le 6$? Justify your answer.

(d) Let h be the function defined by $h(x) = \int_0^{6x^2} g(t)\,dt$. Find $h'(1)$. Show the computations that lead to your answer.

(e) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with height x and base in the region R. The volume of the solid is given by $\int_0^6 A(x)\,dx$. Write an expression for $A(x)$.

(f) What is the volume of the solid described in part (e)? Show the computations that lead to your answer.

(g) Evaluate $\lim\limits_{x\to 6} \frac{g(x)-2x+1}{e^x-e^6}$. Give a reason for your answer.

(h) Find the value of $\int_0^6 \frac{g''(x)}{g'(x)}\,dx$ or show that it does not exist.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

3-1-1b

The response needs to identify $x = 6$ (and no other x-values). This can be done by presenting an ordered pair but in this case, the y-coordinate must be correct.

The following responses would earn the first point:

· $x = 6$

· $(6,\ 11)$

The following responses would not earn the first point:

· $(6,\ 0)$

· $x = -1,\ x = 6,\ x = 7$

Second point:

The justification must discuss g' changing from positive to negative at $x = 6$.

The point cannot be earned for just stating that g' changes sign or changes at $x = 6$.

Note: Only referencing "the slope" or "the derivative" is too ambiguous to earn the point.

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Critical point at $x = 6$
- ☐ Relative maximum with justification

Solution:

g has a critical point where $g'(x) = 0$.

$g'(x) = 0 \ \Rightarrow\ x = 6$

$g'(x)$ changes from positive to negative at $x = 6$.

Therefore, g has a relative maximum at $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response needs to apply the Fundamental Theorem of Calculus.

Here are some examples that earn the first point:

· $\int_0^6 g'(x)\, dx = g(x)\big|_0^6$

· $g(x)\big|_0^6$

· $g(6) - g(0)$

3-1-1b

The first point would not be earned for $\int_0^6 g(x)\ dx = g(x)|_0^6 = g(6) - g(0)$.

Second point:

The response must use the values of $g(6)$and $g(0)$ to earn this point.

This response earns both points: $g(6) - g(0) = 11 - (-7)$.

A response of $11 + 7$ earns the second point, but not the first.

A response of 18 with no supporting work earns no points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

Area $= \int_0^6 g'(x)\ dx = g(6) - g(0)$

$= 11 - (-7) = 18$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must convey that g' is continuous as a consequence of g' being differentiable.

· "Since g' is differentiable, g' is continuous" or "g' is differentiable and therefore continuous"

would earn the first point.

If there is no implication involving being differentiable, then the first point is not earned.

· For example: "g' is differentiable and continuous" does not earn this point.

Second point:

The response needs to be Yes. Further, the response must reference the MVT either directly or by stating the conclusion of the theorem in words. Since g is twice differentiable, the response does not need to say that g' is differentiable.

However, the response must reference the continuity of g' in order to earn this second point. The response must reference the difference quotient (or average rate of change) from the MVT either in words or with an equation.

Here are some examples that earn the second point, but not the first:

· Since g' is differentiable and continuous, then by the MVT there exists a value c such that $g''(c)$ is the average rate of change of g' on $[a,\ b]$.

· Since g' is continuous, then there must exist a value c such that $g''(c)$ is the average rate of change of g' on $[0,\ 6]$.

3-1-1b

· Since g' is continuous, then there must exist a value c such that $g''(c) = \frac{g'(6)-g'(0)}{6-0}$.

These responses do not earn any points in part (c):

· Since g' is continuous then yes.

· Yes, by MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

g is twice differentiable. $\Rightarrow$ g' is differentiable on $0 \leq x \leq 6$.
$\Rightarrow$ g' is continuous on $0 \leq x \leq 6$.

Therefore, the Mean Value Theorem can be applied to g' on the interval $0 \leq x \leq 6$ to guarantee that there exists a value of c, such that $g''(c)$ equals the average rate of change of g' over the interval $0 \leq x \leq 6$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First two points:

These two points may be awarded in either order, or simultaneously.

We must see $g\left(6x^2\right)$ for the FTC point and $12x$ for the chain rule point. If there is a $g(0)$ present as part of the derivative of h or in the calculation of $h'(1)$, then the FTC point is not earned.

Examples:

· $h'(x) = g\left(6x^2\right) \cdot 12x$ earns the FTC point and chain rule point.

· $g\left(6x^2\right)$ earns the FTC point but not the chain rule point.

· $g\left(6x^2\right) - g(0)$ does not earn either point.

· $g\left(6x^2\right) \cdot 12x - g(0)$ earns the chain rule point but not the FTC point.

Third point:

To be eligible for this point, the response must have already earned both the FTC point and the chain rule point. In addition, the response must use the value of $g(6)$. Answer must be our answer.

There is an alternate version of the FTC that is correct: $g\left(6x^2\right) \cdot 12x - g(0) \cdot 0$ would earn at least the first two points.

0	1	2	3

3-1-1b

The student response accurately includes **all three** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Chain rule
- ☐ Answer

Solution:

$h'(x) = g\left(6x^2\right) \cdot 12x$

$h'(1) = g\left(6 \cdot 1^2\right) \cdot 12 \cdot 1 = g(6) \cdot 12 = 11 \cdot 12 = 132$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Both of these responses earn this point:

· $A(x) = x \cdot g'(x)$

· $x \cdot g'(x)$

Some responses give the formula for $A(x)$ indirectly. In this situation, the connection must be clear and correct.

This response earns the point: $\int_0^6 A(x)\, dx = \int_0^6 x \cdot g'(x)\, dx.$

These responses do not earn the point:

· $\int_0^6 x \cdot g'(x)\, dx$

· $\int_0^6 A(x)\, dx = A(x) = x \cdot g'(x)$

If the correct expression is used in part (f) but is never shown in part (e), the response does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$A(x) = x \cdot g'(x)$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

3-1-1b

The integration by parts point is earned for correctly performing integration by parts regardless of the limits of integration (those are part of the answer point). This is like an “indefinite integration by parts” point. Details of the $u \cdot v'$ decomposition are not required.

Here are some example of responses that earn the first point:

- $\int_0^6 x\,g'(x)\,dx = x\,g(x)|_0^6 - \int_0^6 g(x)\,dx$
- $\int x\,g'(x)\,dx = x\,g(x) - \int g(x)\,dx$

Second point:

A numerical answer must be presented, although it does not need to be simplified.

$6 \cdot g(6) - 20.4$ does not earn the answer point.

$6 \cdot 11 - 0 - 20.4$ is the minimal computation that must be shown in order to earn both points. If the 0 is missing from this expression, the first point for integration by parts is not earned.

If the response uses an expression from part (e) that was incorrect, then we can read for both points in part (f) only if the integration requires integration by parts.

To be eligible for the answer point the integration by parts point must be earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$\begin{array}{ll} u = x & dv = g'(x)\,dx \\ du = dx & v = g(x) \end{array}$$

$$\int_0^6 x \cdot g'(x)\,dx = x \cdot g(x)|_0^6 - \int_0^6 g(x)\,dx$$

$$= 6 \cdot g(6) - 0 \cdot g(0) - \int_0^6 g(x)\,dx$$
$$= 6 \cdot 11 - 0 \cdot (-7) - 20.4 = 45.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

To earn the first point, the response must convey that both the numerator and the denominator approach 0.

Any response that includes “$\text{limit} = \frac{0}{0}$” does not earn the first point.

3-1-1b

Presentations, such as "$\to \frac{0}{0}$" or "$\text{limit} = \text{indeterminate form } \frac{0}{0}$" can earn the first point as long as the response does not have "$\text{limit} = \frac{0}{0}$".

If the response writes the limit as a quotient and puts an arrow from the numerator to 0 and a separate arrow from the denominator to 0 then the point is earned.

A response can earn the point by evaluating the numerator at $x = 6$, and the denominator at $x = 6$ without limit notation because both the numerator and denominator are continuous functions.

Second point:

The point is earned for attempting to apply L'Hospital's Rule by differentiating the numerator and the denominator.

The point is not earned for applying the quotient rule.

Errors in differentiating will not earn the answer point.

The second point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Each of these responses earns the second point:

- $\lim\limits_{x \to 6} \frac{g\prime(x) - 2}{e^x}$
- $\lim\limits_{x \to 6} \frac{g\prime(x) - 2}{e^x - e^6}$
- $\lim\limits_{x \to 6} \frac{g(x) - 2x - 1}{e^x - e^6} = \frac{g\prime(x) - 2}{e^x}$

Note: If the response shown in this last bullet is equated to our answer, the response earns only the second point, because of equating terms that are not equivalent. If the correct answer is given but not set equal to this expression, the response earns the second two points.

Third point:

The answer must be correct. The answer does not need to be simplified, but if it is, it must be correct. Either -0.004 or -0.005 are acceptable.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ g' is continuous. $\Rightarrow$ g is continuous. $\Rightarrow$ $\lim\limits_{x \to 6} g(x) = g(6)$ and $\lim\limits_{x \to 6} g'(x) = g'(6)$.

$$\lim_{x \to 6} (g(x) - 2x + 1) = g(6) - 12 + 1 = 11 - 12 + 1 = 0$$

$$\lim_{x \to 6} \left(e^x - e^6\right) = e^6 - e^6 = 0$$

3-1-1b

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to 6}\frac{g(x)-2x+1}{e^x-e^6}=\lim_{x\to 6}\frac{g'(x)-2}{e^x}$$

$$=\frac{g'(6)-2}{e^6}=\frac{0-2}{e^6}=-\frac{2}{e^6}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must include a correct one-sided limit that defines the improper integral in order to earn this point. Here is an example:

$$\cdot\lim_{b\to 6^-}\int_0^b \frac{g''(x)}{g'(x)}\,dx$$

Second point:

The point is earned for finding the antiderivative of $\frac{g''(x)}{g'(x)}$. No limit notation is required and limits of integration do not affect this point. This point can be earned if the first was not earned.

This response earns the second point: $\int_0^6 \frac{g''(x)}{g'(x)}\,dx=\ln|g'(x)|\,|_{x=0}^{x=6}$

Note that details of the substitution do not need to be provided.

The antiderivative must have absolute values.

Third point:

The point is earned for correctly evaluating the one sided limit of the anti-derivative.

This response earns the third point: $\lim_{b\to 6^-}\ln|g'(b)|-\ln 4$ does not exist.

A response of $-\infty$ earns the point.

Note: If there is no limit in part (h) then the response could only earn the antiderivative point if that is found correctly. It could not earn the first or third points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Improper integral
- ☐ Antiderivative
- ☐ Answer with reason

Solution:

$$\int_0^6 \frac{g''(x)}{g'(x)}\,dx=\lim_{b\to 6^-}\int_0^b \frac{g''(x)}{g'(x)}\,dx$$

3-1-1b

$$= \lim_{b \to 6^-} [\ln |g'(x)|]_0^b = \lim_{b \to 6^-} [\ln g'(b) - \ln g'(0)]$$
$$= \lim_{b \to 6^-} [\ln g'(b) - \ln 4]$$

Because $\lim_{b \to 6^-} \ln g'(b)$ does not exist, the value of the integral does not exist.

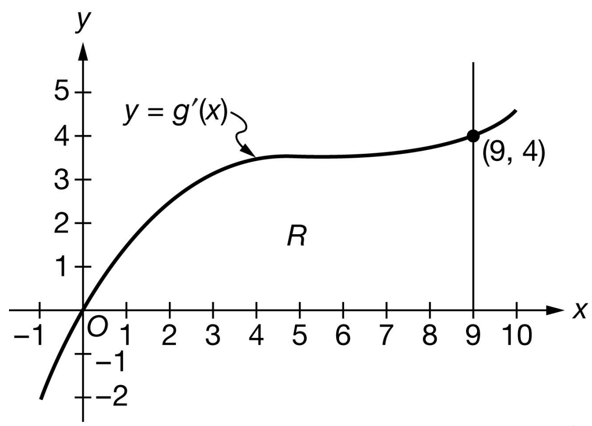
3-1-1b

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of g', the derivative of the twice-differentiable function g, is shown above for $-1 < x < 10$. Let R be the region in the first quadrant bounded by the graph of g', the x-axis, and the vertical line $x = 9$. It is known that $g(0) = -7$, $g(9) = 20$, and $\int_0^9 g(x)\,dx = 39.6$.

(a) Find all values of x in the interval $-1 < x < 10$, if any, at which g has a critical point. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find the area of region R. Show the computations that lead to your answer.

(c) Must there exist a value of c, for $0 < c < 9$, such that $g''(c)$ is equal to the average rate of change of g' over the interval $0 \le x \le 9$? Justify your answer.

(d) Let h be the function defined by $h(x) = \int_0^{x^2} g(t)\,dt$. Find $h'(3)$. Show the computations that lead to your answer.

(e) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with height x and base in the region R. The volume of the solid is given by $\int_0^9 A(x)\,dx$. Write an expression for $A(x)$.

(f) What is the volume of the solid described in part (e)? Show the computations that lead to your answer.

(g) Evaluate $\lim_{x \to 0} \frac{g(x) - 5x + 7}{e^x - 1}$. Give a reason for your answer.

(h) Find the value of $\int_0^9 \frac{g''(x)}{g'(x)}\,dx$ or show that it does not exist.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The response needs to identify $x = 0$ (and no other x-values). This can be done by presenting an ordered pair but in this case, the y-coordinate must be correct.

The following responses would earn the first point:

3-1-1b

· $x = 0$

· $(0, \ -7)$

The following responses would not earn the first point:

· $(0, \ 0)$

· $x = -1, x = 0, x = 10$

Second point:

The justification must discuss g' changing from negative to positive at $x = 0$.

The point cannot be earned for just stating that g' changes sign or changes at $x = 0$.

Note: Only referencing "the slope" or "the derivative" is too ambiguous to earn the point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Critical point at $x = 0$
- [ ] Relative minimum with justification

Solution:

g has a critical point where $g'(x) = 0$.

$g'(x) = 0 \Rightarrow x = 0$

$g'(x)$ changes from negative to positive at $x = 0$.

Therefore, g has a relative minimum at $x = 0$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response needs to apply the Fundamental Theorem of Calculus.

Here are some examples that earn the first point:

· $\int_0^9 g'(x)\, dx = g(x)\big|_0^9$

· $g(x)\big|_0^9$

· $g(9) - g(0)$

The first point would not be earned for $\int_0^9 g(x)\, dx = g(x)\big|_0^9 = g(9) - g(0)$.

Second point:

3-1-1b

The response must use the values of $g\left(9\right)$ and $g\left(0\right)$ to earn this point.

This response earns both points: $g\left(9\right)-g\left(0\right)=20-\left(-7\right)$.

A response of $20+7$ earns the second point, but not the first.

A response of 27 with no supporting work earns no points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$$\text{Area}=\int_{0}^{9}g'\left(x\right)\,dx=g\left(9\right)-g\left(0\right)$$

$$=20-\left(-7\right)=27$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must convey that g' is continuous as a consequence of g' being differentiable.

· "Since g' is differentiable, g' is continuous" or "g' is differentiable and therefore continuous" would earn the first point.

If there is no implication involving g' being differentiable, then the first point is not earned.

· For example: "g' is differentiable and continuous" does not earn this point.

Second point:

The response needs to be Yes. Further, the response must reference the MVT either directly or by stating the conclusion of the theorem in words. Since g is twice differentiable, the response does not need to say that g' is differentiable.

However, the response must reference the continuity of g' in order to earn this second point. The response must reference the difference quotient (or average rate of change) from the MVT either in words or with an equation.

Here are some examples that earn the second point, but not the first:

· Since g' is differentiable and continuous, then by the MVT there exists a value c such that $g''\left(c\right)$ is the average rate of change of g' on $\left[a,\ b\right]$.

· Since g' is continuous, then there must exist a value c such that $g''\left(c\right)$ is the average rate of change of g' on $\left[0,\ 9\right]$.

· Since g' is continuous, then there must exist a value c such that $g''\left(c\right)=\frac{g'\left(9\right)-g'\left(0\right)}{9-0}$.

These responses do not earn any points in part (c):

· Since g' is continuous then yes.

3-1-1b

· Yes, by MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

g is twice differentiable. $\Rightarrow$ g' is differentiable on $0 \le x \le 9$.
$\Rightarrow$ g' is continuous on $0 \le x \le 9$.

Therefore, the Mean Value Theorem can be applied to g' on the interval $0 \le x \le 9$ to guarantee that there exists a value of c, for $0 < c < 9$, such that $g''(c)$ equals the average rate of change of g' over the interval $0 \le x \le 9$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First two points:

These two points may be awarded in either order, or simultaneously.

We must see $g\left(x^2\right)$ for the FTC point and $2x$ for the chain rule point. If there is a $g(0)$ present as part of the derivative of h or in the calculation of $h'(3)$, then the FTC point is not earned.

Examples:

· $h'(x) = g\left(x^2\right) \cdot 2x$ earns the FTC point and chain rule point.

· $g\left(x^2\right)$ earns the FTC point but not the chain rule point.

· $g\left(x^2\right) - g(0)$ does not earn either point.

· $g\left(x^2\right) \cdot 2x - g(0)$ earns the chain rule point but not the FTC point.

Third point:

To be eligible for this point, the response must have already earned both the FTC point and the chain rule point. In addition, the response must use the value of $g(9)$. Answer must be our answer.

There is an alternate version of the FTC that is correct: $g\left(x^2\right) \cdot 2x - g(0) \cdot 0$ would earn at least the first two points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus
- [ ] Chain rule

3-1-1b

☐ Answer

Solution:

$h'(x) = g(x^2) \cdot 2x$

$h'(3) = g(3^2) \cdot 2 \cdot 3 = g(9) \cdot 6 = 20 \cdot 6 = 120$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Both of these responses earn this point:

· $A(x) = x \cdot g\prime(x)$

· $x \cdot g\prime(x)$

Some responses give the formula for $A(x)$ indirectly. In this situation, the connection must be clear and correct.

This response earns the point: $\int_0^9 A(x)\, dx = \int_0^9 x \cdot g'(x)\, dx.$

These responses do not earn the point:

· $\int_0^9 x \cdot g'(x)\, dx$

· $\int_0^9 A(x)\, dx = A(x) = x \cdot g\prime(x)$

If the correct expression is used in part (f) but is never shown in part (e), the response does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

$A(x) = x \cdot g\prime(x)$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The integration by parts point is earned for correctly performing integration by parts regardless of the limits of integration (those are part of the answer point). This is like an “indefinite integration by parts” point. Details of the $u \cdot v\prime$ decomposition are not required.

Here are some example of responses that earn the first point:

3-1-1b

- $\int_0^9 x\, g\prime\,(x)\ dx = x\, g\,(x)|_0^9 - \int_0^9 g\,(x)\ dx$
- $\int x\, g\prime\,(x)\ dx = x\, g\,(x) - \int g\,(x)\ dx$

Second point:

A numerical answer must be presented, although it does not need to be simplified.

$9 \cdot g\,(9) - 39.6$ does not earn the answer point.

$9 \cdot 20 - 0 - 39.6$ is the minimal computation that must be shown in order to earn both points. If the 0 is missing from this expression, the first point for integration by parts is not earned.

If the response uses an expression from part (e) that was incorrect, then we can read for both points in part (f) only if the integration requires integration by parts.

To be eligible for the answer point the integration by parts point must be earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Integration by parts
- ☐ Answer

Solution:

$$\begin{array}{ll} u = x & dv = g'\,(x)\ dx \\ du = dx & v = g\,(x) \end{array}$$

$$\int_0^9 x \cdot g'\,(x)\ dx = x \cdot g\,(x)|_0^9 - \int_0^9 g\,(x)\ dx$$

$$= 9 \cdot g\,(9) - 0 \cdot g\,(0) - \int_0^9 g\,(x)\ dx$$
$$= 9 \cdot 20 - 0 \cdot\ (-7) - 39.6 = 140.4$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

To earn the first point, the response must convey that both the numerator and the denominator approach 0.

Any response that includes “$\text{limit} = \frac{0}{0}$” does not earn the first point.

Presentations, such as “$\rightarrow \frac{0}{0}$” or “$\text{limit} = \text{indeterminate form } \frac{0}{0}$” can earn the first point as long as the response does not have “$\text{limit} = \frac{0}{0}$”.

If the response writes the limit as a quotient and puts an arrow from the numerator to 0 and a separate arrow from the denominator to 0 then the point is earned.

3-1-1b

A response can earn the point by evaluating the numerator at $x = 0$, and the denominator at $x = 0$ without limit notation because both the numerator and denominator are continuous functions.

Second point:

The point is earned for attempting to apply L'Hospital's Rule by differentiating the numerator and the denominator.

The point is not earned for applying the quotient rule.

Errors in differentiating will not earn the answer point.

The second point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Each of these responses earns the second point:

- $\lim\limits_{x\to 0} \frac{g\prime(x)-5}{e^x}$
- $\lim\limits_{x\to 0} \frac{g\prime(x)-5}{e^x-1}$
- $\lim\limits_{x\to 0} \frac{g(x)-5x+7}{e^x-1} = \frac{g\prime(x)-5}{e^x}$

Note: If the response shown in this last bullet is equated to our answer, the response earns only the second point, because of equating terms that are not equivalent. If the correct answer is given but not set equal to this expression, the response earns the second two points.

Third point:

The answer must be correct. The answer does not need to be simplified, but if it is, it must be correct.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ g' is continuous. $\Rightarrow$ g is continuous. $\Rightarrow$ $\lim\limits_{x\to 0} g(x) = g(0)$ and $\lim\limits_{x\to 0} g'(x) = g'(0)$.

$\lim\limits_{x\to 0} (g(x) - 5x + 7) = g(0) - 0 + 7 = -7 - 0 + 7 = 0$

$\lim\limits_{x\to 0} (e^x - 1) = e^0 - 1 = 1 - 1 = 0$

Therefore, L'Hospital's Rule can be applied.

$\lim\limits_{x\to 0} \frac{g(x)-5x+7}{e^x-1} = \lim\limits_{x\to 0} \frac{g'(x)-5}{e^x}$

$= \frac{g'(0)-5}{e^0} = \frac{0-5}{1} = -5$

3-1-1b

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must include a correct one-sided limit that defines the improper integral in order to earn this point. Here is an example:

$$\cdot \lim_{b\to 0^+} \int_b^9 \frac{g''(x)}{g'(x)}\,dx$$

Second point:

The point is earned for finding the antiderivative of $\frac{g''(x)}{g'(x)}$. No limit notation is required and limits of integration do not affect this point. This point can be earned if the first was not earned.

This response earns the second point: $\int_0^9 \frac{g''(x)}{g'(x)}\,dx = \ln|g'(x)||_{x=0}^{x=9}$

Note that details of the substitution do not need to be provided.

The antiderivative must have absolute values.

Third point:

The point is earned for correctly evaluating the one sided limit of the anti-derivative.

This response earns the third point: $\ln 4 - \lim_{b\to 0^+} \ln|g\prime(b)|$ does not exist.

A response of ∞ earns the point.

Note: If there is no limit in part (h) then the response could only earn the antiderivative point if that is found correctly. It could not earn the first or third points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Improper integral
- ☐ Antiderivative
- ☐ Answer with reason

Solution:

$$\int_0^9 \frac{g''(x)}{g'(x)}\,dx = \lim_{b\to 0^+} \int_b^9 \frac{g''(x)}{g'(x)}\,dx$$

$$= \lim_{b\to 0^+} [\ln|g'(x)|]_b^9 = \lim_{b\to 0^+} [\ln g'(9) - \ln g'(b)]$$

$$= \lim_{b\to 0^+} [\ln 4 - \ln g'(b)]$$

Because $\lim_{b\to 0^+} \ln g'(b)$ does not exist, the value of the integral does not exist.

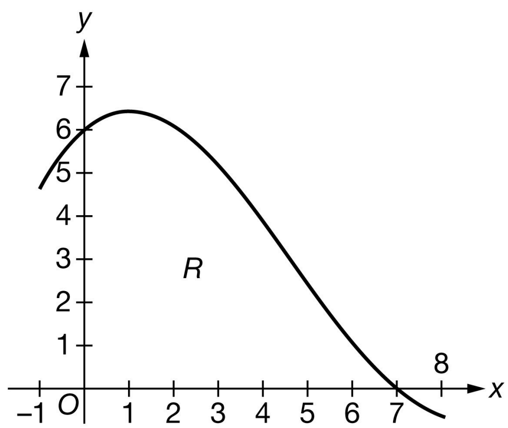
3-1-1b

18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

The graph of g', the derivative of the twice-differentiable function g, is shown above for $-1 < x < 8$. Let R be the region in the first quadrant bounded by the graph of g' and the x-axis from $x = 0$ to $x = 7$. It is known that $g(0) = -16$, $g(7) = 12$, and $\int_0^7 g(x)\,dx = 15.4$.

(a) Find all values of x in the interval $-1 < x < 8$, if any, at which g has a critical point. Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.

(b) Find the area of region R. Show the computations that lead to your answer.

(c) Must there exist a value of c, for $0 < c < 7$, such that $g''(c)$ is equal to the average rate of change of g' over the interval $0 \le x \le 7$? Justify your answer.

(d) Let h be the function defined by $h(x) = \int_0^{x^3-1} g(t)\,dt$. Find $h'(2)$. Show the computations that lead to your answer.

(e) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with height x and base in the region R. The volume of the solid is given by $\int_0^7 A(x)\,dx$. Write an expression for $A(x)$.

(f) What is the volume of the solid described in part (e)? Show the computations that lead to your answer.

(g) Evaluate $\lim_{x\to 0} \frac{g(x)-x+16}{\sin x}$. Give a reason for your answer.

(h) Find the value of $\int_0^7 \frac{g''(x)}{g'(x)}\,dx$ or show that it does not exist.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

First point:

The response needs to identify $x = 7$ (and no other x-values). This can be done by presenting an ordered pair but in this case, the y-coordinate must be correct.

3-1-1b

The following responses would earn the first point:

· $x = 7$

· $(7,\ 12)$

The following responses would not earn the first point:

· $(7,\ 0)$

· $x = -1, x = 7, x = 8$

Second point:

The justification must discuss g' changing from positive to negative at $x = 7$.

The point cannot be earned for just stating that g' changes sign or changes at $x = 7$.

Note: Only referencing "the slope" or "the derivative" is too ambiguous to earn the point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Critical point at $x = 7$
- [ ] Relative maximum with justification

Solution:

g has a critical point where $g'\ (x) = 0$.

$g'\ (x) = 0 \ \Rightarrow \ x = 7$

$g'\ (x)$ changes from positive to negative at $x = 7$.

Therefore, g has a relative maximum at $x = 7$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response needs to apply the Fundamental Theorem of Calculus.

Here are some examples that earn the first point:

· $\displaystyle\int_0^7 g'\ (x)\ dx = g\ (x)|_0^7$

· $g\ (x)|_0^7$

· $g\ (7) - g\ (0)$

The first point would not be earned for $\displaystyle\int_0^7 g\ (x)\ dx = g\ (x)|_0^7 = g\ (7) - g\ (0)$.

3-1-1b

Second point:

The response must use the values of $g\,(7)$ and $g\,(0)$ to earn this point.

This response earns both points: $g\,(7) - g\,(0) = 12 - (-16)$.

A response of $12 + 16$ earns the second point, but not the first.

A response of 28 with no supporting work earns no points.

0	1 ✓

The student response accurately includes **one** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$\text{Area} = \int_0^7 g'\,(x)\;dx = g\,(7) - g\,(0)$$

$$= 12 - (-16) = 28$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must convey that g' is continuous as a consequence of g' being differentiable.

· "Since g' is differentiable, g' is continuous" or "g' is differentiable and therefore continuous" would earn the first point.

If there is no implication involving g' being differentiable, then the first point is not earned.

· For example: "g' is differentiable and continuous" does not earn this point.

Second point:

The response needs to be Yes. Further, the response must reference the MVT either directly or by stating the conclusion of the theorem in words. Since g is twice differentiable, the response does not need to say that g' is differentiable.

However, the response must reference the continuity of g' in order to earn this second point. The response must reference the difference quotient (or average rate of change) from the MVT either in words or with an equation.

Here are some examples that earn the second point, but not the first:

· Since g' is differentiable and continuous, then by the MVT there exists a value c such that is the average rate of change of g' on $[a,\ b]$.

· Since g' is continuous, then there must exist a value c such that $g''\,(c)$ is the average rate of change of g' on $[0,\ 7]$.

· Since g' is continuous, then there must exist a value c such that $g''\,(c) = \frac{g'(7) - g'(0)}{7-0}$.

These responses do not earn any points in part (c):

3-1-1b

· Since g' is continuous then yes.

· Yes, by MVT.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Conditions
- [ ] Conclusion using Mean Value Theorem

Solution:

g is twice differentiable. $\Rightarrow$ g' is differentiable on $0 \leq x \leq 7$.
$\Rightarrow$ g' is continuous on $0 \leq x \leq 7$.

Therefore, the Mean Value Theorem can be applied to g' on the interval $0 \leq x \leq 7$ to guarantee that there exists a value of c, for $0 < c < 7$, such that $g''(c)$ equals the average rate of change of g' over the interval $0 \leq x \leq 7$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First two points:

These two points may be awarded in either order, or simultaneously.

We must see $g\left(x^3-1\right)$ for the FTC point and $3x^2$ for the chain rule point. If there is a $g(0)$ present as part of the derivative of h or in the calculation of $h'(2)$, then the FTC point is not earned.

Examples:

· $h'(x) = g\left(x^3-1\right) \cdot 3x^2$ earns the FTC point and chain rule point.

· $g\left(x^3-1\right)$ earns the FTC point but not the chain rule point.

· $g\left(x^3-1\right) - g(0)$ does not earn either point.

· $g\left(x^3-1\right) \cdot 3x^2 - g(0)$ earns the chain rule point but not the FTC point.

Third point:

To be eligible for this point, the response must have already earned both the FTC point and the chain rule point. In addition, the response must use the value of $g(7)$. Answer must be our answer.

There is an alternate version of the FTC that is correct: $g\left(x^3-1\right) \cdot 3x^2 - g(0) \cdot 0$ would earn at least the first two points.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Fundamental Theorem of Calculus

3-1-1b

- [ ] Chain rule
- [ ] Answer

Solution:

$h'(x) = g\left(x^3 - 1\right) \cdot 3x^2$

$h'(2) = g\left(2^3 - 1\right) \cdot 3 \cdot 2^2 = g(7) \cdot 3 \cdot 4 = 12 \cdot 12 = 144$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Both of these responses earn this point:

· $A(x) = x \cdot g'(x)$

· $x \cdot g'(x)$

Some responses give the formula for $A(x)$ indirectly. In this situation, the connection must be clear and correct.

This response earns the point: $\int_0^7 A(x)\,dx = \int_0^7 x \cdot g'(x)\,dx$.

These responses do not earn the point:

· $\int_0^7 x \cdot g'(x)\,dx$

· $\int_0^7 A(x)\,dx = A(x) = x \cdot g'(x)$

If the correct expression is used in part (f) but is never shown in part (e), the response does not earn this point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$A(x) = x \cdot g'(x)$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The integration by parts point is earned for correctly performing integration by parts regardless of the limits of integration (those are part of the answer point). This is like an “indefinite integration by parts” point. Details of the $u \cdot v'$ decomposition are not required.

Here are some example of responses that earn the first point:

3-1-1b

- $\int_0^7 x\,g'(x)\ dx = x\,g(x)|_0^7 - \int_0^7 g(x)\ dx$
- $\int x\,g'(x)\ dx = x\,g(x) - \int g(x)\ dx$

Second point:

A numerical answer must be presented, although it does not need to be simplified.

$7 \cdot g(7) - 15.4$ does not earn the answer point.

$7 \cdot 12 - 0 - 15.4$ is the minimal computation that must be shown in order to earn both points. If the 0 is missing from this expression, the first point for integration by parts is not earned.

If the response uses an expression from part (e) that was incorrect, then we can read for both points in part (f) only if the integration requires integration by parts.

To be eligible for the answer point the integration by parts point must be earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- [ ] Integration by parts
- [ ] Answer

Solution:

$$u = x \qquad dv = g'(x)\ dx$$
$$du = dx \quad v = g(x)$$

$$\int_0^7 x \cdot g'(x)\ dx = x \cdot g(x)|_0^7 - \int_0^7 g(x)\ dx$$

$$= 7 \cdot g(7) - 0 \cdot g(0) - \int_0^7 g(x)\ dx$$
$$= 7 \cdot 12 - 0 \cdot (-16) - 15.4 = 68.6$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

To earn the first point, the response must convey that both the numerator and the denominator approach 0.

Any response that includes "$\text{limit} = \frac{0}{0}$" does not earn the first point.

Presentations, such as "$\to \frac{0}{0}$" or "$\text{limit} = \text{indeterminate form } \frac{0}{0}$" can earn the first point as long as the response does not have "$\text{limit} = \frac{0}{0}$".

If the response writes the limit as a quotient and puts an arrow from the numerator to 0 and a separate arrow from the denominator to 0 then the point is earned.

3-1-1b

A response can earn the point by evaluating the numerator at $x=0$, and the denominator at $x=0$ without limit notation because both the numerator and denominator are continuous functions.

Second point:

The point is earned for attempting to apply L'Hospital's Rule by differentiating the numerator and the denominator.

The point is not earned for applying the quotient rule.

Errors in differentiating will not earn the answer point.

The second point may be earned even if the limit notation is missing from some expressions, but it cannot be earned if the response never expresses anything we recognize as a limit.

Each of these responses earns the second point:

- $\lim_{x\to 0}\frac{g\prime(x)-1}{\cos\ x}$
- $\lim_{x\to 0}\frac{g\prime(x)-x}{\cos\ x}$
- $\lim_{x\to 0}\frac{g(x)-x-2}{\sin\ x}=\frac{g'(x)-1}{\cos\ x}$

Note: If the response shown in this last bullet is equated to our answer, the response earns only the second point, because of equating terms that are not equivalent. If the correct answer is given but not set equal to this expression, the response earns the second two points.

Third point:

The answer must be correct. The answer does not need to be simplified, but if it is, it must be correct.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ g' is continuous. $\Rightarrow$ g is continuous. $\Rightarrow$ $\lim_{x\to 0}g\left(x\right)=g\left(0\right)$ and $\lim_{x\to 0}g'\left(x\right)=g'\left(0\right)$.

$\lim_{x\to 0}\left(g\left(x\right)-x+16\right)=g\left(0\right)-0+16=-16-0+16=0$

$\lim_{x\to 0}\sin\ x=\sin\ 0=0$

Therefore, L'Hospital's Rule can be applied.

$\lim_{x\to 0}\frac{g(x)-x+16}{\sin\ x}=\lim_{x\to 0}\frac{g'(x)-1}{\cos\ x}$

$=\frac{g'(0)-1}{\cos\ 0}=\frac{6-1}{1}=\frac{5}{1}=5$

3-1-1b

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point:

The response must include a correct one-sided limit that defines the improper integral in order to earn this point. Here is an example:

- $\lim_{b \to 7^-} \int_0^b \frac{g''(x)}{g'(x)}\, dx$

Second point:

The point is earned for finding the antiderivative of $\frac{g''(x)}{g'(x)}$. No limit notation is required and limits of integration do not affect this point. This point can be earned if the first was not earned.

This response earns the second point: $\int_0^7 \frac{g''(x)}{g'(x)}\, dx = \ln|g'(x)|\,|_{x=0}^{x=7}$

Note that details of the substitution do not need to be provided.

The antiderivative must have absolute values.

Third point:

The point is earned for correctly evaluating the one sided limit of the anti-derivative.

This response earns the third point: $\lim_{b \to 7^-} \ln|g'(b)| - \ln 6$ does not exist.

A response of $-\infty$ earns the point.

Note: If there is no limit in part (h) then the response could only earn the antiderivative point if that is found correctly. It could not earn the first or third points.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Improper integral
- ☐ Antiderivative
- ☐ Answer with reason

Solution:

$$\int_0^7 \frac{g''(x)}{g'(x)}\, dx = \lim_{b \to 7^-} \int_0^b \frac{g''(x)}{g'(x)}\, dx$$

$$= \lim_{b \to 7^-} [\ln|g'(x)|]_0^b = \lim_{b \to 7^-} [\ln g'(b) - \ln g'(0)]$$

$$= \lim_{b \to 7^-} [\ln g'(b) - \ln 6]$$

Because $\lim_{b \to 7^-} \ln g'(b)$ does not exist, the value of the integral does not exist.

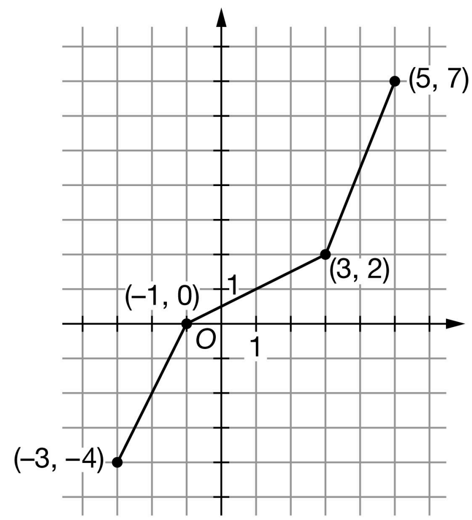
3-1-1b

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-3 \leq x \leq 5$ consists of three line segments, as shown in the figure above. Let k be the function defined by $k(x) = \int_3^x g(t)\,dt$ for $-3 \leq x \leq 5$.

(a) For the function k, is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.

(b) Find the absolute minimum value of k on the interval $-3 \leq x \leq 5$. Justify your answer.

(c) On what open intervals contained in $-3 < x < 5$ is the graph of k both increasing and concave up? Give a reason for your answer.

(d) Find the value of $k''(-1)$, or explain why it does not exist.

(e) Find $\lim_{x \to 3} \frac{k(x)}{x+3}$. Show the computations that lead to your answer.

(f) Let h be the function defined by $h(x) = k(x^2)$ Find the value of $h'\left(\sqrt{3}\right)$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as $\mathrm{sqrt}(3)$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to g. The obvious is $k' = g$. A response of local minimum with the explanation: “g (or the graph) changes from negative to positive” is an implied connection. A response of local minimum with explanation: “k' changes from negative to positive,” without the connection of $k' = g$ is not acceptable.

Simply saying “k changes from decreasing to increasing” is not acceptable.

In this situation, the Second Derivative Test $k''(-1) > 0$ is an incorrect statement and a response with this reason does not earn the point.

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0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of k because $k'(x) = g(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (g is negative on $(-3, -1)$ and positive on $(-1, 5)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ AND an area calculation in trying to find $k\,(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y-values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

A common error that shows an attempt to find area is $k\,(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words “must check the endpoints” is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y-values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

· $k\,(-3) = 8$ (adding the triangle areas)

· $k\,(5) = 5$ (finding the area of the triangle and forgetting the rectangle)

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· These can be used to earn the second point.

A correct argument that eliminates $x = 5$ by stating $k(5)$ is positive will earn the second point as long as $x = -3$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -3$, $x = -1$, and $x = 5$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Identifies unique critical point
- [ ] Justifies critical point as location of absolute minimum
- [ ] Answer with justification
- [ ] Considers $x = -1$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-3 \leq x \leq 5$ where $k'(x) = g(x) = 0$, $x = -1$ is the only critical point of k on the interval $-3 \leq x \leq 5$.

Because $x = -1$ is also the location of a relative minimum for k, $k(-1)$ is the absolute minimum for k on $-3 \leq x \leq 5$.

$$k(-1) = \int_3^{-1} g(t)dt = -\int_{-1}^{3} g(t)dt = -4$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $k'(x) = g(x) = 0$.

$$k(-1) = \int_3^{-1} g(t)dt = -\int_{-1}^{3} g(t)dt = -4$$

$$k(-3) = \int_3^{-3} g(t)dt = -\int_{-3}^{3} g(t)dt = 0$$

$$k(5) = \int_3^{5} g(t)dt = 9$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Part C

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-1, 5)$ or inequality $-1 < x < 5$.

We will allow $[-1, 5]$, $(-1, 5]$, $[-1, 5)$, or an equivalent inequality.

Presenting $(-1, 3) \cup (3, 5)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The "$\cup$" is not necessary. Be careful, one bracket on either 3 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons "k' is positive and increasing" or "k' is positive and k'' is positive" will earn the second point if the connection $k' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function k, then the second point is not earned.

The interval $(-1, 7)$ with the correct justification will earn one of the two points. (The response took the y-coordinate of the right-hand point of the interval $(5, 7)$). This is considered a copy error. This is the only copy error that is allowed.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

k is increasing where $k' = g$ is positive.

k is concave up where $k' = g$ is increasing.

The graph of k is both increasing and concave up on the open interval $(-1, 5)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $k''(x) = g'(x)$.

We are not allowing an import of a previous connection between k'' and g'.

The explicit connection, $k''(x) = g'(x)$ earns the first point.

It is not assumed that a response which only presents $k'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g', the differentiability of g, or the characteristic of the graph of g.

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

· limits involving the difference quotient that appears in these scoring guidelines,

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· limits involving g' as the argument of the limit,

· or some correct verbal description of the characteristic of the graph of g at $x = -1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will NOT accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g. The phrase $k''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $k''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Considers $g'(x)$
- [ ] Answer with explanation

Solution:

Because $\lim\limits_{x\to -1^-} \frac{g(x)-g(-1)}{x-(-1)} = 2$ and $\lim\limits_{x\to -1^+} \frac{g(x)-g(-1)}{x-(-1)} = \frac{1}{2}$, $g''(-1) = k''(-1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of k does not need to be stated.

An answer of $\frac{k(3)}{6}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{6}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

3-1-1b

Solution:

k is continuous. $\Rightarrow \lim\limits_{x \to 3} k(x) = k(3)$

$\lim\limits_{x \to 3} \frac{k(x)}{x+3} = \frac{k(3)}{3+3} = \frac{0}{6} = 0$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $h : k'(x^2) \cdot 2x$.

This point is earned for any of $k'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$k'(x^2) = g(x^2)$, $k'(3) = g(3)$, or $k'(3) = 2$.

This point would be earned with a response of $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $2 \cdot 2\sqrt{3}$, or $4\sqrt{3}$. Note, the product need not be simplified.

An answer of $4\sqrt{3}$ with no supporting work earns no points in part (f).

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$h'(x) = k'(x^2) \cdot 2x = g(x^2) \cdot 2x$

$h'\left(\sqrt{3}\right) = g(3) \cdot 2\sqrt{3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$

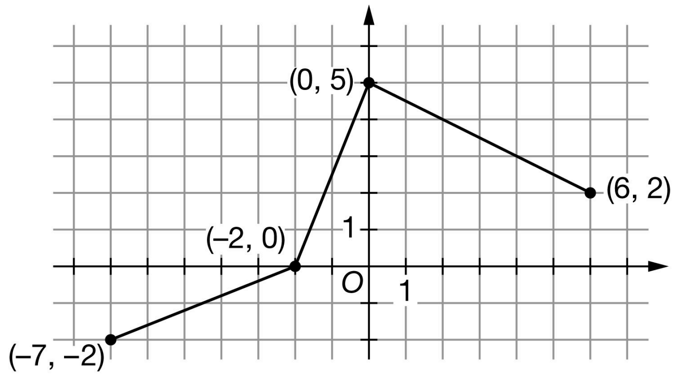
3-1-1b

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the function h on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let r be the function defined by $r(x) = \int_0^x h(t)\,dt$ for $-7 \leq x \leq 6$.

(a) For the function r, is $x = -2$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.

(b) Find the absolute minimum value of r on the interval $-7 \leq x \leq 6$. Justify your answer.

(c) On what open intervals contained in $-7 < x < 6$ is the graph of r both increasing and concave down? Give a reason for your answer.

(d) Find the value of $r''(-2)$, or explain why it does not exist.

(e) Find $\lim_{x \to 0} \frac{r(x)}{x-4}$.Show the computations that lead to your answer.

(f) Let v be the function defined by $v(x) = r(x^2)$. Find the value of $v'\left(\sqrt{3}\right)$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as $\mathrm{sqrt}(3)$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h. The obvious is $r' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " r' changes from negative to positive," without the connection of $r' = h$ is not acceptable.

Simply saying " r changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $r''(2) > 0$ is an incorrect statement, and a response with this reason does not earn the point.

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0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$x = -2$ is the location of a relative minimum of r because $r'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -2$ is the only critical point.(h is negative on $(-7, -2)$ and positive on $(-2, 6)$ is another way to state this.)

Note: Stating that $x = -2$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -2$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -2$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -2$. This consideration is more than just the statement $x = -2$. The response must use $x = -2$ and an area calculation in trying to find $r(-2)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y-values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $h(x)$ values.

A common error that shows an attempt to find area is $r(-2) = 5$. This earns the first point.

The second point is for consideration of the endpoints along with an attempt to find area. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y-values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

- $r(-7) = 10$ (adding the triangle areas)

3-1-1b

$\cdot\, r(6) = 9$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $r(6)$ is positive will earn the second point as long as $x = -7$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -7$, $x = -2$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

Note: A coordinate point is not an acceptable answer.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Identifies unique critical point
- [ ] Justifies critical point as location of absolute minimum
- [ ] Answer with justification
- [ ] Considers $x = -2$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

Because $x = -2$ is the only point on the interval $-7 \le x \le 6$ where $r'(x) = h(x) = 0$, $x = -2$ is the only critical point of r on the interval $-7 \le x \le 6$.

Because $x = -2$ is also the location of a relative minimum for r, $r(-2)$ is the absolute minimum for r on $-7 \le x \le 6$.

$$r(-2) = \int_0^{-2} h(t)\, dt = -\int_{-2}^{0} h(t)\, dt = -5$$

Therefore, the absolute minimum value of r on $-7 \le x \le 6$ is $r(-2) = -5$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -2$ is the only critical point in the interval because it is the only point where $r'(x) = h(x) = 0$.

$$r(-2) = \int_0^{-2} h(t)\, dt = -\int_{-2}^{0} h(t)\, dt = -5$$

$$r(-7) = \int_0^{-7} h(t)\, dt = -\int_{-7}^{0} h(t)\, dt = 0$$

$$r(6) = \int_0^{6} h(t)\, dt = 21$$

Therefore, the absolute minimum value of r on $-7 \le x \le 6$ is $r(-2) = -5$.

Part C

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(0,\ 6)$ or inequality $0 < x < 6$.

We will allow $[0,\ 6]$, $(0,\ 6]$, $[0,\ 6)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons r' is positive and decreasing or r' is positive and r'' is negative will earn the second point if the connection $r' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function r, then the second point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

r is increasing where $r' = h$ is positive.

r is concave down where $r' = h$ is decreasing.

The graph of r is both increasing and concave down on the open interval $(0,\ 6)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a consideration of $r''(x) = h'(x)$.

We are not allowing an import of a previous connection between r'' and h'.

The explicit connection $r''(x) = h'(x)$ earns the first point.

It is not assumed that a response which presents $r'(x) = h(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to h', the differentiability of h, or the characteristic of the graph of h.

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

· limits involving the difference quotient that appears in these scoring guidelines,

· limits involving h' as the argument of the limit, or some correct verbal description of the characteristic of the graph of h at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

3-1-1b

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $r''\ (1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $r''\ (1)$ does not exist because there is a cusp on h is acceptable and earns both points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'\ (x)$
- ☐ Answer with explanation

Solution:

Because $\lim\limits_{x\to -2^-}\frac{h(x)-h(-2)}{x-(-2)}=\frac{2}{5}$ and $\lim\limits_{x\to -2^+}\frac{h(x)-h(-2)}{x-(-2)}=\frac{5}{2}$, $h'\ (-2)=r''\ (-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of r does not need to be stated.

An answer of $\frac{r(0)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

r is continuous. $\Rightarrow \lim\limits_{x\to 0} r\ (x)=r\ (0)$

$\lim\limits_{x\to 0}\frac{r(x)}{x-4}=\frac{r(0)}{0-4}=\frac{0}{-4}=0$

3-1-1b

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $v : r'\left(x^2\right) \cdot 2x$.

This point is earned for any of $r'\left(x^2\right) \cdot 2x$, $h\left(x^2\right) \cdot 2x$, $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$r'\left(x^2\right) = h\left(x^2\right)$, $r'(3) = h(3)$, or $r'(3) = \frac{7}{2}$.

This point would be earned with a response of $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $\frac{7}{2} \cdot 2\sqrt{3}$ or $7\sqrt{3}$. Note, the product need not be simplified.

An answer of $7\sqrt{3}$ with no supporting work earns no points in part (f).

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$v'(x) = r'\left(x^2\right) \cdot 2x = h\left(x^2\right) \cdot 2x$$

$$v'\left(\sqrt{3}\right) = h(3) \cdot 2\sqrt{3} = \frac{7}{2} \cdot 2\sqrt{3} = 7\sqrt{3}$$

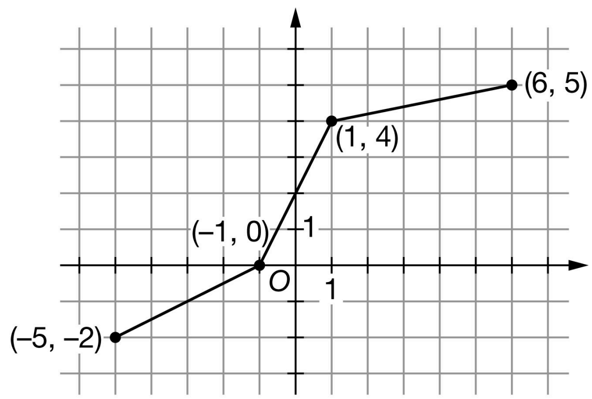
3-1-1b

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h

The graph of the function h on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let p be the function defined by $p(x) = \int_1^x h(t)\,dt$ for $-5 \leq x \leq 6$.

(a) For the function p, is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.

(b) Find the absolute minimum value of p on the interval $-5 \leq x \leq 6$. Justify your answer.

(c) On what open intervals contained in $-5 < x < 6$ is the graph of p both decreasing and concave up? Give a reason for your answer.

(d) Find the value of $p''(1)$, or explain why it does not exist.

(e) Find $\lim\limits_{x\to 1} \frac{p(x)}{x+4}$. Show the computations that lead to your answer.

(f) Let k be the function defined by $k(x) = p(x^2)$. Find the value of $k'\left(\sqrt{2}\right)$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{2}$ can be keyboarded as $\text{sqrt}(2)$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h. The obvious is $p' = h$. A response of local minimum with the explanation: "h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: "p' changes from negative to positive," without the connection of $p' = h$ is not acceptable.

Simply saying "p changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $p''(-1) > 0$ is an incorrect statement and a response with this reason would not earn the point.

3-1-1b

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$x=-1$ is the location of a relative minimum of p because $p'(x)=h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x=-1$ is the only critical point. (h is negative on $(-5,-1)$ and positive on $(-1,6)$ is another way to state this.)

Note: Stating that $x=-1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x=-1$ then produces the absolute minimum. Note that they do not have to repeat the justification of $x=-1$ being a relative minimum from part part (a). If this is not in (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x=-1$. This consideration is more than just the statement $x=-1$. The response must use $x=-1$ and an area calculation in trying to find $p(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y–values of the coordinate points from $h(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $h(x)$ values.

A common error that shows an attempt to find area is $k(-1)=4$. This earns the first point.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y –values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

· $p(-5)=8$ (adding the triangle areas)

· $p(6)=\frac{5}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

3-1-1b

A correct argument that eliminates $x = 6$ by stating $p(6)$ is positive will earn the second point as long as $x = -5$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = -1$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Identifies unique critical point
- [ ] Justifies critical point as location of absolute minimum
- [ ] Answer with justification
- [ ] Considers $x = -1$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-5 \leq x \leq 6$ where $p'(x) = h(x) = 0$, $x = -1$ is the only critical point of p on the interval $-5 \leq x \leq 6$.

Because $x = -1$ is also the location of a relative minimum for p, $p(-1)$ is the absolute minimum for p on $-5 \leq x \leq 6$.

$$p(-1) = \int_1^{-1} h(t)\, dt = -\int_{-1}^{1} h(t)\, dt = -4$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $p'(x) = h(x) = 0$.

$$p(-1) = \int_1^{-1} h(t)\, dt = -\int_{-1}^{1} h(t)\, dt = -4$$

$$p(-5) = \int_1^{-5} h(t)\, dt = -\int_{-5}^{1} h(t)\, dt = 0$$

$$p(6) = \int_1^{6} h(t)\, dt = 22.5$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

3-1-1b

The first point is for the interval $(-5, \ -1)$ or inequality $-5 < x < -1$.

We will allow $[-5, \ -1]$, $(-5, \ -1]$, $[-5, \ -1)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons p' is negative and increasing or p' is negative and p'' is positive will earn the second point if the connection $p' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function p, then the second point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

p is decreasing where $p' = h$ is negative.

p is concave up where $p' = h$ is increasing.

The graph of p is both decreasing and concave up on the open interval $(-5, \ -1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $p''(x) = h'(x)$.

We are not allowing an import of a previous connection between p'' and h'.

The explicit connection, $p'' = h'$, earns the first point.

It is not assumed that a response which only presents $p' = h$ has made that connection.

The implicit connection will be made in the explanation with a connection to h', the differentiability of h, or the characteristic of the graph of h.

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

· limits involving the difference quotient that appears in these scoring guidelines,

· limits involving h' as the argument of the limit,

· or some correct verbal description of the characteristic of the graph of h at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

3-1-1b

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h. The phrase $p''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $p''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

$p''(x) = h'(x)$

Because $\lim\limits_{x \to 1^-} \frac{h(x)-h(1)}{x-1} = 2$, and $\lim\limits_{x \to 1^+} \frac{h(x)-h(1)}{x-1} = \frac{1}{5}$, $h'(1) = p''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of p does not need to be stated.

An answer of $\frac{p(1)}{5}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{5}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors

0	1 ✓

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

p is continuous. $\Rightarrow \lim\limits_{x \to 1} p(x) = p(1)$ $\lim\limits_{x \to 1} \frac{p(x)}{x+4} = \frac{p(1)}{1+4} = \frac{0}{5} = 0$

Part F

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of k: $p'\left(x^2\right) \cdot 2x$.

This point is earned for any of $p'\left(x^2\right) \cdot 2x$, $h\left(x^2\right) \cdot 2x$, $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$p'\left(x^2\right) = h\left(x^2\right)$, $p'(2) = h(2)$, or $p'(2) = \frac{21}{5}$.

This point would be earned with a response of $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, even with no preceding work.

The third point is for either $\frac{21}{5} \cdot 2\sqrt{2}$ or $\frac{42}{5}\sqrt{2}$. Note, the product need not be simplified.

An answer of $\frac{42}{5}\sqrt{2}$ with no supporting work earns no points in part (f).

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = p'\left(x^2\right) \cdot 2x = h\left(x^2\right) \cdot 2x$$

$$k'\left(\sqrt{2}\right) = h(2) \cdot 2\sqrt{2} = \frac{21}{5} \cdot 2\sqrt{2} = \left(\frac{42}{5}\right)\sqrt{2}$$

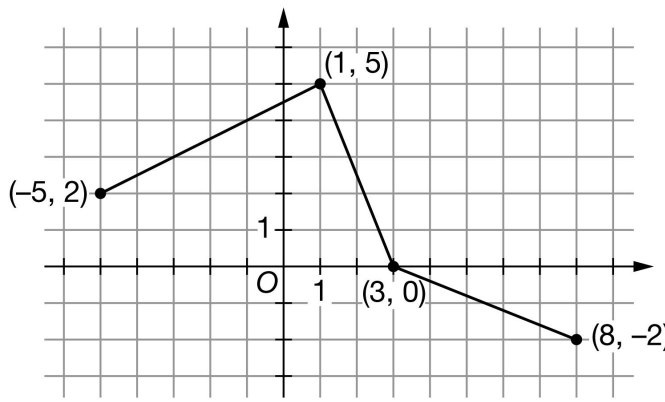
3-1-1b

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of three line segments, as shown in the figure above. Let m be the function defined by $m(x) = \int_1^x g(t)\,dt$ for $-5 \leq x \leq 8$.

(a) For the function m, is $x = 3$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.

(b) Find the absolute maximum value of m on the interval $-5 \leq x \leq 8$. Justify your answer.

(c) On what open intervals contained in $-5 < x < 8$ is the graph of m both increasing and concave down? Give a reason for your answer.

(d) Find the value of $m''(1)$, or explain why it does not exist.

(e) Find $\lim_{x \to 1} \frac{m(x)}{x+1}$. Show the computations that lead to your answer.

(f) Let k be the function defined by $k(x) = m(x^2)$. Find the value of $k'(1)$ or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to g. The obvious is $m' = g$. A response of local maximum with the explanation: " g (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " m' changes from positive to negative" without the connection of $m' = g$ is not acceptable.

Simply saying " m changes from increasing to decreasing" is not acceptable.

In this situation, the Second Derivative Test $m''(3) < 0$ is an incorrect statement and the response with this reason does not earn the point.

3-1-1b

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$x = 3$ is the location of a relative maximum of m because $m'(x) = g(x)$ changes from positive to negative there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 3$ is the only critical point. (g is positive on $(-5,\ 3)$ and negative on $(3,\ 8)$ is another way to state this.)

Note: Stating that $x = 3$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 3$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 3$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 3.$ This consideration is more than just the statement $x = 3.$ The response must use $x = 3$ and an area calculation in trying to find $m(3).$

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $g(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

· $m(-5) = -9$ (finding the area of the triangle and forgetting the rectangle)

· $m(8) = 10$ (adding the triangle areas). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $m(-5)$ is negative will earn the second point as long as $x = 8$ is also addressed.

3-1-1b

Note: A Candidates Test chart/table that includes $x = -5$, $x = 3$, and $x = 8$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

NOTE: A coordinate point is not an acceptable answer.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Identifies unique critical point
- [ ] Justifies critical point as location of absolute maximum
- [ ] Answer with justification
- [ ] Considers $x = 3$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

Because $x = 3$ is the only point on the interval $-5 \leq x \leq 8$ where $m'(x) = g(x) = 0$, $x = 3$ is the only critical point of m on the interval $-5 \leq x \leq 8$.

Because $x = 3$ is also the location of a relative maximum for m, $m(3)$ is the absolute maximum for m in $-5 \leq x \leq 8$.

$$m(3) = \int_1^3 g(t)\, dt = 5$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 3$ is the only critical point in the interval because it is the only point where $m'(x) = g(x) = 0$.

$$m(3) = \int_1^3 g(t)\, dt = 5$$

$$m(-5) = \int_1^{-5} g(t)\, dt = -\int_{-5}^{1} g(t)\, dt = -21$$

$$m(8) = \int_1^8 g(t)\, dt = 0$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(1, 3)$ or inequality $1 < x < 3$.

3-1-1b

We will allow $[1,\ 3]$, $(1,\ 3]$, $[1,\ 3)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons m' is positive and decreasing or m' is positive and m'' is negative, will earn the second point if the connection $m' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function m, then the second point is not earned.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

m is increasing where $m' = g$ is positive.

m is concave down where $m' = g$ is decreasing.

The graph of m is increasing and concave down on the open interval $(1,\ 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $m''\ (x) = g'\ (x)$.

We are not allowing an import of a previous connection between m'' and g'.

The explicit connection, $m''\ (x) = g'\ (x)$, earns the first point.

It is not assumed that a response which only presents $m'\ (x) = g\ (x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g', the differentiability of g, or the characteristic of the graph of g.

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

· limits involving the difference quotient that appears in these scoring guidelines,

· limits involving g' as the argument of the limit,

· or some correct verbal description of the characteristic of the graph of g at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'\ (x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute

3-1-1b

value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g. The phrase $m''(1)$ of 1does not exist because there is a cusp will not earn this point. Whereas, the phrase $m''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

$m''(x) = g'(x)$

Because $\lim_{x\to 1^-} \frac{g(x)-g(1)}{x-1} = \frac{1}{2}$ and $\lim_{x\to 1^+} \frac{g(x)-g(1)}{x-1} = -\frac{5}{2}$, $g'(1) = m''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of m does not need to be stated.

An answer of $\frac{m(1)}{2}$ will not earn the point,

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{2}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors.

0	1

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

m is continuous. $\Rightarrow \lim_{x\to 1} m(x) = m(1)$

$$\lim_{x\to 1} \frac{m(x)}{x+1} = \frac{m(1)}{1+1} = \frac{0}{2} = 0$$

Part F

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : m'\left(x^2\right) \cdot 2x$.

This point is earned for any of $m'\left(x^2\right) \cdot 2x$, $g\left(x^2\right) \cdot 2x$, $g(1) \cdot 2$, or $5 \cdot 2$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$m'\left(x^2\right) = g\left(x^2\right)$, $m'(1) = g(1)$, or $m'(1) = 5$.

This point would be earned with a response of $g(1) \cdot 2$, or $5 \cdot 2$, even with no preceding work.

The third point is for either $5 \cdot 2$ or 10. Note, the product need not be simplified.

An answer of 10 with no supporting work earns no points in part (f).

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$k'(x) = m'\left(x^2\right) \cdot 2x = g\left(x^2\right) \cdot 2x$

$k'(1) = g(1) \cdot 2 = 5 \cdot 2 = 10$

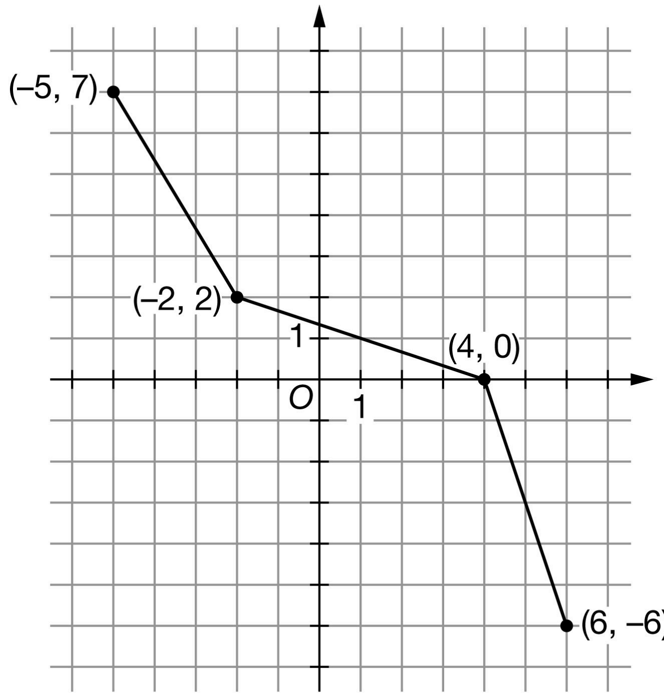
3-1-1b

23. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

The graph of the function f on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^{x} f(t)\,dt$ for $-5 \leq x \leq 6$.

(a) For the function h, is $x = 4$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.

(b) Find the absolute maximum value of h on the interval $-5 \leq x \leq 6$. Justify your answer.

(c) On what open intervals contained in $-5 < x < 6$ is the graph of h both increasing and concave down? Give a reason for your answer.

(d) Find the value of $h''(-2)$, or explain why it does not exist.

(e) Find $\lim\limits_{x \to -2} \frac{h(x)}{x-2}$. Show the computations that lead to your answer.

(f) Let k be the function defined by $k(x) = h(x^2)$. Find the value of $k'\left(\sqrt{5}\right)$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{5}$ can be keyboarded as $\mathrm{sqrt}(5)$)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

3-1-1b

The response must include the wording local maximum or relative maximum. We also need to see a connection to h. The obvious is $h' = f$. A response of local maximum with the explanation: " f (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " h' changes from positive to negative," without the connection of $h' = f$ will not earn the point.

Simply saying " h. changes from increasing to decreasing" is not enough to earn the point.

In this situation, appealing to the Second Derivative Test by claiming that $h''(4) < 0$ is an incorrect statement, so a response with this reason does not earn the point.

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the location of a relative maximum of h because $h'(x) = f(x)$ changes from positive to negative at this point.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 4$ is the only critical point. (f is positive on $(-5,\ 4)$ and negative on $(4,\ 6)$ is another way to state this.)

Note: Stating that $x = 4$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 4$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 4$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct value of the maximum.

The presentation of a coordinate point $(4,\ 6)$ as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 4$. This consideration is more than just the statement $x = 4$. The response must use $x = 4$ and an area calculation in trying to find $h(4)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find areas for $h(-5)$ and $h(6)$.

The response cannot simply write the integrals with the correct upper limits. Attempts at both areas must be made.

3-1-1b

Using the y -values of the coordinate points from $f\left(x\right)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f\left(x\right)$ values.

Common errors that show an attempt to find area:

· $h\left(6\right)=12$ (adding the triangle areas)

· A$h\left(-5\right)=-\frac{15}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x=-5$ by stating $h\left(-5\right)$ is negative will earn the second point as long as $x=6$ is also addressed.

Note: A Candidates Test chart/table that includes $x=-5$, $x=4$, and $x=6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point $\left(4,\ 6\right)$ is not an acceptable answer.

			✓
0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x=4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x=4$ is the only point on the interval $-5\leq x\leq 6$ where $h'\left(x\right)=f\left(x\right)=0$, $x=4$ is the only critical point of h on the interval $-5\leq x\leq 6$.

Because $x=4$ is also the location of a relative maximum of h, $h(4)$ is the absolute maximum for h on $-5\leq x\leq 6$.

$$h\left(4\right)=\int_{-2}^{4}f\left(t\right)\,dt=6$$

Therefore, the absolute maximum value of h on $-5\leq x\leq 6$ is $h\left(4\right)=6$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x=4$ is the only critical point in the interval because it is the only point where $h'\left(x\right)=f\left(x\right)=0$.

$$h\left(4\right)=\int_{-2}^{4}f\left(t\right)\,dt=6$$

$$h\left(-5\right)=\int_{-2}^{-5}f\left(t\right)\,dt=-\int_{-5}^{-2}f\left(t\right)\,dt=-13.5$$

3-1-1b

$h(6) = \int_{-2}^{6} f(t)\ dt = 0$

Therefore, the absolute maximum value of h on $-5 \le x \le 6$ is $h(4) = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(-5,\ 4)$ or inequality $-5 < x < 4$.

We will allow $[-5,\ 4]$, $(-5,\ 4]$, $[-5,\ 4)$, or an equivalent inequality.

Presenting $(-5,\ -2) \cup (-2,\ 4)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The " $\cup$ " is not necessary. Be careful, one bracket on either -2 is actually the entire interval and is eligible for both points.
The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons " h' is positive and decreasing" or " h' is positive and h'' is negative" will earn the second point if the connection $h' = f$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function h, then the second point is not earned.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

The graph of h is both increasing and concave down on the open interval $(-5,\ 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $h''(x) = f'(x)$. .

We are not allowing an import of a previous connection between h'' and f'.

The explicit connection, $h''(x) = f'(x)$, earns the first point.

It is not assumed that a response which only presents $h'(x) = f(x)$ has made the connection between h'' and f'.

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

· limits involving the difference quotient that appear in these scoring guidelines,

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· limits involving h' as the argument of the limit,

· or some correct verbal description of the characteristic of the graph of f at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h''(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: f' does not exist because f (or "the graph") has a/is a (blank). Fill in the blank with corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of f. The phrase $h''(-2)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $h''(-2)$ does not exist because there is a cusp on f is acceptable and earns both points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(x)$
- ☐ Answer with explanation

Solution:

$h''(x) = f'(x)$

Because A $\lim\limits_{x \to -2^-} \frac{f(x)-f(-2)}{x-(-2)} = -\frac{5}{3}$ and $\lim\limits_{x \to -2^+} \frac{f(x)-f(-2)}{x-(-2)} = -\frac{1}{3}$, $f'(-2) = h''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of h does not need to be stated.

An answer of $\frac{h(-2)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard limit notation errors.

0	1

The student response accurately includes the criteria below.

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- [ ] Answer

Solution:

h is continuous. $\Rightarrow \lim_{x \to -2} h(x) = h(-2)$

$$\lim_{x \to -2} \frac{h(x)}{x-2} = \frac{h(-2)}{-2-2} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : h'\left(x^2\right) \cdot 2x$.

This point is earned for any of of $h'\left(x^2\right) \cdot 2x$, $f\left(x^2\right) \cdot 2x$, $f(5) \cdot 2\sqrt{5}$, or $-3 \cdot 2\sqrt{5}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but could still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$h'\left(x^2\right) = f\left(x^2\right)$, $h'(5) = f(5)$, or $h'(5) = -3$.

This point would be earned with a response of $f(5) \cdot 2\sqrt{5}$ or $-3 \cdot 2\sqrt{5}$ even with no preceding work.

The third point is for either $-3 \cdot 2\sqrt{5}$ or $-6\sqrt{5}$. Note, the product need not be simplified.

An answer of $-6\sqrt{5}$ with no supporting work earns no points in part (f).

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Chain rule
- [ ] Fundamental Theorem of Calculus
- [ ] Answer

Solution:

$$k'(x) = h'\left(x^2\right) \cdot 2x = f\left(x^2\right) \cdot 2x$$

$$k'\left(\sqrt{5}\right) = f(5) \cdot 2\sqrt{5} = -3 \cdot 2\sqrt{5} = -6\sqrt{5}$$

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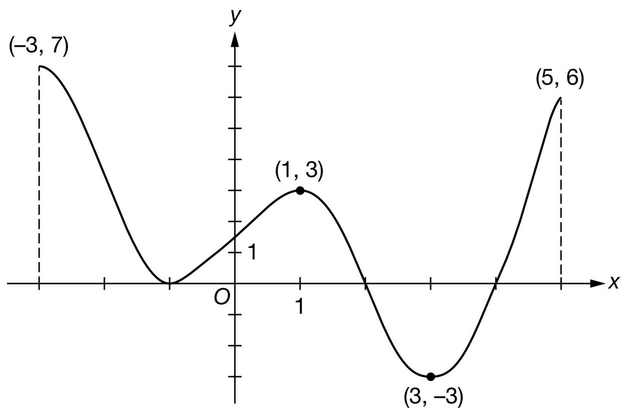
24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

The figure above shows the graph of g', the derivative of a twice-differentiable function g, on the closed interval $-3 \le x \le 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x-axis and the graph of g' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 7, 5, 4, and 3, respectively. The function g is defined for all real numbers and satisfies $g(3) = 8$ and $g(5) = 9$.

(a) Find the x-coordinate of each critical point of g on the open interval $-3 < x < 5$.

(b) Identify the critical point from part (a) which corresponds to a relative minimum of g. Give a reason for your answer.

(c) Find the value of g at the x-coordinate identified in part (b). Show the computations that lead to your answer.

(d) Find the absolute minimum value of g on $-3 \le x \le 5$. Justify your answer.

(e) On what open intervals is g both decreasing and concave down? Give a reason for your answer.

(f) Explain why there must be a value of r, for $3 < r < 5$, such that $g(r) = 8.5$.

(g) Find $\lim_{x \to 5^-} \frac{g(x)-9}{x^2-25}$. Give a reason for your answer.

(h) The function f is defined by $f(x) = (g(x))^3$. Find $f'(3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1$, $x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -1$, $x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(2, 0)$, $(4, 0)$, we will read the x -coordinates, and disregard the y-coordinates.

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0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x-coordinate of the local minimum as 4 and give the correct reason of "g' changes from negative to positive" to earn the point.

Insufficient reasons, which do not earn the point, include:

· Saying that "g switches from decreasing to increasing at $x = 4$"

· Saying that "$g'(4)$ changes signs from negative to positive"

If the response talks about "the graph" without specifying which graph, we will interpret "the graph" here (and throughout the rest of the problem) as being the graph of g', the one given in the stem of the problem.

A response indicating that "$g'(x)$ is negative for $x < 4$ and positive for $x > 4$ " is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

$x = 4$ is the only location on $-3 < x < 5$ for which this is true, so $x = 4$ is the only location of a relative minimum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

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The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 2$ is selected as the location of the local minimum in part (b). Note that $g(-3) = -2$, $g(-1) = 5$, and $g(2) = 10$. No other x-coordinates will be read for consistency.

The response $g(4) = g(5) - 3 = 6$ earns both points.

The responses of both $9 - 3 = 6$ and $g(4) = 6$ do not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_4^3 g'(x)dx = 2$ then

· $g(4) = g(3) - \int_4^3 g'(x)\,dx = 8 - 2 = 6$ earns both points, because it's possible to find $\int_4^3 g'(x)\,dx = 2$ using the areas and initial conditions given.

· any response using perceived symmetry about $x = 3$ can earn at most the first point, for example $g(4) = g(3) - \int_4^3 g'(x)\,dx = 8 - \frac{1}{2} \cdot 4 = 6$ earns just the first point.

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $g(4) = g(5) + \int_5^4 g'(x)\,dx$
- ☐ Answer

Solution:

$$g(4) = g(5) + \int_5^4 g'(x)\,dx$$

$$= g(5) - \int_4^5 g'(x)\,dx = 9 - 3 = 6$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word "considers" means that there needs to be more than just a mention of the x-value.

For $x = 4$, the response must either state its y-value or reference it as a minimum.

For the endpoints, the y-values must be presented, but need not be correct.

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Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y-values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y-values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $g(5) = 9$ by indicating that g' is positive on the interval $(4, 5)$, thus eliminating 9 as a possible absolute minimum.

A complete list of given or computable values for $g(x)$ is given in the table below.

x	$g(x)$
−3	−2
1	5
2	10
3	8
4	6
5	9

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$g(-3) = g(5) + \int_5^{-3} g'(x)\,dx$$

$$= g(5) - \int_{-3}^{5} g'(x)\,dx = 9 - (7 + 5 - 4 + 3) = -2$$

$g(4) = 6$

$g(5) = 9$

So the absolute minimum value of g on $-3 \le x \le 5$ is $g(-3) = -2$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[2, 3]$, $[2, 3)$, $(2, 3]$, and $(2, 3)$ all earn the first point.

The reason "g' is decreasing" can be expressed by either "g'' is negative" or "the slope of g' is negative."

We will overlook the false implication that $g' < 0$ on $[2, 3]$.

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The second point can be earned without earning the first point only if the given answer is a subset of $[2, 3]$.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

g is decreasing where g' is negative.

g is concave down where g' is decreasing.

g is both decreasing and concave down on the open interval $(2, 3)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3, 5]$, or the point is not earned.

The y-values of g need to be mentioned by either referring to the numerical values of 8 and 9, or mentioning $g(3)$ and $g(5)$.

The response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, the statement "since g is differentiable it's continuous and 8.5 is between 8 and 9 so such a point exists" would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

0	1

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

g is differentiable. $\Rightarrow g$ is continuous.

$g(3) = 8 < 8.5 < 9 = g(5)$

By the Intermediate Value Theorem, there must be a value of r, for $3 < r < 5$, such that $g(r) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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The response does not need to mention the continuity of g' or g.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g(5) - 9 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g(5) = 9$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

- The statement $\lim\limits_{x\to5^-} \frac{g(x)-9}{x^2-25} = \frac{g(5)-9}{5^2-25} \begin{smallmatrix}\to\\ \to\end{smallmatrix} \frac{0}{0}$ earns the point.
- The false equation $\lim\limits_{x\to5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim\limits_{x\to5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x} = \frac{3}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim\limits_{x\to5^-} \frac{g(x)-9}{x^2-25} = \frac{0}{0}$ and $\lim\limits_{x\to5^-} \frac{g(x)-9}{x^2-25} = \frac{g'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{3}{5}$, or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ that both g' and g are continuous. $\Rightarrow$ $\lim\limits_{x\to5^-} g(x) = g(5)$ and $\lim\limits_{x\to5^-} g'(x) = g'(5)$.

$$\lim_{x\to5^-} (g(x) - 9) = g(5) - 9 = 9 - 9 = 0$$

$$\lim_{x\to5^-} \left(x^2 - 25\right) = 5^2 - 25 = 0$$

Therefore, L'Hospital's Rule can be applied.

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$$\lim_{x\to 5^-}\frac{g(x)-9}{x^2-25}=\lim_{x\to 5^-}\frac{g'(x)}{2x}$$

$$=\frac{g'(5)}{2\cdot 5}=\frac{6}{10}=\frac{3}{5}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3\cdot(g(x))^2$ or $3\cdot(g(3))^2$.

The chain rule point can be earned by seeing an expression involving g but not g') times $g'(3)$.

All three points will be awarded for seeing only $3\cdot 8^2\cdot(-3)$, but $3\cdot 64\cdot(-3)$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$f'(x)=3(g(x))^2\cdot g'(x)$$

$$f'(3)=3(g(3))^2\cdot g'(3)=3\cdot 8^2\cdot(-3)=-576$$

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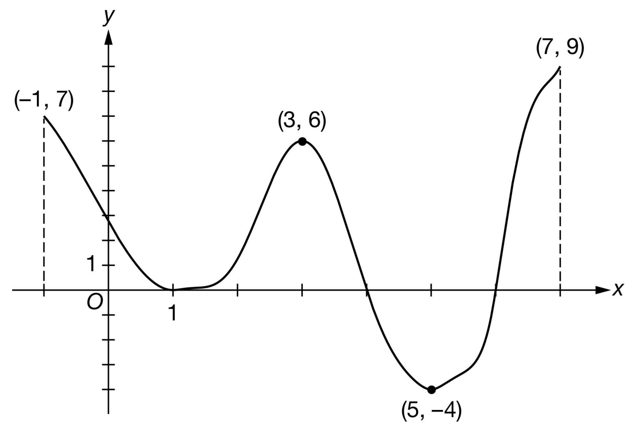
25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h'

The figure above shows the graph of h', the derivative of a twice-differentiable function h, on the closed interval $-1 \le x \le 7$. The graph of h' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions bounded by the x-axis and the graph of h' on the intervals $[-1, 1]$, $[1, 4]$, $[4, 6]$, and $[6, 7]$ are 6, 8, 5.6, and 6, respectively. The function h is defined for all real numbers and satisfies $h(5) = 5$ and $h(7) = 8$.

(a) Find the x-coordinate of each critical point of h on the open interval $-1 < x < 7$.

(b) Identify the critical point from part (a) which corresponds to a relative minimum of h. Give a reason for your answer.

(c) Find the value of h at the x-coordinate identified in part (b). Show the computations that lead to your answer.

(d) Find the absolute minimum value of h on $-1 \le x \le 7$. Justify your answer.

(e) On what open intervals is h both increasing and concave up? Give a reason for your answer.

(f) Explain why there must be a value of p, for $5 < p < 7$, such that $h(p) = 6.5$.

(g) Find $\lim\limits_{x \to 7^-} \frac{h(x)-8}{3x-21}$. Give a reason for your answer.

(h) The function a is defined by $a(x) = (h(x))^2$. Find $a'(5)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = 1$, $x = 4$, and $x = 6$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1,\ 7)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1,\ 7)$. The point is still earned provided the three critical points, $x = 1$, $x = 4$, and $x = 6$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1,\ 0)$, $(4,\ 0)$, $(6,\ 0)$, we will read the x -coordinates, and disregard the y-coordinates.

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0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'(x) = 0$ at $x = 1$, $x = 4$, and $x = 6$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x-coordinate of the local minimum as 6 and give the correct reason of "h' changes from negative to positive" to earn the point.

Insufficient reasons, which do not earn the point, include:

· Saying that "h switches from decreasing to increasing at $x = 6$."

· Saying that "h' (6) changes signs from negative to positive"

If the response talks about "the graph" without specifying which graph, we will interpret "the graph" here (and throughout the rest of the problem) as being the graph of h', the one given in the stem of the problem.

A response indicating that "h' (x) is negative for $x < 6$ and positive for $x > 6$" is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

A relative minimum value of h occurs where h' (x) changes from negative to positive values.

$x = 6$ is the only location on $-1 < x < 7$ for which this is true, so $x = 6$ is the only location of a relative minimum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

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Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -1$, $x = 1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $h(-1) = -6.4$, $h(1) = -0.4$, and $h(4) = 7.6$. No other x-coordinates will be read for consistency.

The response $h(6) = h(7) - 6 = 2$ earns both points.

The responses of both $8 - 6 = 2$ and $h(6) = 2$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $h(6) = h(7) - \int_6^7 h'(x)\,dx = 8 - 6 = 2$ earns both points.

Special Case: If in part (b), a response picks $x = 5$ or $x = 7$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $h(6) = h(7) + \int_7^6 h'(x)\,dx$
- ☐ Answer

Solution:

$$h(6) = h(7) + \int_7^6 h'(x)\,dx$$
$$= h(7) - \int_6^7 h'(x)\,dx = 8 - 6 = 2$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word "considers" means that there needs to be more than just a mention of the x-value.

For $x = 6$, the response must either state its y-value or reference it as a minimum.

For the endpoints, the y-values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y-values in the table of candidates are correct.

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If non-critical numbers are in the list of candidates, their y-values cannot be computed (except for $x = 5$), and thus the third point cannot be earned.

It is possible to not refer to $h\left(7\right) = 8$ by indicating that h' is positive on the interval $\left(6,\ 7\right)$, thus eliminating 8 as a possible absolute minimum.

A complete list of given or computable values for $h\left(x\right)$ is given in the table below

x	$h(x)$
-1	-6.4
1	-0.4
4	7.6
5	5
6	2
7	8

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = 6$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$h\left(-1\right) = h\left(7\right) + \int_{7}^{-1} h'\left(x\right)\, dx$$

$$= h\left(7\right) - \int_{-1}^{7} h'\left(x\right)\, dx = 8 - \left(6 + 8 - 5.6 + 6\right) = -6.4$$

$h\left(6\right) = 2$

$h\left(7\right) = 8$

Therefore, the absolute minimum value of h on $-1 \leq x \leq 7$ is $h\left(-1\right) = -6.4$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example "$\left(1,\ 3\right]$ and $\left[6,\ 7\right)$" earns the first point.

The reason "h' is increasing" can be expressed by either "h'' is positive" or "the slope of h' is positive."

We will overlook the false implication that $h' > 0$ on $[1, 3]$ and $[6, 7]$.

The second point can be earned without earning the first point only if the given answer is a subset of $\left[1,\ 3\right] \cup \left[6,\ 7\right]$.

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0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where h' is positive.

h is concave up where h' is increasing.

h is both increasing and concave up on the open intervals $(1,\ 3)$ and $(6,\ 7)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[5,\ 7]$, or the point is not earned.

The y-values of h need to be mentioned by either referring to the numerical values of 5 and 8, or mentioning $h\ (5)$ and $h\ (7)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since h is differentiable it’s continuous and 6.5 is between 5 and 8 such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

0	1

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$h\ (5) = 5 < 6.5 < 8 = h\ (7)$

By the Intermediate Value Theorem, there must be a value of p, for $5 < p < 7$, such that $h\ (p) = 6.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h.

3-1-1b

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h\left(7\right)-8=0$ and $3\cdot7-21=0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h\left(7\right)=8$ and $3\cdot7=21$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

- The statement $\lim\limits_{x\to7^-}\frac{h(x)-8}{3x-21}=\frac{h(7)-8\to0}{3\cdot7-21\to0}$ earns the point.
- The false equation $\lim\limits_{x\to7^-}\frac{h(x)-8}{3x-21}=\frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim\limits_{x\to7^-}\frac{h(x)-8}{3x-21}=\frac{h'(x)}{3}=\frac{9}{3}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim\limits_{x\to7^-}\frac{h(x)-8}{3x-21}=\frac{0}{0}$ and $\lim\limits_{x\to7^-}\frac{h(x)-8}{3x-21}=\frac{h'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of 3, or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow$ $\lim\limits_{x\to7^-}h\left(x\right)=h\left(7\right)$ and $\lim\limits_{x\to7^-}h'\left(x\right)=h'\left(7\right)$.

$$\lim_{x\to7^-}\left(h\left(x\right)-8\right)=h\left(7\right)-8=8-8=0$$

$$\lim_{x\to7^-}\left(3x-21\right)=3\cdot7-21=0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to7^-}\frac{h(x)-8}{3x-21}=\lim_{x\to7^-}\frac{h'(x)}{3}$$

3-1-1b

$= \frac{h'(7)}{3} = \frac{9}{3} = 3$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The Power rule point can be earned by just seeing $2 \cdot h(x)$ or $2 \cdot h(5)$.

The chain rule point can be earned by seeing an expression involving h but not h') times $h'(5)$.

All three points will be awarded for seeing only $2 \cdot 5 \cdot (-4)$, but $10 \cdot (-4)$, $2 \cdot (-20)$, or $-8 \cdot 5$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.

✓

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$a'(x) = 2h(x) \cdot h'(x)$

$a'(5) = 2 \cdot h(5) \cdot h'(5) = 2 \cdot 5 \cdot (-4) = -40$

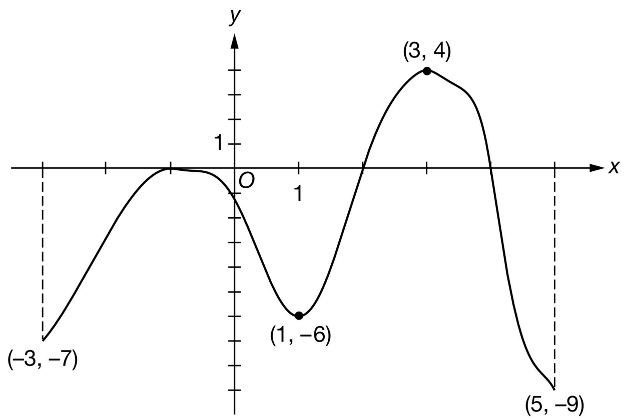
3-1-1b

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f'

The figure above shows the graph of f', the derivative of a twice-differentiable function f, on the closed interval $-3 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x-axis and the graph of f' on the intervals $[-3, -1]$, $[-1, 2]$, $[2, 4]$, and $[4, 5]$ are 6, 8, 5.6, and 6, respectively. The function f is defined for all real numbers and satisfies $f(3) = 10$ and $f(5) = 7$.

(a) Find the x-coordinate of each critical point of f on the open interval $-3 < x < 5$.

(b) Identify the critical point from part (a) which corresponds to a relative minimum of f. Give a reason for your answer.

(c) Find the value of f at the x-coordinate identified in part (b). Show the computations that lead to your answer.

(d) Find the absolute minimum value of f on $-3 \leq x \leq 5$. Justify your answer.

(e) On what open intervals is f both decreasing and concave down? Give a reason for your answer.

(f) Explain why there must be a value of d, for $3 < d < 5$, such that $f(d) = 8.5$.

(g) Find $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25}$. Give a reason for your answer.

(h) The function m is defined by $m(x) = (f(x))^2$. Find $m'(3)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -1, \ x = 2$, and $x = 4$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, \ 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, \ 5)$. The point is still earned provided the three critical points, $x = -1, \ x = 2$, and $x = 4$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, \ 0), \ (2, \ 0), \ (4, \ 0)$, we will read the x -coordinates, and disregard the y -coordinates.

3-1-1b

0	1

The student response accurately includes the criteria below.

☐ Answer

Solution:

$f'(x) = 0$ at $x = -1$, $x = 2$, and $x = 4$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local minimum as 2 and give the correct reason of " f' changes from negative to positive" to earn the point.

Insufficient reasons, which do not earn the point, include:

· Saying that " f switches from decreasing to increasing at $x = 2$ "

· Saying that " $f'(2)$ changes signs from negative to positive"

If the response talks about "the graph" without specifying which graph, we will interpret "the graph" here (and throughout the rest of the problem) as being the graph of f', the one given in the stem of the problem.

A response indicating that " $f'(x)$ is negative for $x < 2$ and positive for $x > 2$ " is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

A relative minimum value of f occurs where $f'(x)$ changes from negative to positive values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative minimum value of f on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

3-1-1b

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -3$, $x = -1$, or $x = 4$ is selected as the location of the local minimum in part (b). Note that $f(-3) = 21.4$, $f(-1) = 15.4$, and $f(4) = 13.$ No other x -coordinates will be read for consistency.

The response $f(2) = f(5) - (5.6 - 6) = 7.4$ earns both points.

The responses of both $7 - (-0.4) = 7.4$ and $f(2) = 7.4$ do not earn the first point, but earn the second point, provided no other work is presented.

The string of equalities $f(2) = f(5) - \int_2^5 f'(x)\, dx = 7 - (-0.4) = 7.4$ earns both points.

Special Case: If in part (b), a response picks $x = 3$ or $x = 5$ as the location of the minimum, then there is no work to do in part (c). This response will earn zero points.

✓

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ $f(2) = f(5) + \int_5^2 f'(x)\, dx$
- ☐ Answer

Solution:

$$f(2) = f(5) + \int_5^2 f'(x)\, dx = f(5) - \int_2^5 f'(x)\, dx$$

$$= f(5) - \int_2^4 f'(x)\, dx - \int_4^5 f'(x)\, dx = 7 - 5.6 - (-6) = 7.4$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a minimum.

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

3-1-1b

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 3$), and thus the third point cannot be earned.

It is possible to not refer to $f(-3) = 21.4$ by indicating that f' is negative on the interval $(-3, \ -1)$, thus eliminating 21.4 as a possible absolute minimum.

A complete list of given or computable values for $f(x)$ is given in the table below.

x	$f(x)$
−3	21.4
−1	15.4
2	7.4
3	10
4	13
5	7

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Considers $x = 2$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$$f(-3) = f(5) + \int_{5}^{-3} f'(x)\, dx$$

$$= f(5) - \int_{-3}^{5} f'(x)\, dx = 7 - (-6 + 5.6 - 8 - 6) = 21.4$$

$f(2) = 7.4$

$f(5) = 7$

So the absolute minimum value of f on $-3 \le x \le 5$ is

$f(5) = 7.$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, for example " $(-1, \ 1]$ and $[4, \ 5)$ " earns the first point.

The reason " f' is decreasing" can be expressed by either " f'' is negative" or "the slope of f' is negative."

We will overlook the false implication that $f' < 0$ on $[-1, \ 1]$ and $[4, \ 5]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, \ 1] \cup [4, \ 5]$.

3-1-1b

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

f is decreasing where f' is negative.

f is concave down where f' is decreasing.

f is both decreasing and concave down on the open intervals $(-1,\ 1)$ and $(4,\ 5)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that f is continuous because f is differentiable or f is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[3,\ 5]$, or the point is not earned.

The y -values of f need to be mentioned by either referring to the numerical values of 7 and 10, or mentioning $f\ (3)$ and $f\ (5)$.

The response is eligible to earn the point without explicitly stating “Intermediate Value Theorem” or “IVT.” For example, the statement “since f is differentiable it’s continuous and 8.5 is between 7 and 10 so such a point exists” would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

0	1

The student response accurately includes the criteria below.

- ☐ Explanation using Intermediate Value Theorem

Solution:

f is differentiable. $\Rightarrow f$ is continuous.

$f\ (5) = 7 < 8.5 < 10 = f\ (3)$

By the Intermediate Value Theorem, there must be a value of d, for $3 < d < 5$, such that $f\ (d) = 8.5$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of f' or f.

3-1-1b

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $2f\,(5) - 14 = 0$ and $5^2 - 25 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $2f\,(5) = 14$ and $5^2 = 25$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f(5)-14 \to 0}{5^2-25 \quad \to 0}$ earns the point.

· The false equation $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x} = \frac{-9}{5}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \frac{0}{0}$ and $\lim\limits_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \frac{2f'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{9}{5}$, or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

f is twice differentiable. $\Rightarrow$ both f' and f are continuous. $\Rightarrow$ $\lim\limits_{x \to 5^-} f\,(x) = f\,(5)$ and $\lim\limits_{x \to 5^-} f\,(x) = f\,(5)$.

$$\lim_{x \to 5^-} (2f\,(x) - 14) = 2 \cdot f\,(5) - 14 = 14 - 14 = 0$$

$$\lim_{x \to 5^-} \left(x^2 - 25\right) = 5^2 - 5 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \to 5^-} \frac{2f(x)-14}{x^2-25} = \lim_{x \to 5^-} \frac{2f'(x)}{2x}$$

3-1-1b

$= \frac{2f'(5)}{2 \cdot 5} = \frac{-9}{5}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2 \cdot f(x)$ or $2 \cdot f(3)$.

The chain rule point can be earned by seeing an expression involving f but not f') times $f'(3)$.

All three points will be awarded for seeing only $2 \cdot 10 \cdot 4$, but $20 \cdot 4$, $2 \cdot 40$, or $8 \cdot 10$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$m'(x) = 2f(x) \cdot f'(x)$

$m'(3) = 2 \cdot f(3) \cdot f'(3) = 2 \cdot 10 \cdot 4 = 80$

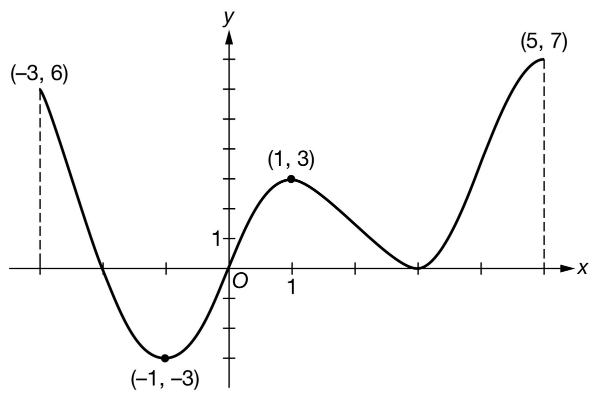
3-1-1b

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

The figure above shows the graph of g', the derivative of a twice-differentiable function g, on the closed interval $-3 \le x \le 5$. The graph of g' has horizontal tangent lines at $x = -1$, $x = 1$, and $x = 3$. The areas of the regions bounded by the x-axis and the graph of g' on the intervals $[-3, -2]$, $[-2, 0]$, $[0, 3]$, and $[3, 5]$ are 3, 4, 5, and 7, respectively. The function g is defined for all real numbers and satisfies $g(1) = 3$ and $g(5) = 13$.

(a) Find the x-coordinate of each critical point of g on the open interval $-3 < x < 5$.

(b) Identify the critical point from part (a) which corresponds to a relative maximum of g. Give a reason for your answer.

(c) Find the value of g at the x-coordinate identified in part (b). Show the computations that lead to your answer.

(d) Find the absolute maximum value of g on $-3 \le x \le 5$. Justify your answer.

(e) On what open intervals is g both decreasing and concave up? Give a reason for your answer.

(f) Explain why there must be a value of a, for $1 < a < 5$, such that $g(a) = 10$.

(g) Find $\lim_{x \to 5^-} \frac{g(x)-13}{3x-15}$. Give a reason for your answer.

(h) The function r is defined by $r(x) = (g(x))^3$. Find $r'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -2$, $x = 0$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 5)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 5)$. The point is still earned provided the three critical points, $x = -2$, $x = 0$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-2, 0)$, $(0, 0)$, $(3, 0)$, we will read the x -coordinates, and disregard the y-coordinates.

3-1-1b

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$g'(x) = 0$ at $x = -2$, $x = 0$, and $x = 3$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x-coordinate of the local maximum as -2 and give the correct reason of "g' changes from positive to negative" to earn the point.

Insufficient reasons, which do not earn the point, include:

· Saying that "g switches from increasing to decreasing at $x = -2$"

· Saying that "$g'(-2)$ changes signs from positive to negative"

If the response talks about "the graph" without specifying which graph, we will interpret "the graph" here (and throughout the rest of the problem) as being the graph of g', the one given in the stem of the problem.

A response indicating that "$g'(x)$ is positive for $x < -2$ and negative for $x > -2$" is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

A relative maximum value of g occurs where $g'(x)$ changes from positive to negative values.

$x = -2$ is the only location on $-3 < x < 5$ for which this is true, so $x = -2$ is the only location of a relative maximum value of g on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

3-1-1b

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x=-3,\ \ x=0$, or $x=3$ is selected as the location of the local maximum in part (b). Note that $g(-3)=2$, $g(0)=1$, and $g(3)=6$. No other x-coordinates will be read for consistency.

The response $g(-2)=g(5)-(-4+5+7)=5$ earns both points.

The response of both $13-8=5$ and $g(-2)=5$ does not earn the first point, but earns the second point, provided no other work is presented.

The string of equalities $g(-2)=g(5)-\int_{-2}^{5} g'(x)\,dx=13-8=5$ earns both points.

Special Case: If in part (b), a response picks $x=1$ or $x=5$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $g(-2)=g(5)+\int_{5}^{-2} g'(x)\,dx$
- ☐ Answer

Solution:

$$g(-2)=g(5)+\int_{5}^{-2} g'(x)\,dx$$

$$=g(5)-\int_{-2}^{5} g'(x)\,dx=13-(7+5-4)=5$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word “considers” means that there needs to be more than just a mention of the x-value.

For $x=-2$, the response must either state its y-value or reference it as a maximum.

For the endpoints, the y-values must be presented but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y-values in the table of candidates are correct.

3-1-1b

If non-critical numbers are in the list of candidates, their y-values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $g(-3) = 2$ by indicating that g' is positive on the interval $(-3, -2)$, thus eliminating 2 as a possible absolute maximum.

A complete list of given or computable values for $g(x)$ is given in the table below.

x	$g(x)$
−3	2
−2	5
0	1
1	3
3	6
5	13

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$g(-3) = g(5) + \int_5^{-3} g'(x)\,dx$$

$$= g(5) - \int_{-3}^{5} g'(x)\,dx = 13 - (7 + 5 - 4 + 3) = 2$$

$$g(-2) = 5$$

$$g(5) = 13$$

So the absolute maximum value of g on $-3 \le x \le 5$ is $g(5) = 13$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 0]$, $[-1, 0)$, $(-1, 0]$, $(-1, 0)$ all earn the first point.

The reason "g' is increasing" can be expressed by either "g'' is positive" or "the slope of g' is positive."

We will overlook the false implication that $g' < 0$ on $[-1, 0]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[-1, 0]$.

3-1-1b

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

g is decreasing where g' is negative.

g is concave up where g' is increasing.

g is both decreasing and concave up on the open interval $(-1,\ 0)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that g is continuous because g is differentiable or g is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1,\ 5]$, or the point is not earned.

The y-values of g need to be mentioned by either referring to the numerical values of 3 and 13, or mentioning $g\,(1)$ and $g\,(5)$.

The response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, the statement "since g is differentiable it's continuous and 10 is between 3 and 13 so such a point exists" would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

g is differentiable. $\Rightarrow$ g is continuous.

$g\,(1) = 3 < 10 < 13 = g\,(5)$

By the Intermediate Value Theorem, there must be a value of a, for $1 < a < 5$, such that $g\,(a) = 10$.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of g' or g.

3-1-1b

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $g\left(5\right)-13=0$ and $3\cdot5-15=0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $g\left(5\right)=13$ and $3\cdot5=15$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

· The statement $\lim\limits_{x\to5^-}\frac{g(x)-13}{3x-15}=\frac{g(5)-13\to0}{3\cdot5-15\to0}$ earns the point.

· The false equation $\lim\limits_{x\to5^-}\frac{g(x)-13}{3x-15}=\frac{0}{0}$ does not earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim\limits_{x\to5^-}\frac{g(x)-13}{3x-15}=\frac{g'(x)}{3}=\frac{7}{3}$, will earn the second point, but not the third point; note that even just one of two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim\limits_{x\to5^-}\frac{g(x)-13}{3x-15}=\frac{0}{0}$ and $\lim\limits_{x\to5^-}\frac{g(x)-13}{3x-15}=\frac{g'(x)}{3}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $\frac{7}{3}$, or its numeric equivalent.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g is twice differentiable. $\Rightarrow$ both g' and g are continuous. $\Rightarrow$ $\lim\limits_{x\to5^-}g\left(x\right)=g\left(5\right)$ and $\lim\limits_{x\to5^-}g'\left(x\right)=g'\left(5\right)$.

$$\lim_{x\to5^-}\left(g\left(x\right)-13\right)=g\left(5\right)-13=13-13=0$$

$$\lim_{x\to5^-}\left(3x-15\right)=3\cdot5-15=0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to5^-}\frac{g(x)-13}{3x-15}=\lim_{x\to5^-}\frac{g'(x)}{3}$$

3-1-1b

$= \frac{g'(5)}{3} = \frac{7}{3}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $3 \cdot (g(x))^2$ or $3 \cdot (g(1))^2$.

The chain rule point can be earned by seeing an expression involving g but not g') times $g'(1)$.

All three points will be awarded for seeing only $3 \cdot 3^2 \cdot 3$, but $3 \cdot 9 \cdot 3$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Power rule
- [ ] Chain rule
- [ ] Answer

Solution:

$r'(x) = 3(g(x))^2 \cdot g'(x)$

$r'(1) = 3(g(1))^2 \cdot g'(1) = 3 \cdot 3^2 \cdot 3 = 81$

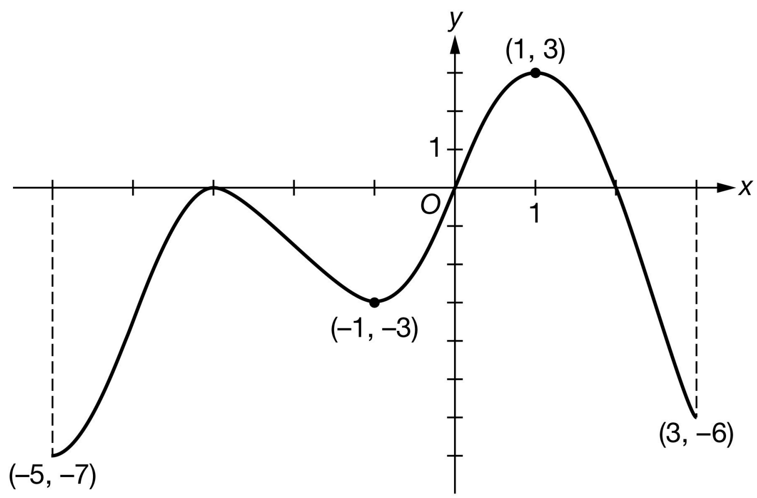
3-1-1b

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of h'

The figure above shows the graph of h', the derivative of a twice-differentiable function h, on the closed interval $-5 \le x \le 3$. The graph of h' has horizontal tangent lines at $x = -3$, $x = -1$, and $x = 1$. The areas of the regions bounded by the x-axis and the graph of h' on the intervals $[-5, -3]$, $[-3, 0]$, $[0, 2]$, and $[2, 3]$ are 7, 5, 4, and 3, respectively. The function h is defined for all real numbers and satisfies $h(1) = 12$ and $h(3) = 11$.

(a) Find the x-coordinate of each critical point of h on the open interval $-5 < x < 3$.

(b) Identify the critical point from part (a) which corresponds to a relative maximum of h. Give a reason for your answer.

(c) Find the value of h at the x-coordinate identified in part (b). Show the computations that lead to your answer.

(d) Find the absolute maximum value of h on $-5 \le x \le 3$. Justify your answer.

(e) On what open intervals is h both increasing and concave down? Give a reason for your answer.

(f) Explain why there must be a value of b, for $1 < b < 3$, such that $h(b) = 11.5$.

(g) Find $\lim\limits_{x \to 3^-} \frac{h(x) - 11}{x^2 - 9}$. Give a reason for your answer.

(h) The function p is defined by $p(x) = (h(x))^2$. Find $p'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must present all three critical values: $x = -3$, $x = 0$, and $x = 2$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 3)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 3)$. The point is still earned provided the three critical points, $x = -3$, $x = 0$, and $x = 2$ are correctly listed.

3-1-1b

If the critical points are given as ordered pairs, e.g., $(-3,\ 0),\ \ (0,\ 0),\ \ (2,\ 0)$ we will read the x -coordinates, and disregard the y -coordinates.

0	1

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

$h'\ (x) = 0$ at $x = -3$, $x = 0$, and $x = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to identify the x -coordinate of the local maximum as 2 and give the correct reason of " h' changes from positive to negative" to earn the point.

Insufficient reasons, which do not earn the point, include:

· Saying that " h switches from increasing to decreasing at $x = 2$ "

· Saying that " $h'\ (2)$ changes signs from positive to negative"

If the response talks about "the graph" without specifying which graph, we will interpret "the graph" here (and throughout the rest of the problem) as being the graph of h', the one given in the stem of the problem.

A response indicating that " $h'\ (x)$ is positive for $x < 2$ and negative for $x > 2$ " is allowed to give this statement as a reason to earn the point.

A sign chart by itself is not sufficient to earn the point. A sign chart must be accompanied by sufficient written explanation of what it means.

0	1

The student response accurately includes the criteria below.

- [ ] Answer with reason

Solution:

A relative maximum value of h occurs where $h'\ (x)$ changes from positive to negative values.

$x = 2$ is the only location on $-3 < x < 5$ for which this is true, so $x = 2$ is the only location of a relative maximum value of h on this interval.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First Point:

3-1-1b

Philosophy: There is a correct connection made between the values of the function given and the areas given in the stem of the problem that will lead to the answer.

The first point is earned if a given value of the function is combined correctly with an integral or integrals leading to the answer. These integrals can be expressed in words.

The first point is earned if a given value of the function is combined correctly with the numerical areas given in the stem of the problem.

Second Point:

The second point is earned by the correct answer or by a consistent answer only when $x = -5,\ \ x = -3$, or $x = 0$ is selected as the location of the local maximum in part (b). Note that $h\ (-5) = 22$, $h\ (-3) = 15$, and $h\ (0) = 10$. No other x -coordinates will be read for consistency.

The response $h\ (2) = h\ (3) - (-3) = 14$ earns both points.

The response of both $11 - (-3) = 14$ and $h\ (2) = 14$ does not earn the first point, but earns the second point, provided no other work is presented.

Special Case: If a response uses $\int_1^2 h'\ (x)\ dx = 2$ then

· $h\ (2) = h\ (1) + \int_1^2 h'\ (x)\ dx = 12 + 2 = 14$ earns both points, because it's possible to find $\int_1^2 h'\ (x)\ dx = 2$ using the areas and initial conditions given.

· any response using perceived symmetry about $x = 1$ can earn at most the first point, for example $h\ (2) = h\ (1) + \int_1^2 h'\ (x)\ dx = 12 + \frac{1}{2} \cdot 4 = 14$ earns just the first point.

Special Case: If in part (b), a response picks $x = 1$ or $x = 3$ as the location of the maximum, then there is no work to do in part (c). This response will earn zero points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $h\ (2) = h\ (3) + \int_3^2 h'\ (x)\ dx$
- ☐ Answer

Solution:

$$h\ (2) = h\ (3) + \int_3^2 h'\ (x)\ dx$$

$$= h\ (3) - \int_2^3 h'\ (x)\ dx = 11 - (-3) = 14$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First and Second Points: The word "considers" means that there needs to be more than just a mention of the x -value.

For $x = 2$, the response must either state its y -value or reference it as a maximum.

3-1-1b

For the endpoints, the y -values must be presented, but need not be correct.

Third Point:

If a Candidates Test is being conducted, the third point is awarded only if all y -values in the table of candidates are correct.

If non-critical numbers are in the list of candidates, their y -values cannot be computed (except for $x = 1$), and thus the third point cannot be earned.

It is possible to not refer to $h(3) = 11$ by indicating that h' is negative on the interval $(2,\ 3)$, thus eliminating 11 as a possible absolute maximum.

A complete list of given or computable values for $h(x)$ is given in the table below

x	$h(x)$
−5	22
−3	15
0	10
1	12
2	14
3	11

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Considers $x = 2$
- [ ] Considers both endpoints
- [ ] Answer with justification

Solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$$h(-5) = h(3) + \int_3^{-5} h'(x)\,dx$$

$$= h(3) - \int_{-5}^{3} h'(x)\,dx = 11 - (-7 - 5 + 4 - 3) = 22$$

$h(2) = 14$

$h(3) = 11$

So the absolute maximum value of h on $-5 \le x \le 3$ is $h(-5) = 22$.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1,\ 2]$, $[1,\ 2)$, $(1,\ 2]$, and $(1,\ 2)$ all earn the first point.

The reason " h' is decreasing" can be expressed by either " h'' is negative" or "the slope of h' is negative."

3-1-1b

We will overlook the false implication that $h' > 0$ on $[1,\ 2]$.

The second point can be earned without earning the first point only if the given answer is a subset of $[1,\ 2]$.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Answer
- [ ] Reason

Solution:

h is increasing where h' is positive.

h is concave down where h' is decreasing.

h is both increasing and concave down on the open interval $(1,\ 2)$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response needs to state that h is continuous because h is differentiable or h is twice differentiable; that is, the response needs to explicitly state this implication.

A response need not mention the interval of continuity. If it does, the interval stated must include the closed interval $[1,\ 3]$, or the point is not earned.

The y -values of h need to be mentioned by either referring to the numerical values of 11 and 12, or mentioning $h\,(1)$ and $h\,(3)$

The response is eligible to earn the point without explicitly stating "Intermediate Value Theorem" or "IVT." For example, the statement "since h is differentiable it's continuous and 11.5 is between 11 and 12 so such a point exists" would earn the point.

If the response states the conditions of IVT but calls it MVT, the point is not earned.

0	1

The student response accurately includes the criteria below.

- [ ] Explanation using Intermediate Value Theorem

Solution:

h is differentiable. $\Rightarrow h$ is continuous.

$h\,(3) = 11 < 11.5 < 12 = h\,(1)$

By the Intermediate Value Theorem, there must be a value of b, for $1 < b < 3$, such that $h\,(b) = 11.5$.

Part G

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response does not need to mention the continuity of h' or h.

The first point requires the evaluation of the numerator and denominator separately, both producing a value of 0.

The point can be earned by saying $h\,(3) - 11 = 0$ and $3^2 - 9 = 0$.

All arithmetic/algebraic expressions must be completely simplified to 0. For example, $h\,(3) = 11$ and $3^2 = 9$ are not enough to earn the first point.

The statement that the limit gives the indeterminate form $\frac{0}{0}$ earns the first point.

Arrows indicating that the numerator and denominator both go to zero are okay as long as there is no equating of the limit to $\frac{0}{0}$. For example,

- The statement $\lim\limits_{x\to3^-} \frac{h(x)-11}{x^2-9} = \frac{h(3)-11}{3^2-9} \begin{smallmatrix}\to 0\\ \to 0\end{smallmatrix}$ earns the point.
- The false equation $\lim\limits_{x\to3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ earn the point.

The second point is earned if there is an apparent attempt at computing the derivatives of the numerator and denominator, even if the derivatives are incorrect.

If the attempted derivatives are incorrect, the response will earn the second point, but not the third point.

Any incorrect statement of equality here, e.g., $\lim\limits_{x\to3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x} = \frac{-6}{6}$, will earn the second point, but not the third point; note that even just one of the two false equalities in the preceding example results in not earning the third point.

Equating multiple non-equivalent expressions may be eligible to earn one of the first two points (even when more than one communication error occurs). For example, a response with both statements $\lim\limits_{x\to3^-} \frac{h(x)-11}{x^2-9} = \frac{0}{0}$ and $\lim\limits_{x\to3^-} \frac{h(x)-11}{x^2-9} = \frac{h'(x)}{2x}$ does not earn the first point, earns the second point, and is eligible for the third point (with the correct answer).

The third point is earned only for the answer of $-\frac{6}{6}$ or its numeric equivalent.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h is twice differentiable. $\Rightarrow$ both h' and h are continuous. $\Rightarrow$ $\lim\limits_{x\to3^-} h\,(x) = h\,(3)$ and $\lim\limits_{x\to3^-} h'\,(x) = h'\,(3)$.

$$\lim_{x\to3^-} (h\,(x) - 11) = h\,(3) - 11 = 11 - 11 = 0$$

$$\lim_{x\to3^-} (x^2 - 9) = 3^2 - 9 = 0$$

3-1-1b

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x\to3^-}\frac{h(x)-11}{x^2-9}=\lim_{x\to3^-}\frac{h'(x)}{2x}$$

$$=\frac{h'(3)}{2\cdot3}=\frac{-6}{6}=-1$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The power rule point can be earned by just seeing $2\cdot h(x)$ or $2\cdot h(1)$.

The chain rule point can be earned by seeing an expression involving h but not h') times $h'(1)$.

All three points will be awarded for seeing only $2\cdot 12\cdot 3$, but $24\cdot 3$, $2\cdot 36$, or $6\cdot 12$ will only earn two points.

Only the correct answer will earn the answer point. If an error is made in applying the power rule or the chain rule, the response is not eligible to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Power rule
- ☐ Chain rule
- ☐ Answer

Solution:

$p'(x)=2h(x)\cdot h'(x)$

$p'(1)=2\cdot h(1)\cdot h'(1)=2\cdot 12\cdot 3=72$

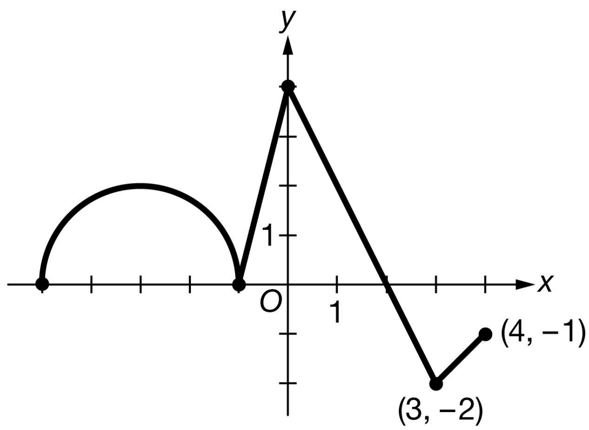
3-1-1b

29. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g'

Let g be a function that is defined on the closed interval $-5 \le x \le 4$ with $g(0) = 5$. The graph of g', the derivative of g, consists of a semicircle and three line segments, as shown above.

(a) On what intervals, if any, is g decreasing? Give a reason for your answer.

(b) Find the average rate of change of g on the interval $[-3, 3]$. Is there a value c, $-3 < c < 3$, for which $g'(c)$ is equal to the average rate of change of g on the interval $[-3, 3]$? Justify your answer.

(c) Find the x-coordinate of each point of inflection of the graph of g on the open interval $(-5, 4)$. Justify your answer.

(d) Let $H(x) = e^{3x} g(x)$. Find the value of $H'(0)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not have to say that g is continuous to earn the point.

A response with correct endpoints can use open, closed, or half-open interval notation to earn the point.

0	1

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

g is decreasing on the interval $[2,\ 4]$ because g is continuous and $g'(x) < 0$ on $(2,\ 4]$.

3-1-1b

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for declaring that the average rate of change is either $\frac{g(3)-g(-3)}{3-(-3)}$ or $\frac{\int_{-3}^{3} g'(x)\,dx}{6}$. The difference quotient and the definite integral must contain $\frac{1}{6}$ to earn the point.

The second point is earned for finding the value of the average rate of change, $\frac{\pi+5}{6}$ or equivalent.

The third point is for stating that g is continuous because g is differentiable (on $[-3,\ 3]$). Stating that g is both differentiable and continuous on the interval is not enough to earn this point.

The fourth point is for applying either the Mean Value Theorem (MVT) and concluding "yes," or for applying the Intermediate Value Theorem (IVT) and concluding "yes." Neither theorem needs to be named if the appropriate conditions are checked and the correct conclusion is reached.

If a response does not appropriately cite/apply the MVT or IVT, the response does not earn the fourth point.

0	1	2	3	4 ✓

The student response accurately includes **all four** of the criteria below.

- ☐ Difference quotient or definite integral
- ☐ Average rate of change
- ☐ Differentiable implies continuous
- ☐ Conclusion using Mean Value Theorem
  – OR –
  Conclusion using Intermediate Value Theorem

Solution:

Average rate of change $= \frac{g(3)-g(-3)}{3-(-3)} = \frac{\int_{-3}^{3} g'(x)\,dx}{6}$

$= \frac{\pi+6-1}{6} = \frac{\pi+5}{6}$

g is differentiable on $[-3,\ 3]$. $\Rightarrow g$ is continuous on $[-3,\ 3]$.

The Mean Value Theorem guarantees that there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

– OR –

g' is continuous on $[-3,\ 3]$.

The range of g' on the interval $[-3,\ 3]$ is $-2 \le g'(x) \le 4$. Because $-2 < \frac{\pi+5}{6} < 4$, there exists a value of c in the interval $(-3,\ 3)$ such that $g'(c)$ is equal to the average rate of change of g on the interval $[-3,\ 3]$.

Part C

3-1-1b

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must identify all four answers.

A response that provides a correct identification and justification for just two (or three) of the four answers earns 1 of the 2 points.

The second point can be earned by correctly discussing the signs of the slopes of the graph of g'.

The second point cannot be earned by the use of vague or undefined terms such as "it" or "the graph" or "the derivative."

The second point cannot be earned by only discussing the concavity of the graph of g.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Answers
- ☐ Justification

Solution:

The graph of g will have a point of inflection where the graph of g' changes from increasing to decreasing or from decreasing to increasing. This occurs at $x = -3, \ \ x = -1, \ \ x = 0$, and $x = 3$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for using the product rule and chain rule to correctly find $H'(x)$ or $H'(0)$.

The second point is earned for correctly using the values of $g(0)$ and $g'(0)$ in the calculation of $H'(0)$. This point can be earned without the first point if there is a single error in finding $H'(x)$.

A response of $3 \cdot 5 + 4$ is the minimal response that earns both points.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ $H'(x)$
- ☐ Answer

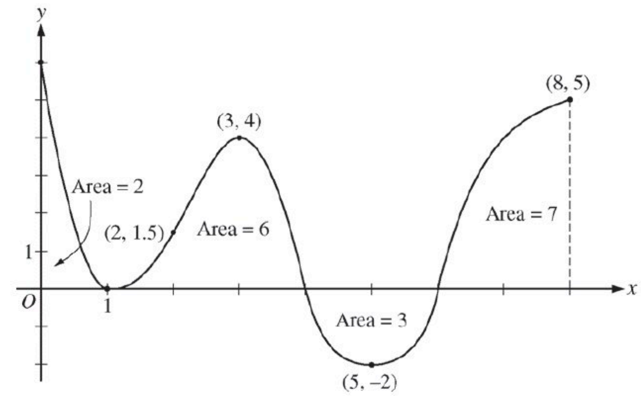
Solution:

$H'(x) = 3e^{3x} \cdot g(x) + e^{3x} \cdot g'(x)$

$H'(0) = 3 \cdot g(0) + g'(0) = 3 \cdot 5 + 4 = 19$

3-1-1b

30.



Graph of f'

The figure above shows the graph of f' the derivative of a twice-differentiable function f, on the closed interval $0 \le x \le 8$. The graph of f' has horizontal tangent lines at x=1, x=3, and x=5. The areas of the regions between the graph of f' and the x-axis are labeled in the figure. The function f is defined for all real numbers and satisfied $f(8)=4$.

The function g is defined by $g(x) = (f(x))^3$. If $f(3) = -5/2$, find the slope of the line tangent to the graph of g at $x = 3$.

Part D

2 points are earned for $g'(x)$

1 point is earned for the answer

$g\prime(x) \;=\; 3[f\,(x)]^2.\,f\prime(x)$

$g\,\prime\;(3) = 3\;[f\,(3)]^{\,2}\cdot f\prime(3) = 3\,(\text{-}\frac{5}{2}\,)^{\,2}\cdot 4 = 75$

0	1	2	3

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the answer

$g\prime(x) \;=\; 3[f\,(x)]^2.\,f\prime(x)$

$g\,\prime\;(3) = 3\;[f\,(3)]^{\,2}\cdot f\prime(3) = 3\,(\text{-}\frac{5}{2}\,)^{\,2}\cdot 4 = 75$

31. Let h be a differentiable function, and let f be the function defined by $f(x) = h(x^2 - 3)$. Which of the following is equal to $f'(2)$?

(A) $h'(1)$

(B) $4h'(1)$

(C) $4h'(2)$

(D) $h'(4)$

(E) $4h'(4)$

3-1-1b

32. If $f(x) = (\ln x)^2$, then $f''\left(\sqrt{e}\right) =$

(A) $\frac{1}{e}$

(B) $\frac{2}{e}$

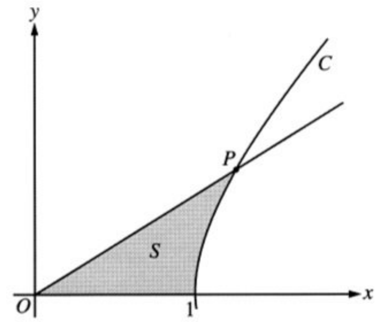
(C) $\frac{1}{2\sqrt{e}}$

(D) $\frac{1}{\sqrt{e}}$

(E) $\frac{2}{\sqrt{e}}$

3-1-1b

33.



The figure above shows the graphs of the line $x = \frac{5}{3}y$ and the curve C given by $x = \sqrt{1+y^2}$. Let S be the shaded region bounded by the two graphs and the x-axis. The line and the curve intersect at point P.

Find the coordinates of point P and the value of $\frac{dx}{dy}$ for curve C at point P.

Part A

1 point is earned for the correct coordinates of P

At P, $\frac{5}{3}y = \sqrt{1+y^2}$, so $y = \frac{3}{4}$.

Since $x = \frac{5}{3}y$, $x = \frac{5}{4}$.

1 point is earned for the correct answer for $\frac{dx}{dy}$ at P

$\frac{dx}{dy} = \frac{y}{\sqrt{1+y^2}} = \frac{y}{x}$. At P, $\frac{dx}{dy} = \frac{3/4}{5/4} = \frac{3}{5}$.

0	1	2

The student response earns two of the following points:

1 point is earned for the correct coordinates of P

At P, $\frac{5}{3}y = \sqrt{1+y^2}$, so $y = \frac{3}{4}$.

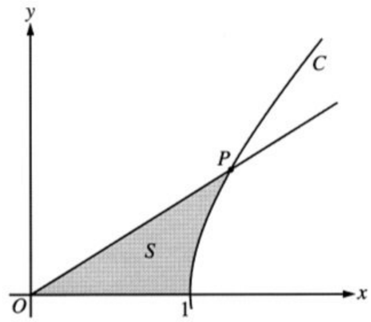
Since $x = \frac{5}{3}y$, $x = \frac{5}{4}$.

1 point is earned for the correct answer for $\frac{dx}{dy}$ at P

$\frac{dx}{dy} = \frac{y}{\sqrt{1+y^2}} = \frac{y}{x}$. At P, $\frac{dx}{dy} = \frac{3/4}{5/4} = \frac{3}{5}$.

3-1-1b

34.



The figure above shows the graphs of the line $x = \frac{5}{3}y$ and the curve C given by $x = \sqrt{1+y^2}$. Let S be the shaded region bounded by the two graphs and the x-axis. The line and the curve intersect at point P.

(a) Find the coordinates of point P and the value of $\frac{dx}{dy}$ for curve C at point P.

(b) Set up and evaluate an integral expression with respect to y that gives the area of S.

(c) Curve C is a part of the curve $x^2 - y^2 = 1$. Show that $x^2 - y^2 = 1$ can be written as the polar equation $r^2 = \frac{1}{\cos^2\theta - \sin^2\theta}$.

(d) Use the polar equation given in part (c) to set up an integral expression with respect to the polar angle θ that represents the area of S.

Part A

1 point is earned for the correct coordinates of P

At P, $\frac{5}{3}y = \sqrt{1+y^2}$, so $y = \frac{3}{4}$.

Since $x = \frac{5}{3}y$, $x = \frac{5}{4}$.

1 point is earned for the correct answer for $\frac{dx}{dy}$ at P

$$\frac{dx}{dy} = \frac{y}{\sqrt{1+y^2}} = \frac{y}{x}. \text{ At } P,\ \frac{dx}{dy} = \frac{3/4}{5/4} = \frac{3}{5}.$$

0	1	2

The student response earns all of the following points:

1 point is earned for the correct coordinates of P

At P, $\frac{5}{3}y = \sqrt{1+y^2}$, so $y = \frac{3}{4}$.

Since $x = \frac{5}{3}y$, $x = \frac{5}{4}$.

1 point is earned for the correct answer for $\frac{dx}{dy}$ at P

$$\frac{dx}{dy} = \frac{y}{\sqrt{1+y^2}} = \frac{y}{x}. \text{ At } P,\ \frac{dx}{dy} = \frac{3/4}{5/4} = \frac{3}{5}.$$

3-1-1b

Part B

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{Area} = \int_0^{3/4} \left(\sqrt{1+y^2} - \frac{5}{3}y\right) dy$$

1 point is earned for the correct answer

= 0.346 or 0.347

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{Area} = \int_0^{3/4} \left(\sqrt{1+y^2} - \frac{5}{3}y\right) dy$$

1 point is earned for the correct answer

= 0.346 or 0.347

Part C

1 point is earned for the correct substitutions $x = r\cos\theta$ and $y = r\sin\theta$ into $x^2 - y^2 = 1$

$x = r\cos\theta$; $y = r\sin\theta$

$$x^2 - y^2 = 1 \Rightarrow r^2\cos^2\theta - r^2\sin^2\theta = 1$$

1 point is earned for correctly isolating r^2

$$r^2 = \frac{1}{\cos^2\theta - \sin^2\theta}$$

0	1	2 ✓

The student response earns all of the following points:

3-1-1b

1 point is earned for the correct substitutions $x = r\cos\theta$ and $y = r\sin\theta$ into $x^2 - y^2 = 1$

$x = r\cos\theta$; $y = r\sin\theta$

$$x^2 - y^2 = 1 \Rightarrow r^2\cos^2\theta - r^2\sin^2\theta = 1$$

1 point is earned for correctly isolating r^2

$$r^2 = \frac{1}{\cos^2\theta - \sin^2\theta}$$

Part D

1 point is earned for the correct limits

Let β be the angle that segment OP makes with the x-axis. Then $\tan\beta = \frac{y}{x} = \frac{3/4}{5/4} = \frac{3}{5}$.

1 point is earned for the correct integrand and constant

$$\text{Area} = \int_0^{\tan^{-1}(3/5)} \frac{1}{2}r^2\,d\theta$$

$$= \frac{1}{2}\int_0^{\tan^{-1}(3/5)} \frac{1}{\cos^2\theta - \sin^2\theta}\,d\theta$$

0	1	2

The student response earns all of the following points:

1 point is earned for the correct limits

Let β be the angle that segment OP makes with the x-axis. Then $\tan\beta = \frac{y}{x} = \frac{3/4}{5/4} = \frac{3}{5}$.

1 point is earned for the correct integrand and constant

$$\text{Area} = \int_0^{\tan^{-1}(3/5)} \frac{1}{2}r^2\,d\theta$$

$$= \frac{1}{2}\int_0^{\tan^{-1}(3/5)} \frac{1}{\cos^2\theta - \sin^2\theta}\,d\theta$$

3-1-1b

35. If f is the function given by $f(x) = e^{x/3}$, which of the following is an equation of the line tangent to the graph of f at the point $(3 \ln 4, 4)$?

(A) $y-4=\frac{4}{3}(x-3\ln 4)$

(B) $y-4=4(x-3\ln 4)$

(C) $y-4=12(x-3\ln 4)$

(D) $y-3\ln 4=4(x-4)$

36. The table shown gives values of f, f', g, and g' at selected values of x.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	0	5	2	-5
2	4	3	7	-1
3	2	-3	5	8
4	0	-7	-2	-4

If $h(x)=g(f(2x))$, then $h'(1)=$

(A) 16

(B) 8

(C) -12

(D) -24

Answer D

Correct. The chain rule needs to be used twice in performing the differentiation, as follows.

$$h'(x)=g'(f(2x))\cdot\frac{d}{dx}f(2x)=g'(f(2x))\cdot f'(2x)\cdot\frac{d}{dx}(2x)=g'(f(2x))\cdot f'(2x)\cdot 2$$

$$h'(1)=g'(f(2))\cdot f'(2)\cdot 2=g'(4)\cdot 3\cdot 2=-4\cdot 6=-24$$

3-1-1b

37.

x	-2	$-2 < x < -1$	-1	$-1 < x < 1$	1	$1 < x < 3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x. Values of f, f', g, and g' for various values of x are given in the table above.

The function h is defined by $h(x)=\ln(f(x))$. Find $h'(3)$. Show the computations that lead to your answer.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

3-1-1b

38.

x	-2	$-2<x<-1$	-1	$-1<x<1$	1	$1<x<3$	3
$f(x)$	12	Positive	8	Positive	2	Positive	7
$f'(x)$	-5	Negative	0	Negative	0	Positive	$\frac{1}{2}$
$g(x)$	-1	Negative	0	Positive	3	Positive	1
$g'(x)$	2	Positive	$\frac{3}{2}$	Positive	0	Negative	-2

The twice-differentiable functions f and g are defined for all real numbers x. Values of f, f', g, and g' for various values of x are given in the table above.

(a) Find the x-coordinate of each relative minimum of f on the interval [-2, 3]. Justify your answers.

(b) Explain why there must be a value c, for $-1 < c < 1$, such that $f''(c) = 0$.

(c) The function h is defined by $h(x) = \ln(f(x))$. Find $h'(3)$. Show the computations that lead to your answer.

(d) Evaluate $\int_{-2}^{3} f'(g(x))g'(x)\,dx$.

Part A

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.

0	1 ✓

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 1$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a relative minimum at $x = 1$.

Part B

1 point is earned for $f'(1) - f'(-1) = 0$

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable ⇒ f' is continuous on the interval $-1 \le x \le 1$

$$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c) = 0$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $f'(1) - f'(-1) = 0$

3-1-1b

1 point is earned for the explanation, using Mean Value Theorem

f' is differentiable ⇒ f' is continuous on the interval $-1 \le x \le 1$

$$\frac{f'(1)-f'(-1)}{1-(-1)} = \frac{0-0}{2} = 0$$

Therefore, by the Mean Value Theorem, there is at least one value c, $-1 < c < 1$, such that $f''(c) = 0$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

0	1	2	3 ✓

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{1}{f(x)} \cdot f'(x)$$
$$h'(3) = \frac{1}{f(3)} \cdot f'(3) = \frac{1}{7} \cdot \frac{1}{2} = \frac{1}{14}$$

Part D

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_{-2}^{3} f'(g(x))g'(x)\,dx &= \left[f(g(x))\right]_{x=-2}^{x=3} \\ &= f(g(3)) - f(g(-2)) \\ &= f(1) - f(-1) \\ &= 2 - 8 = -6\end{aligned}$$

0	1	2	3 ✓

The student response earns all of the following points:

3-1-1b

2 points are earned for stating Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_{-2}^{3} f'(g(x))g'(x)\,dx &= [f(g(x))]_{x=-2}^{x=3}\\ &= f(g(3)) - f(g(-2))\\ &= f(1) - f(-1)\\ &= 2 - 8 = -6\end{aligned}$$

39. If $\frac{d}{dx}f(x) = g(x)$ and if $h(x) = x^2$, then $\frac{d}{dx}(f(h(x))) =$

(A) $g(x^2)$

(B) $2xg(x)$

(C) $g'(x)$

(D) $2xg(x^2)$ ✓

(E) $x^2g(x^2)$

40. The slope of the line tangent to the graph of $y = \ln(1 - x)$ at x = -1 is

(A) -1

(B) -1/2 ✓

(C) 1/2

(D) ln 2

(E) 1

41. The slope of the line tangent to the graph of y = ln(1-x) at x = -1 is

(A) -1

(B) -1/2 ✓

(C) 1/2

(D) ln2

(E) 1

42. If $\frac{d}{dx}(f(x)) = g(x)$ and $\frac{d}{dx}(g(x)) = f(x^2)$, then $\frac{d^2}{dx^2}(f(x^3)) =$

(A) $f(x^6)$

(B) $g(x^3)$

(C) $3x^2g(x^3)$

(D) $9x^4f(x^6) + 6x\,g(x^3)$ ✓

(E) $f(x^6) + g(x^3)$

43. If *f* and *g* are twice differentiable and if $h(x) = f(g(x))$, then $h''(x) =$

(A) $f''(g(x))[g'(x)]^2 + f'(g(x))g''(x)$ ✓

(B) $f''(g(x))g'(x) + f'(g(x))g''(x)$

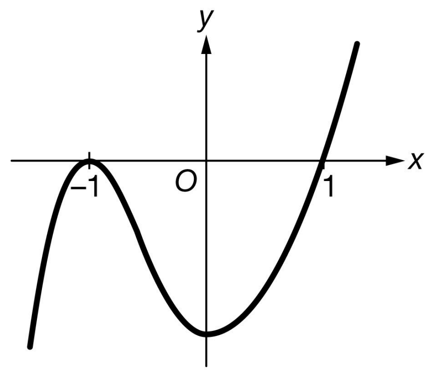
(C) $f''(g(x))[g'(x)]^2$

(D) $f''(g(x))g''(x)$

(E) $f''(g(x))$

3-1-1b

44.



Graph of f

The figure above shows the graph of a differentiable function f. If $g(x) = f\left(x^2\right)$, which of the following statements is true?

(A) $g'(-1) < 0$

(B) $g'(-1) > 0$

(C) $g'(1) < 0$

(D) $g'(1) = 0$

Answer A

Correct. By the chain rule, $g'(x) = 2xf'\left(x^2\right) \Rightarrow g'(-1) = -2f'(1)$. The graph of f is increasing for $x > 0$, and therefore $f'(1) > 0$. Therefore, $g'(-1) < 0$.

3-1-1b

45. Let f be the function given by $f(x) = \frac{x}{\sqrt{x^2-4}}$.

Find $f'(x)$

Part D

1 point is earned for derivative

2 point is earned for solution

$$f\prime(x) = \frac{\sqrt{x2-4-x\left[1/2(x2-4)^{-1/2}2x\right]}}{x2-4}$$
$$= \frac{\sqrt{x^2-4}-\frac{x2}{\sqrt{x^2-4}}}{x^2-4}$$
$$= \frac{-4}{(x^2-4)^{3/2}}$$

0	1	2	3

The student response earns three of the following points:

1 point is earned for derivative

2 point is earned for solution

$$f\prime(x) = \frac{\sqrt{x2-4-x\left[1/2(x2-4)^{-1/2}2x\right]}}{x2-4}$$
$$= \frac{\sqrt{x^2-4}-\frac{x2}{\sqrt{x^2-4}}}{x^2-4}$$
$$= \frac{-4}{(x^2-4)^{3/2}}$$

46. $\int \frac{x^2+1}{\left(x^3+3x-5\right)^3} dx =$

(A) $-\frac{3}{2} \cdot \frac{1}{\left(3x^2+3\right)^2} + C$

(B) $-\frac{1}{6} \cdot \frac{1}{\left(3x^2+3\right)^2} + C$

(C) $-\frac{3}{2} \cdot \frac{1}{\left(x^3+3x-5\right)^2} + C$

(D) $-\frac{1}{6} \cdot \frac{1}{\left(x^3+3x-5\right)^2} + C$

Answer D

Correct. Starting with the substitution $u = x^3 + 3x - 5$, $u = x^3 + 3x - 5 \Rightarrow \frac{du}{dx} = 3x^2 + 3 = 3\left(x^2+1\right) \Rightarrow dx = \frac{du}{3(x^2+1)}$. Substituting for $x^3 + 3x - 5$ and for dx gives $\int \frac{x^2+1}{(x^3+3x-5)^3} dx = \int \frac{1}{u^3} \cdot \frac{1}{3} du = \frac{1}{3} \cdot \left(-\frac{1}{2u^2}\right) + C = -\frac{1}{6} \cdot \frac{1}{(x^3+3x-5)^2} + C$.

3-1-1b

47. $\frac{d}{dx}\left(2\left(\sin\sqrt{x}\right)^2\right) =$

(A) $4\cos\left(\frac{1}{2\sqrt{x}}\right)$

(B) $4\sin\sqrt{x}\,\cos\sqrt{x}$

(C) $\frac{2\sin\sqrt{x}}{\sqrt{x}}$

(D) $\frac{2\sin\sqrt{x}\cos\sqrt{x}}{\sqrt{x}}$

Answer D

Correct. The chain rule must be used twice for this composition of three functions.

$\frac{d}{dx}\left(2\left(\sin\sqrt{x}\right)^2\right) = 2\cdot 2\left(\sin\sqrt{x}\right)\cdot\left(\frac{d}{dx}\left(\sin\sqrt{x}\right)\right)$

$= 2\cdot 2(\sin\sqrt{x})\cdot\left(\cos\sqrt{x}\cdot\frac{d}{dx}\left(\sqrt{x}\right)\right)$

$= 2\cdot 2\left(\sin\sqrt{x}\right)\cdot\cos\sqrt{x}\cdot\frac{1}{2\sqrt{x}}$

$= \frac{2\sin\sqrt{x}\cos\sqrt{x}}{\sqrt{x}}$

48. If $f(x) = \sqrt[3]{5x^2-7}$, then $f'(x) =$

(A) $\frac{10x}{3}\left(5x^2-7\right)^{-\frac{2}{3}}$

(B) $\frac{1}{3}\left(5x^2-7\right)^{-\frac{2}{3}}$

(C) $\frac{3}{4}\left(5x^2-7\right)^{\frac{4}{3}}$

(D) $\frac{1}{3}(10x)^{-\frac{2}{3}}$

Answer A

Correct. A combination of the power rule and the chain rule is used to compute the derivative.

$f'(x) = \frac{d}{dx}\left(5x^2-7\right)^{\frac{1}{3}} = \frac{1}{3}\left(5x^2-7\right)^{-\frac{2}{3}}\cdot\frac{d}{dx}\left(5x^2-7\right) = \frac{1}{3}\left(5x^2-7\right)^{-\frac{2}{3}}\cdot(10x) = \frac{10x}{3}\left(5x^2-7\right)^{-\frac{2}{3}}$

3-1-1b

49. If $g(x) = \ln x$ and f is a differentiable function of x, which of the following is equivalent to the derivative of $f(g(x))$ with respect to x ?

(A) $f'\left(\frac{1}{x}\right)$

(B) $\frac{f'(x)}{x}$

(C) $f'(\ln x)$

(D) $\frac{f'(\ln x)}{x}$

Answer D

Correct. The chain rule provides a way to differentiate a composition of functions. The "outside" function is $f(x)$ and the "inside" function is $g(x) = \ln x$. Then $g'(x) = \frac{1}{x}$ and the chain rule can be used, as follows.
$\frac{d}{dx} f(\ln x) = \frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) = f'(\ln x) \cdot \frac{1}{x}$

50. For which of the following functions is the chain rule an appropriate method to find the derivative with respect to x ?

I. $y = \sin\left(3x^2\right)$
II. $y = e^x \tan x$
III. $y = \frac{1}{8x^4 - 2x}$

(A) I only

(B) II only

(C) III only

(D) I and III only

Answer D

Correct. Each of the functions in I and III is a composition of component functions.
In I, let $f(x) = \sin x$ and let $g(x) = 3x^2$. Then $\sin\left(3x^2\right) = f(g(x))$, and the chain rule would give
$\frac{d}{dx} \sin\left(3x^2\right) = f'(g(x))g'(x) = \cos\left(3x^2\right)6x$.
In III, let $f(x) = \frac{1}{x} = x^{-1}$ and $g(x) = 8x^4 - 2x$. Then $\frac{1}{8x^4 - 2x} = f(g(x))$, and the chain rule would give
$\frac{d}{dx} \frac{1}{8x^4 - 2x} = f'(g(x))g'(x) = -\frac{1}{\left(8x^4 - 2x\right)^2} \cdot \left(32x^3 - 2\right)$.
In II, the function $y = e^x \tan x$ is the product of the exponential function and the tangent function and is not the composition of two component functions. The product rule would be used to find the derivative, and the chain rule would not be appropriate.

3-1-1b

51. Let f be a differentiable function. If $h(x) = (1 + f(3x))^2$, which of the following gives a correct process for finding $h'(x)$?

(A) $h'(x) = 2(1 + f(3x))$

(B) $h'(x) = 2(1 + f(3x)) \cdot f'(3x)$

(C) $h'(x) = 2(1 + f(3x)) \cdot f'(x)$

(D) $h'(x) = 2(1 + f(3x)) \cdot f'(3x) \cdot 3$

Answer D

Correct. This requires using the chain rule twice to compute the derivative.
$h'(x) = 2(1 + f(3x)) \cdot \left(\frac{d}{dx}(1 + f(3x))\right) = 2\left(1 + f(3x)\right)\left(f'(3x) \cdot \frac{d}{dx}(3x)\right) = 2(1 + f(3x)) \cdot f'(3x) \cdot 3$

52. Let f be the function defined by $f(x) = e^{h(x)}$, where h is a differentiable function. Which of the following is equivalent to the derivative of f with respect to x ?

(A) $e^{h(x)}$

(B) $e^{h'(x)}$

(C) $h'(x)e^{h(x)}$

(D) $h(x)e^{h(x)-1}$

Answer C

Correct. The function f is the composition of the exponential function e^x with the function h. The chain rule provides a way to differentiate a composite function. The “outside” function is $g(x) = e^x$, and the “inside” function is $h(x)$. Then $g'(x) = e^x$, and the chain rule can be used, as follows.
$\frac{d}{dx}e^{h(x)} = \frac{d}{dx}g(h(x)) = g'(h(x)) \cdot h'(x) = e^{h(x)} \cdot h'(x)$

53. Let $f(x) = x^3$ and $g(x) = \frac{x}{x-1}$. If h is the function defined by $h(x) = f(g(x))$, which of the following gives a correct expression for $h'(x)$?

(A) $3(g(x))^2 = 3\left(\frac{x}{x-1}\right)^2$

(B) $3(g'(x))^2 = 3\left(\frac{(x-1)-x}{(x-1)^2}\right)^2$

(C) $3(g(x))^2 g'(x) = 3\left(\frac{x}{x-1}\right)^2 \cdot \frac{(x-1)-x}{(x-1)^2}$

(D) $(g'(x))^3 = \left(\frac{(x-1)-x}{(x-1)^2}\right)^3$

Answer C

Correct. The function h is the composition of the “outside” function $f(x) = x^3$ and the “inside” function $g(x) = \frac{x}{x-1}$. The appropriate differentiation rule to apply is therefore the chain rule. Differentiating the “inside” function also requires use of the quotient rule, as follows.
$h'(x) = \frac{d}{dx}f(g(x)) = f'(g(x))g'(x) = 3(g(x))^2 g'(x) = 3\left(\frac{x}{x-1}\right)^2 \cdot \frac{(x-1)-x}{(x-1)^2}$

3-1-1b

54. If $f(x) = \left(e^{3x} + \sin(2x)\right)^4$, then $f'(x) =$

(A) $4\left(3e^{3x} + 2\cos(2x)\right)^3$

(B) $4\left(e^{3x} + \sin(2x)\right)^3 \left(e^{3x} + \cos(2x)\right)$

(C) $4\left(e^{3x} + \sin(2x)\right)^3 \left(3e^{3x} + 2\sin(2x)\right)$

(D) $4\left(e^{3x} + \sin(2x)\right)^3 \left(3e^{3x} + 2\cos(2x)\right)$

Answer D

Correct. Both the power rule and the chain rule must be used to compute the derivative. The chain rule is used several times, as follows.
$f'(x) = 4\left(e^{3x} + \sin(2x)\right)^3 \cdot \frac{d}{dx}\left(e^{3x} + \sin(2x)\right) = 4\left(e^{3x} + \sin(2x)\right)^3 \left(3e^{3x} + 2\cos(2x)\right)$

55. If $g(x) = 2\ln(x+1)$ and f is a differentiable function of x, which of the following is equivalent to the derivative of $f(g(x))$ with respect to x ?

(A) $f'\left(\frac{2}{x+1}\right)$

(B) $\frac{2f'(x)}{x+1}$

(C) $f'(2\ln(x+1))$

(D) $\frac{2f'(2\ln(x+1))}{x+1}$

Answer D

Correct. The chain rule provides a way to differentiate a composition of functions. The "outside" function is $f(x)$ and the "inside" function is $g(x) = 2\ln(x+1)$. Then $g'(x) = \frac{2}{x+1}$ and the chain rule can be used, as follows.
$\frac{d}{dx} f(2\ln(x+1)) = \frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x) = f'(2\ln(x+1)) \cdot \frac{2}{x+1}$

3-1-1b

56. For which of the following functions is the chain rule an appropriate method to find the derivative with respect to x ?

I. $y = \cos\left(\sqrt{x}+1\right)$
II. $y = 2^x \sin x$
III. $y = \frac{20}{40x^2-1}$

(A) I only
(B) II only
(C) III only
(D) I and III only ✓

Answer D

Correct. Each of the functions in I and III is a composition of component functions.
In I, let $f(x) = \cos x$ and let $g(x) = \sqrt{x}+1$. Then $\cos\left(\sqrt{x}+1\right) = f(g(x))$, and the chain rule would give $\frac{d}{dx}\cos\left(\sqrt{x}+1\right) = f'(g(x))g'(x) = -\sin\left(\sqrt{x}+1\right)\cdot\left(\frac{1}{2\sqrt{x}}\right)$.
In III, let $f(x) = \frac{20}{x} = 20x^{-1}$ and $g(x) = 40x^2-1$. Then $\frac{20}{40x^2-1} = f(g(x))$, and the chain rule would give $\frac{d}{dx}\frac{20}{40x^2-1} = f'(g(x))g'(x) = -\frac{20}{(40x^2-1)^2}\cdot(80x)$.
In II, the function $y = 2^x \sin x$ is the product of an exponential function and the sine function and is not the composition of two component functions. The product rule would be used to find the derivative, and the chain rule would not be appropriate.

57. Let f be a differentiable function. If $h(x) = (2+f(\sin x))^3$, which of the following gives a correct process for finding $h'(x)$?

(A) $h'(x) = 3(2+f(\sin x))^2$
(B) $h'(x) = 3(2+f(\sin x))^2 \cdot f'(\sin x)$
(C) $h'(x) = 3(2+f(\sin x))^2 \cdot f'(\cos x)$
(D) $h'(x) = 3(2+f(\sin x))^2 \cdot f'(\sin x)\cdot\cos x$ ✓

Answer D

Correct. This requires using the chain rule twice to compute the derivative.
$$h'(x) = 3(2+f(\sin x))^2\cdot\left(\frac{d}{dx}(2+f(\sin x))\right) = 3(2+f(\sin x))^2\cdot\left(f'(\sin x)\cdot\frac{d}{dx}(\sin x)\right) = 3(2+f(\sin x))^2\cdot f'(\sin x)\cdot\cos x$$

3-1-1b

58. Let f be the function defined by $f(x) = \sin(h(x))$, where h is a differentiable function. Which of the following is equivalent to the derivative of f with respect to x ?

(A) $\cos(h(x))$

(B) $\cos(h\prime(x))$

(C) $\cos(h(x))h\prime(x)$ ✓

(D) $\sin(h(x))h\prime(x)$

Answer C

Correct. The function f is the composition of the trigonometric function $\sin x$ with the function h. The chain rule provides a way to differentiate a composite function. The "outside" function is $g(x) = \sin x$, and the "inside" function is $h(x)$. Then $g'(x) = \cos x$, and the chain rule can be used, as follows.
$\frac{d}{dx}\sin(h(x)) = \frac{d}{dx}g(h(x)) = g'(h(x)) \cdot h'(x) = \cos(h(x)) \cdot h'(x)$

59. Let $f(x) = 5x^4$ and $g(x) = e^{2x} + x$. If h is the function defined by $h(x) = f(g(x))$, which of the following gives a correct expression for $h'(x)$?

(A) $20(g(x))^3 = 20\left(e^{2x} + x\right)^3$

(B) $20(g'(x))^3 = 20\left(2e^{2x} + 1\right)^3$

(C) $20(g(x))^3 \cdot g'(x) = 20\left(e^{2x} + x\right)^3 \cdot \left(2e^{2x} + 1\right)$ ✓

(D) $5(g'(x))^4 = 5\left(2e^{2x} + 1\right)^4$

Answer C

Correct. The function h is the composition of the "outside" function $f(x) = 5x^4$ and the "inside" function $g(x) = e^{2x} + x$. The appropriate differentiation rule to apply is therefore the chain rule.
$h'(x) = \frac{d}{dx}f(g(x)) = f'(g(x))g'(x) = 20(g(x))^3 g'(x) = 20\left(e^{2x} + x\right)^3 \cdot \left(2e^{2x} + 1\right)$

3-1-1b

60. If $f(x) = \left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^3$, then $f'(x) =$

(A) $3\left(-\frac{1}{2\sqrt{x}}\sin\left(\sqrt{x}\right) - \frac{2}{x}\right)^2$

(B) $3\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^2 \cdot \left(-\sin\left(\sqrt{x}\right) - \frac{1}{x^2}\right)$

(C) $3\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^2 \cdot \left(\frac{1}{2\sqrt{x}}\cos\left(\sqrt{x}\right) - \frac{2}{x}\right)$

(D) $3\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^2 \cdot \left(-\frac{1}{2\sqrt{x}}\sin\left(\sqrt{x}\right) - \frac{2}{x}\right)$

Answer D

Correct. Both the power rule and the chain rule must be used to compute the derivative. The chain rule is used several times, as follows.

$f'(x) = 3\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^2 \cdot \frac{d}{dx}\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right) = 3\left(\cos\left(\sqrt{x}\right) - \ln\left(x^2\right)\right)^2 \left(-\frac{1}{2\sqrt{x}}\sin\left(\sqrt{x}\right) - \frac{2}{x}\right)$

3-1-1b

61. Let f be the function defined by $f(x)=(1+\tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.

Write an equation for the line tangent to the graph of f at the point where $x=0$.

Part A

1 point is earned for $f(0)=1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$f(0)=1$
$f'(x)=\frac{3}{2}(1+\tan(x))^{1/2}\left(\sec^2(x)\right)$
$f'(0)=\frac{3}{2}$
$y-1=\frac{3}{2}x$ or $y=\frac{3}{2}x+1$

0	1	2	3	4	5 ✓

The student response earns five of the following points:

1 point is earned for $f(0)=1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$f(0)=1$
$f'(x)=\frac{3}{2}(1+\tan(x))^{1/2}\left(\sec^2(x)\right)$
$f'(0)=\frac{3}{2}$
$y-1=\frac{3}{2}x$ or $y=\frac{3}{2}x+1$

3-1-1b

62. Let f be the function defined by $f(x) = (1 + \tan x)^{3/2}$ for $-\pi/4 < x < \pi/2$.

(a) Write an equation for the line tangent to the graph of f at the point where $x = 0$.

(b) Using the equation found in part (a), approximate $f(0.02)$.

(c) Let f^{-1} denote the inverse function of f. Write an expression that gives $f^{-1}(x)$ for all x in the domain of f^{-1}.

Part A

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$f(0) = 1$

$$f'(x) = \frac{3}{2}(1+\tan(x))^{\frac{1}{2}}\left(\sec^2(x)\right)$$
$$f'(0) = \frac{3}{2}$$
$$\boxed{y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1}$$

0	1	2	3	4	5

The student response earns all of the following points:

1 point is earned for $f(0) = 1$

2 points are earned for $f'(x)$

1 point is earned for $f'(0)$

1 point is earned for equation of line. Must use viable $f(0)$ and $f'(0)$

Note: max 2/5 for answer only

$f(0) = 1$

$$f'(x) = \frac{3}{2}(1+\tan(x))^{\frac{1}{2}}\left(\sec^2(x)\right)$$
$$f'(0) = \frac{3}{2}$$
$$\boxed{y - 1 = \frac{3}{2}x \quad \text{or} \quad y = \frac{3}{2}x + 1}$$

Part B

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

3-1-1b

$f(0.02) \approx \boxed{\frac{3}{2}(0.02)+1}$

0	1

The student response earns all of the following points:

1 point is earned for evaluating their equation from part (a) at $x = 0.02$

$f(0.02) \approx \boxed{\frac{3}{2}(0.02)+1}$

Part C

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for tan y

1 point is earned for solving for y

$$x = (1+\tan y)^{3/2}$$
$$x^{2/3} = 1+\tan y$$
$$\tan y = x^{2/3} - 1$$
$$\boxed{\begin{array}{l} y = \arctan\left(x^{2/3}-1\right) \\ \qquad \text{OR} \\ f^{-1}(x) = \arctan\left(x^{2/3}-1\right) \end{array}}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for reversing variables anywhere in process

1 point is earned for solving for tan y

1 point is earned for solving for y

$$x = (1+\tan y)^{3/2}$$
$$x^{2/3} = 1+\tan y$$
$$\tan y = x^{2/3} - 1$$
$$\boxed{\begin{array}{l} y = \arctan\left(x^{2/3}-1\right) \\ \qquad \text{OR} \\ f^{-1}(x) = \arctan\left(x^{2/3}-1\right) \end{array}}$$

3-1-1b

63. If $f(x) = \sin\left(x^2 + \pi\right)$, then $f'\left(\sqrt{2\pi}\right) =$

(A) $-2\sqrt{2\pi}$ ✓

(B) -2

(C) -1

(D) $\cos\left(2\sqrt{2\pi}\right)$

64. If $f(x) = \sqrt{x^2 - 4}$ and $g(x) = 3x - 2$, then the derivative of $f(g(x))$ at $x = 3$ is

(A) $\frac{7}{\sqrt{5}}$ ✓

(B) $\frac{14}{\sqrt{5}}$

(C) $\frac{18}{\sqrt{5}}$

(D) $\frac{15}{\sqrt{21}}$

(E) $\frac{30}{\sqrt{21}}$